



THÈSE DE DOCTORAT
DE L'UNIVERSITÉ PSL

Préparée à l'École Supérieure de Physique
et de Chimie Industrielles de la ville de Paris

**Le titre du manuscrit de thèse pour obtenir le titre de
docteur**

Soutenue par

Prénom Nom

Le xx mois 202x

École doctorale n°xxx

Titre de l'école doctorale

Spécialité

Acoustique

Préparée au

Laboratoire de Recherche

Composition du jury :

Prénom NOM 1

Affiliation

*Président du jury
Examineur*

Prénom NOM 2

Affiliation

Rapporteur

Prénom NOM 3

Affiliation

Rapporteur

Prénom NOM 4

Affiliation

Rapporteur

Prénom NOM 5

Affiliation

Rapporteur

Prénom NOM 6

DR Affiliation

Directeur de thèse

Remerciements

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Mots clés : mot clé 1, mot clé 2, mot clé 3, mot clé 4

Abstract

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Keywords : keyword 1, keyword 2, keyword 3, keyword 4

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Chapitre 1

Introduction

Objectifs

Objectif rapide du chapitre Sed mattis, erat sit amet gravida malesuada, elit augue egestas diam, tempus scelerisque nunc nisl vitae libero. Sed consequat feugiat massa. Nunc porta, eros in eleifend varius, erat leo rutrum dui, non convallis lectus orci ut nibh. Sed lorem massa, nonummy quis, egestas id, condimentum at, nisl. Maecenas at nibh. Aliquam et augue at nunc pellentesque ullamcorper. Duis nisl nibh, laoreet suscipit, convallis ut, rutrum id, enim. Phasellus odio. Nulla nulla elit, molestie non, scelerisque at, vestibulum eu, nulla. Ut odio nisl, facilisis id, mollis et, scelerisque nec, enim. Aenean sem leo, pellentesque sit amet, scelerisque sit amet, vehicula pellentesque, sapien.

1.1 Quelques conseils d'utilisation

1.1.1 Organisation du projet

Fichier maître à compiler Le fichier `main.tex` contient tous les sous-fichiers à inclure. C'est l'organisation du document.

Configuration Le dossier `config/` contient les fichiers de paramétrage du document, les macros, ainsi que le package pour la couverture PSL (section 1.1.2).

Chapitres Ils sont situés dans des dossiers indépendants, avec des dossiers de figures associés (`.../fig/`). L'auto-complétion permet de trouver les figures associées au chapitre voulu : `ch1/fig/xxxx.jpeg`.

Autres fichiers et pages Les autres fichiers de texte (conclusion, remerciements), et information (info, liste des publications, glossaire) sont situés dans le dossier `others/`.

1.1.2 Couvertures par Pierre Guillou pour PSL

Seules les couvertures sont imposées par PSL. Le fichier ici a été réalisée par [Pierre Guillou](https://pierre.guillou.net/psl-cover/2018/)¹ et est disponible sur le site dans les [ressources du site de PSL](https://collegedoctoral.psl.eu/doctorat-psl/espace-ressources/)² (Version 1.2 - 20 juillet 2019). A noter que le président du jury doit être sur la première ligne de la liste du jury.

1.1.3 Illustrations

Overleaf propose l'auto-complétion pour les figures de figures, ce qui est assez rapide et efficace. Un label est ajouté pour chaque figure afin de la citer après. La figure suivant est référencée par le label `fig1_cir` car c'est une figure du chapitre 1. Pour une table du chapitre 1, on pourra utiliser `tab1_power`. La figure [fig. 1.1a] illustre ce propos, pour plus de détails voir le chapitre 3.

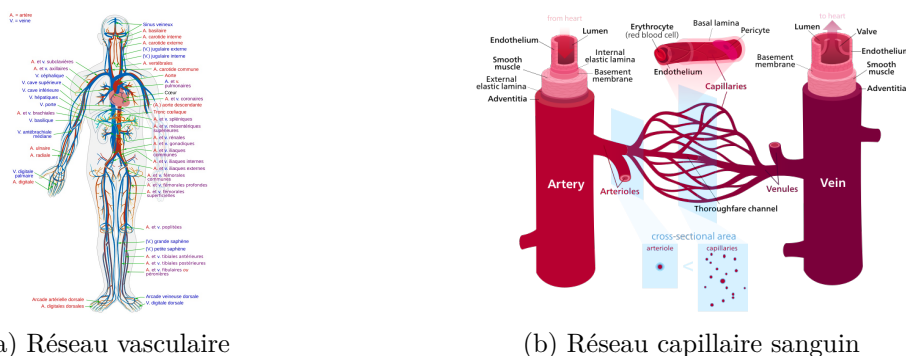


FIGURE 1.1 – (a) Le réseau vasculaire est composé des réseaux artériel et veineux. (b) Les capillaires sanguins réalisent la jonction entre les deux réseaux. [Wikimedia]

1. <https://pierre.guillou.net/psl-cover/2018/>
 2. <https://collegedoctoral.psl.eu/doctorat-psl/espace-ressources/>

1.2 Une deuxième section

1.2.1 Acronymes

C'est assez pratique d'utiliser des acronymes, mais il faut les définir avant.

Ici, un commentaire

A faire

Ici on va utiliser le GPU ou la *Random Access Memory*, ou encore *Solid-State Drive (SSD)*.

1.2.2 Quelques citations

1.2.2.1 Avec la commande Cite

On peut citer des références [ueda__tissue__2020; thomas__microbubble__2013] pour appuyer le propos. La table des matières est réalisée avec biblatex.

1.2.2.2 En synchronisant Zotero

Zotero peut être synchronisé pour faire la gestion de la bibliographie. Il ne suffit que de mettre à jour le .bib de temps en temps.

Chapitre 2

Literature review: calibrating comparisons between child and artificial language learners

Objectifs

Establish when comparisons between children and artificial language learners support developmental inference by jointly calibrating the learners' input, evaluation outcomes, developmental trajectories, and uncertainty.

2.1 Scope and inferential framework

Understanding how children acquire language from limited experience is a central problem in developmental science, but the relevant learning conditions cannot be manipulated freely in children. Researchers cannot randomly assign infants to radically different linguistic environments, control the timing of every construction they hear, or intervene directly on the learning architecture they bring to the task. Computational models complement observational and experimental work because they instantiate explicit learning hypotheses and can be exposed to controlled input. In this role, a model can address a learnability question: whether a specified learning procedure is sufficient to acquire a target pattern from a specified body of experience [Chater, 2006; Perfors, 2011; Dupoux, 2018].

This logic predates contemporary large language models. Early connectionist models were deliberately constructed as cognitive simulations and used to test whether apparently rule-governed behavior could emerge from pattern learning [Rumelhart, 1986; Elman, 1990]. Recurrent networks were subsequently used to study phenomena such as agreement and auxiliary inversion under distributional learning accounts [Elman, 1991; Lewis, 2001; Real, 2004]. Modern language models differ in an important respect: they are generally optimized for engineering objectives, trained on large collections of adult-written text, and not designed as process models of development [Brown, 2020; Warstadt, 2022]. Their strong performance on syntactic, semantic, and reasoning benchmarks is therefore scientifically suggestive, but it does not by itself establish that they learn as children do [Linzen, 2021; Mahowald, 2024].

The inference from a successful simulation is asymmetric. If a model learns a phenomenon under conditions that adequately approximate the child’s learning problem, its success provides evidence of *sufficiency*: the represented input and learning biases can, in principle, produce the observed outcome. Failure is less decisive. It may reveal an inadequate learning mechanism, but it may instead arise from missing modalities, an unsuitable objective, an inappropriate evaluation, or input unlike that available to the child. Human learners may also exploit working-memory constraints, perceptual grounding, or social scaffolding that the model does not represent [Tomasello, 2003; Warstadt, 2022]. A model failure consequently does not establish that a phenomenon is unlearnable from experience, nor does model success establish mechanistic identity.

This chapter develops a calibrated-comparison framework around two necessary dimensions. *Outcome alignment* asks whether the model and the child are evaluated on the same construct, using stimuli, response contrasts, and developmental summaries that support a common interpretation. *Input alignment* asks whether the model’s training experience approximates child-available input in amount, register, modality, interactional structure, and temporal organization. The two dimensions are related but not interchangeable. A model trained on child-directed speech but evaluated only by an engineering benchmark provides little direct evidence about child behavior. Conversely, a model tested with developmental stimuli after training on web-scale adult text answers a different learnability question from the one faced by a child.

Architecture is treated here as an experimental variable, not as a third quantity that can

be matched by inspection. At Marr’s algorithmic level, recurrent, attentional, multimodal, and retrieval-augmented systems embody different hypotheses about memory, representation, and inductive bias [Marr, 1982]. Yet there is no independently validated specification of the algorithm used by the developing child against which an architecture can simply be checked. Architectural comparisons become interpretable when input and outcomes are held sufficiently constant: a systematic difference can then identify which properties are sufficient or insufficient under those conditions. Claims of architectural plausibility made before these controls remain undetermined.

The review is organized by the methodological decisions that determine those inferences. Section 2.2 examines task affordances, categorical and graded metrics, developmental trajectories, and measurement uncertainty. Section 2.3 examines the quality, quantity, and temporal organization of model input. Section 2.4 then states reporting requirements and identifies the empirical gaps addressed by the remainder of this thesis. The review is concept-driven and selective rather than a systematic or meta-analytic census. The study tables therefore map representative paradigms across domains; they should not be read as exhaustive inventories of every developmental experiment or model benchmark.

2.2 Outcome alignment

2.2.1 Task affordances across linguistic domains

Outcome alignment is stronger than topical similarity. Giving a child and a model stimuli about “syntax” is not sufficient if the child’s dependent variable indexes preference after familiarization while the model’s score is the probability of an isolated sentence. A calibrated comparison should specify at least four correspondences: the linguistic construct being probed, the information available in the stimulus, the decision or response that produces the dependent variable, and the point in development or training at which it is observed. Where any one of these changes, the interpretation must be qualified.

Child competence is usually inferred through an overt behavior—sucking rate, head turning, visual fixation, pointing, or production—after sensory encoding, attention, decision making, and motor execution. A model usually exposes a probability or hidden representation, which is converted into an experimental outcome by an additional *task model*: a distance function, probe, classifier, prompting protocol, or scoring rule. The task model is part of the hypothesis. Layer choice, pooling, prompt wording, and decision threshold can change the apparent competence even when the underlying trained model is fixed. Table 2.1 uses four deliberately conservative alignment categories: *close* when matched stimuli and decisions permit trial-level comparison; *partial* when construct and contrast are shared but response generation differs; *indirect* when only a broader theoretical construct is shared; and *none established* when the literature lacks a defensible counterpart.

TABLE 2.1 – Alignment of representative developmental and model evaluation paradigms. Ratings describe the current measurement correspondence, not the importance of the linguistic domain.

Domain	Child measure	Model measure	Alignment	Representative evidence
Phonology	Habituation, high-amplitude sucking, or conditioned head turning to an acoustic contrast	Distance-based ABX discrimination over selected model representations	Partial	[Eimas, 1971 ; Werker, 1984 ; Schatz, 2013 ; Matusevych, 2023]
Morphology	Contextualized elicitation of an inflected or derived nonce form	Generation from a nonce lemma and sentence frame, sometimes with explicit feature labels	Partial to close	[Berko, 1958 ; Weissweiler, 2023 ; Anh, 2024]
Word form	Familiarization followed by preference for words versus part-words; spontaneous or reported vocabulary production	Probability-based word/nonword discrimination or generation of forms	Partial	[Saffran, 1996 ; Aslin, 1998 ; Nguyen, 2022]
Word meaning	Spoken word–referent matching by looking or pointing	Forced-choice image–text matching	Close for matched trials	[Bergelson, 2012 ; Lany, 2011 ; Wang, 2025]
Semantic categories	Sorting or response-time effects for category membership and typicality	Geometry or similarity of learned embeddings	Indirect	[Lenci, 2022]
Syntax	Artificial-grammar preference or child production; later explicit judgments	Probability comparison for grammatical minimal pairs	Partial; potentially close for matched discrimination	[Marcus, 1999 ; Warstadt, 2020a ; Hu, 2020]
Sentence meaning	Looking-time, violation-of-expectation, or neural response to semantic anomaly	Sentence probability, surprisal, or semantic minimal-pair accuracy	Partial	[Margoni, 2023 ; Friedrich, 2006 ; Kauf, 2023 ; Michaelov, 2024]
Pragmatics	Action prediction or response in socially grounded belief and communicative tasks	Prompted textual answers to narrated scenarios	Indirect; no grounded production analogue	[Wimmer, 1983 ; Wellman, 2001 ; Kosinski, 2024 ; Strachan, 2024]

Phonological knowledge. Early phonological perception is commonly studied with habituation and dishabituation, high-amplitude sucking, or conditioned head-turn procedures. A change in sucking, looking, or head orientation after a stimulus change is interpreted as discrimination [Eimas, 1971 ; Best, 1988 ; Kuhl, 1992 ; Sundara, 2018]. These paradigms revealed the reorganization of sensitivity to native and non-native contrasts during the first year [Werker, 1984 ; Tsuji, 2014]. The dependent variable, however, reflects more than auditory separation: the infant must encode the signal, judge it sufficiently novel, and produce a measurable response. Familiarity,

fatigue, and task parameters affect whether discrimination is expressed.

Speech models are often evaluated with an ABX task. Given tokens A , B , and X , the task asks whether the representation of X is closer to A or B [Schatz, 2013; Schatz, 2014; Dunbar, 2021; Lavechin, 2023]. This avoids training a downstream phoneme classifier, but it does not eliminate modeling choices. The selected layer and distance metric determine which acoustic or phonological information contributes to the score. ABX is also structurally closer to a three-item psychophysical comparison than to an infant’s novelty response. Comparisons of ABX trajectories with infant contrast sensitivity are therefore informative when languages, contrasts, exposure, and testing intervals are matched, but current correspondence remains partial [Schatz, 2021; Matuskevych, 2023; Poli, 2024].

Production offers a possible closer bridge because acoustic properties can be extracted from both child and generated speech. In practice, child vocalizations involve articulatory development, phonological approximation, and transcription decisions, whereas most computational production work has been evaluated qualitatively or against adult targets. A calibrated production comparison would need common acoustic measures, age-appropriate targets, and an explicit model of how representational knowledge becomes an articulatory output.

Morphological knowledge. Children’s productive morphology is classically probed by eliciting inflected or derived forms of nonce words in meaningful sentence and picture contexts [Berko, 1958; Engelmann, 2019; Michaly, 2025]. Perceptual work can reveal earlier sensitivity to morphological structure—for example, 6- and 8-month-olds distinguish relations between suffixed nonce forms and stems [Kim, 2021]—but that outcome is not the same as productive generalization. The point of the Wug paradigm is that a response cannot be a stored form: it must extend a pattern to a novel stem. Computational evaluations range from supervised paradigm completion, where a model receives a source form and a derivational slot [Cotterell, 2017], to Wug-style generation in which a nonce form appears in a sentence frame [Weissweiler, 2023; Anh, 2024].

Sentence-framed generation is the closest counterpart because the populations can receive parallel contexts and their produced forms can be coded by the same rule. Alignment weakens when the model is given an explicit feature bundle or list of candidate affixes that the child does not see. Discriminative well-formedness judgments are useful diagnostics, but they are not substitutes for an elicited-production comparison unless a corresponding child judgment task is administered. The strongest design therefore matches the context, nonce inventory, response opportunity, and scoring procedure rather than treating every morphology benchmark as equivalent to a Wug test.

Word forms and lexical production. Infant word segmentation is often tested by familiarizing infants with a continuous speech stream and then contrasting words with part-words [Saffran, 1996; Aslin, 1998]. Evidence that infants exploit sequential statistics extends to natural-language materials [Pelucchi, 2009]; infant-directed prosody can also facilitate mapping a novel word form to a referent [Graf Estes, 2013]. Yet the direction of listening preference is not a transparent scale of knowledge. Brief exposure can produce a familiarity preference, whereas

longer or easier exposure can produce a novelty preference [Hunter, 1988; Houston-Price, 2004]. Treating either direction as success requires a theory of how familiarization, complexity, and processing stage determine behavior.

Model benchmarks commonly compare the probability of a word with that of a matched nonword, or evaluate segmentation from speech representations [Nguyen, 2022; Dunbar, 2021]. The target contrast is related to the infant task, but an explicit probability ranking is not the same response as a listening preference. Trial-level comparison is most credible when the same speech materials and foils are used and when the analysis predicts the magnitude and direction of infant behavior rather than coding both possible preferences as binary success.

Expressive vocabulary introduces a different alignment problem. Naturalistic transcripts record child productions after adult interpretation and normalization, while caregiver inventories summarize reported comprehension or production over a fixed list [MacWhinney, 2000b; Fenson, 1994]. Model generation provides directly observable token output, but it depends strongly on the prompt, decoding policy, and sampling opportunities. Comparing vocabulary size alone can therefore conflate lexical knowledge with observation and elicitation. A defensible comparison aligns the opportunity to produce a word, the unit of counting, the lexical normalization policy, and the cumulative exposure available at each measurement point.

Word meaning. Early comprehension is commonly assessed with looking-while-listening or forced-choice procedures: the child hears a word while viewing a target and distractor, and gaze or pointing toward the named referent indexes comprehension [Bergelson, 2012; Lany, 2011; Bergelson, 2017; Barbir, 2023]. Multimodal benchmarks can preserve the same basic decision by requiring a model to match a word or short phrase to one of two images [Wang, 2025]. When the visual alternatives, linguistic prompt, and scoring rule are matched, this is among the closest available child-model comparisons.

Important differences remain. Children’s referential decisions unfold in time and are supported by prosody, joint attention, and expectations about the speaker. Static image-text pairs remove those cues. Conversely, distributional models are often evaluated through lexical-relation classification or similarity among learned representations [Lenci, 2022]; these evaluations do not directly reproduce an infant category-sorting or referent-selection task. Such representations can be explanatory predictors, but they should not be labeled behaviorally aligned without showing that they predict trial-level child responses.

Syntactic structure. Artificial grammar learning exposes infants or children to sequences generated by a hidden regularity and tests their response to novel conforming and violating sequences [Marcus, 1999; Gervain, 2012]. Child production additionally reveals generalization and characteristic errors that an endpoint judgment may conceal. Model evaluations such as BLiMP and SyntaxGym contrast grammatical and minimally different ungrammatical sentences using probability or surprisal [Warstadt, 2020a; Hu, 2020; Marvin, 2018; Salazar, 2020].

The binary contrast creates a useful structural analogy, but equivalence depends on more than the number of choices. Many model benchmarks reflect adult grammatical norms, contain

vocabulary or constructions inappropriate for infants, and provide written strings rather than spoken sequences. Alignment improves when stimuli are developmentally licensed, are presented in a comparable modality, and support the same generalization claim. Production trajectories remain a major gap: probability differences at isolated checkpoints do not yet reproduce the sequence and distribution of children’s spontaneous or elicited syntactic errors.

Sentence meaning. Infant sentence comprehension can be inferred from looking or violation-of-expectation responses when a described event conflicts with a visual or causal outcome [Aslin, 2007 ; Margoni, 2023]. Event-related potentials, including N400 responses to semantic violations, provide a complementary processing measure [Friedrich, 2006 ; Lozano, 2025]. Models are evaluated through plausibility minimal pairs or word-by-word surprisal,

$$I(w_i) = -\log P_{\text{LM}}(w_i \mid w_{<i}),$$

which has been related to adult reading time and N400 amplitude [Hale, 2001 ; Smith, 2013 ; Frank, 2015 ; Michaelov, 2024]. A common anomalous stimulus can support a calibrated contrast, but response magnitude is not automatically commensurable: surprisal is a model-internal quantity, whereas looking and neural signals reflect a larger processing chain. The comparison should test an explicit linking function rather than infer equality from a shared anomaly label.

Pragmatic inference. False-belief and related tasks ask whether children predict an agent’s behavior from that agent’s informational state rather than from reality [Wimmer, 1983 ; Wellman, 2001]. Language models can answer textual versions of these scenarios and sometimes reach high accuracy [Kosinski, 2024 ; Strachan, 2024]. Surface-level task identity is deceptive here. Children act or answer within a socially interpretable event; models usually generate text after a decontextualized narrative, and the answer may be recoverable from recurring linguistic patterns. Existing tasks establish benchmark performance, not equivalence between grounded mental-state reasoning and textual continuation. Pragmatic production—adapting an utterance to a particular interlocutor’s knowledge and communicative needs—has no well-established developmental model analogue.

Across domains, the best current alignment occurs when a common forced-choice structure can be applied to matched perceptual materials, as in word–referent matching. Preference tasks and minimal-pair scores share contrasts but differ in how evidence is expressed. Similarity measures are generally explanatory proxies rather than behavioral equivalents. Production can in principle be scored jointly, but only if elicitation and opportunity are aligned. These distinctions motivate the representative evidence maps in Tables 2.2 and 2.3.

2.2.2 Selected empirical coverage

The following tables replace numerical cross-indexing with direct citations. They are intentionally selective: their purpose is to show which combinations of domain, input, and outcome are represented, and where the human and model literatures still fail to meet at the same task.

TABLE 2.2 – Selected computational studies spanning the outcome domains reviewed in this chapter.

Study	Domain	Model or task	Training or evaluation input	Primary outcome
[Schatz, 2013]	Phonology	ABX evaluation	Speech tokens and learned acoustic representations	Contrast discrimination error
[Schatz, 2014]	Phonology	Noise-sensitive ABX	Controlled speech and acoustic degradation	Robustness of discrimination
[Dunbar, 2017]	Phonology	Zero-resource speech systems	English, French, and Mandarin speech	ABX and representation quality
[Dunbar, 2021]	Phonology / lexical / syntax	ZeroSpeech evaluation suite	Untranscribed speech with lexical and syntactic probes	ABX, spoken-word, and grammar scores
[Lavechin, 2023]	Speech development	BabySLM	Developmentally motivated speech input	Phonetic, lexical, and syntactic scores
[Matuskevych, 2023]	Phonology	Alternative acoustic learners	English, French, and Japanese speech	Contrast-specific developmental trajectories
[Poli, 2024]	Phonology	Contrast-learning model	Speech embedded in broader ambient sound	Trajectory of discrimination
[Cotterell, 2017]	Morphology	Sequence-to-sequence derivation	Source forms with derivational paradigm slots	Derived-form accuracy
[Weissweiler, 2023]	Morphology	Wug-style LM evaluation	Nonce roots in linguistic contexts	Productive inflection
[Anh, 2024]	Morphology	Generative language models	Multilingual Wug-style items	Human-rated generalization accuracy
[Nguyen, 2022]	Word form	Spoken language model probes	Continuous or segmented speech	Lexical discrimination and segmentation
[Vong, 2025]	Word meaning	Multimodal neural learners	Child-specific SAYCam audiovisual streams	Robustness of word–referent learning across children
[Wang, 2025]	Word meaning	Developmental vision–language model	Synthetic child-directed captions and visual scenes	Visual two-word matching
[Lenci, 2022]	Lexical semantics	Semantic-relation classifier	Distributional representations and lexical-relation datasets	Relation-classification performance
[Warstadt, 2020a]	Syntax	BLiMP minimal pairs	Controlled grammatical contrasts	Forced-choice accuracy
[Hu, 2020]	Syntax	SyntaxGym-style targeted evaluation	Sentences instantiating syntactic dependencies	Surprisal interactions and accuracy
[Marvin, 2018]	Syntax	Targeted LM evaluation	Grammatical and ungrammatical sentence pairs	Relative sentence probability
[Salazar, 2020]	Syntax / semantics	Pseudo-log-likelihood scoring	Masked language models and minimal pairs	Sentence-level acceptability
[Hale, 2001]	Sentence processing	Probabilistic parser	Incrementally parsed sentences	Word-by-word surprisal
[Smith, 2013]	Sentence processing	Probabilistic language model	Adult reading materials	Relation between surprisal and reading time
[Frank, 2015]	Sentence meaning	Recurrent language model	Sentence stimuli paired with EEG evidence	Relation between surprisal and N400
[Kauf, 2023]	Sentence meaning	Autoregressive language models	Plausible and implausible event descriptions	Semantic minimal-pair accuracy
[Michaelov, 2024]	Sentence meaning	Large language models	ERP stimulus materials	Surprisal–N400 correspondence
[Kosinski, 2024]	Pragmatics	Prompted language models	Textual false-belief scenarios	Mental-state answer accuracy
[Strachan, 2024]	Pragmatics	Comparative LLM evaluation	Theory-of-mind tasks used with humans	Cross-task accuracy and error profile

TABLE 2.3 – Selected infant and child studies illustrating the behavioral and neural measures discussed in this chapter.

Study	Domain / age	Paradigm	Stimulus or setting	Observed response
[Eimas, 1971]	Phonology / young infants	High-amplitude sucking	Synthetic voice-onset-time contrast	Recovery of sucking response
[Werker, 1984]	Phonology / 6–12 months	Conditioned head turn	Native and non-native consonant contrasts	Detection of sound change
[Best, 1988]	Phonology / infancy	Habituation	Non-native click consonants	Dishabituation or looking recovery
[Kuhl, 1992]	Phonology / 6 months	Conditioned response	Vowel prototypes and variants	Category-sensitive discrimination
[Sundara, 2018]	Phonology / 6–10 months	Infant-controlled habituation	Language-specific coronal contrasts	Looking-time recovery
[Berko, 1958]	Morphology / preschool and school age	Wug elicitation	Pictures and nonce words	Produced plural and past-tense forms
[Engelmann, 2019]	Morphology / preschool children	Elicited production	Real and novel verbs in Finnish and Polish	Inflectional generalization
[Kim, 2021]	Morphology / 6–8 months	Segmentation and recognition	Suffixed nonce forms and bare stems	Sensitivity to morphological relatedness
[Michaly, 2025]	Morphology / school-age children	Comprehension and production	Real- and pseudoword derivational analogies	Accuracy and use of morphological elements
[Saffran, 1996]	Word segmentation / 8 months	Familiarization preference	Artificial continuous speech	Word versus part-word listening
[Aslin, 1998]	Word segmentation / 8 months	Head-turn preference	Artificial speech stream	Statistical segmentation preference
[Pelucchi, 2009]	Word segmentation / 8 months	Familiarization preference	Natural Italian child-directed speech	Word versus part-word listening
[Graf Estes, 2013]	Word learning / 17 months	Habituation word learning	Novel labels in infant- or adult-directed prosody	Learning of label–object mappings
[Bergelson, 2012]	Word meaning / 6–9 months	Looking while listening	Familiar nouns and paired images	Target-directed gaze
[Lany, 2011]	Word meaning / second year	Forced-choice comprehension	Novel word with two referents	Gaze to selected target
[Bergelson, 2017]	Word meaning / infancy	Looking while listening	Named objects in visual scenes	Time course of target gaze
[Barbir, 2023]	Word meaning / toddlers	Forced-choice comprehension	Spoken word and paired pictures	Looking or pointing choice
[Marcus, 1999]	Syntax / 7 months	Artificial grammar learning	Repetition patterns over syllables	Listening preference at test
[Gervain, 2012]	Sequence structure / newborns	Functional near-infrared spectroscopy	Speech sequences varying identity relations and position	Cortical response to identity and position
[Margoni, 2023]	Event and sentence meaning / infancy	Violation of expectation	Possible and impossible event outcomes	Differential looking time
[Friedrich, 2006]	Word / sentence meaning / toddlers	Picture–word semantic violation	Congruent and incongruent spoken items	N400 response

Continued on next page

Table 2.3 continued

Study	Domain / age	Paradigm	Stimulus or setting	Observed response
[Lozano, 2025]	Semantic processing / bilingual toddlers	ERP experiment	Semantically congruent and incongruent materials	Development of N400 effects
[Wimmer, 1983]	Pragmatics / preschool onward	False-belief task	Character, object transfer, and search story	Predicted search location
[Wellman, 2001]	Pragmatics / childhood	Meta-analysis of false-belief tasks	Diverse belief scenarios	Age-related proportion correct

2.2.3 Thresholded evidence and graded performance

The same task can support different claims depending on how competence is operationalized. A *thresholded* or milestone-based analysis asks whether evidence for a specified contrast exceeds a criterion. A *graded* analysis asks how strongly, accurately, or efficiently the learner expresses that contrast. These questions are complementary, but neither metric can stand in for the other.

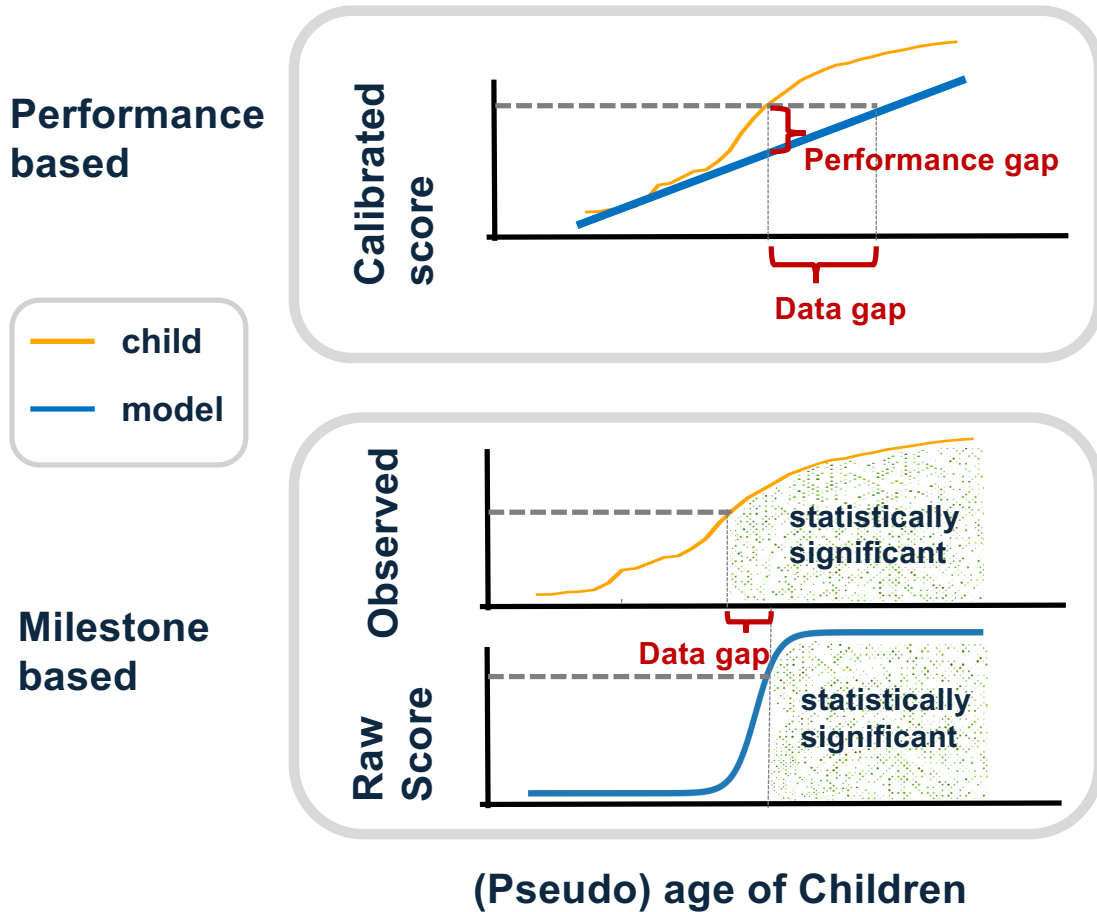


FIGURE 2.1 – Schematic contrast between thresholded and graded evaluation. A milestone analysis treats evidence for a specified competence as present once an explicit statistical criterion is met. A graded analysis retains continuous performance and supports comparison of strength and trajectory. Both depend on the validity of the underlying task.

Thresholded evaluation is common in targeted syntactic work. A model is said to exhibit

a dependency when it assigns higher probability to the predicted member of a controlled contrast reliably above chance, or when a theoretically specified surprisal interaction is present [Linzen, 2016; Marvin, 2018; Hu, 2020; Pearl, 2013]. This formulation clarifies what has been demonstrated: the model has crossed a criterion for one operationally defined construction. It does not imply mastery of syntax more generally, and it does not show that the mechanism producing the effect is the same as a child’s. Results can be sensitive to item construction, probability estimation, and the statistical rule used to declare the milestone [Warstadt, 2020a; Wilcox, 2023].

Graded evaluation reports accuracy, surprisal, perplexity, correlation, or another continuous score [Warstadt, 2023; Hu, 2024]. Continuous measures preserve differences among checkpoints and architectures, but global improvement can obscure a missing generalization. Perplexity may decrease as frequent sequences become more predictable while a particular long-distance dependency remains absent [Hu, 2020]. Likewise, reaching or exceeding an adult aggregate score does not establish that every component competence has been acquired.

For child–model comparison, thresholded analyses are most informative when the criterion corresponds to an independently established developmental effect, while graded analyses are most informative when both populations contribute observations on a shared scale or through an explicit linking model. Reporting both is often preferable: a threshold locates the emergence of a theoretically defined behavior, and the continuous effect size describes how it strengthens. Figure 2.1 illustrates that the two summaries compress different features of the same evidence.

2.2.4 Developmental trajectories

Endpoint similarity is weak evidence for developmental similarity because different learning routes can converge on the same final score. Trajectory analysis instead evaluates performance across exposure and asks about acquisition order, curve shape, error distribution, and sensitivity to changes in the learning environment [Chang, 2022; Evanson, 2023; Ma, 2025]. It links milestone and graded analyses: milestones identify transitions, while continuous measurements reveal whether change is gradual, abrupt, monotonic, or temporarily regressive.

Lexical work illustrates this approach through neural age of acquisition: the point in training at which the model’s evidence for an individual word meets a predeclared criterion. Model acquisition order can then be compared with child age-of-acquisition norms and related to frequency, contextual diversity, and interaction [Chang, 2022; Ma, 2025]. The comparison requires care. A checkpoint is an amount of optimization, not a child’s chronological age; the mapping becomes meaningful only after accounting for the amount and order of input processed and for repeated exposure across epochs.

Curve shape provides additional evidence. Children sometimes show U-shaped development, producing stored irregular forms correctly, then overgeneralizing a productive rule, and finally recovering the exceptions [Marcus, 1992]. If analogous non-monotonic patterns appear in neural learners, they are more informative than endpoint accuracy because they constrain the interaction among item frequency, generalization, and memory. Still, a visually similar curve is not sufficient. The error types, item-level ordering, exposure scale, and conditions producing the reversal must

also agree.

Trajectory analysis also tests whether grammatical phenomena emerge in an order resembling child development. Some model capabilities improve smoothly, whereas others appear more abruptly, and their relative ordering may change with architecture or corpus composition [Evanson, 2023]. Distributional analyses of word-learning trajectories can test whether acquisition order reflects interpretable lexical properties or merely the statistics of a particular corpus [Ficarra, 2025]. Such comparisons are most useful when age norms are precise and the evaluation remains constant across checkpoints; post-hoc selection of thresholds or checkpoints can manufacture apparent stages.

Finally, a developmental trajectory includes the sequence of experience, not only the cumulative token count. Curriculum learning and interaction may change what is learned at a given budget. Evidence that preserving local discourse continuity can matter more than globally sorting child-directed utterances by child age cautions against equating any “simple-to-complex” ordering with the child’s curriculum [Feng, 2024a]. Synthetic variation can improve some morphosyntactic outcomes without producing equivalent gains in semantics or pragmatics [Haga, 2024a]. Trajectory alignment must therefore be reported per domain rather than reduced to one global developmental score.

2.2.5 Measurement uncertainty

Both developmental experiments and model evaluations vary, but their uncertainty enters at different stages. The relevant question is not simply which learner is noisier. It is which components of the observed variance reflect the target knowledge, which reflect the measurement procedure, and which can legitimately be compared.

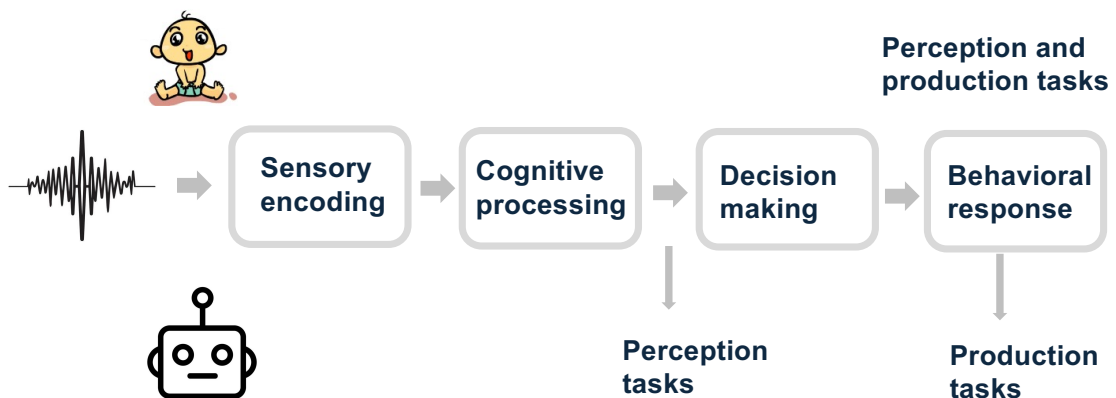


FIGURE 2.2 – Sources of variation across a simplified evaluation pipeline. Infant and child behavior is observed after sensory encoding, cognitive processing, decision or response selection, and motor execution. Many model perception scores are read from representations through a task model before an overt response stage; generated model output is observed later in the pipeline. The comparison must identify, rather than assume away, these distinct measurement stages.

For children, sensory encoding depends on signal quality and hearing; cognitive processing includes the developing representation of interest; decision and response selection depend on

arousal, familiarity, memory, and task interpretation; execution adds oculomotor, articulatory, equipment, and coding error. Some variation is nuisance, but some may be developmental signal: inconsistent responding can reflect a transitional representation rather than random failure [Mitchell, 2022; Kucharskỳ, 2024]. Large collaborative studies document substantial heterogeneity across laboratories, procedures, ages, and individual infants [The ManyBabies Consortium, 2020]. Preferential-looking effects can be robust at the group level while showing limited individual test–retest reliability [Monroy, 2021; DeBolt, 2020; Byers-Heinlein, 2022; Schreiner, 2024].

For models, variability arises from initialization, data order, stochastic optimization, dropout, preprocessing, hardware and numerical nondeterminism, and sampling at inference [Srivastava, 2014; Dodge, 2020; Summers, 2021]. Conditional on a fixed trained model and a deterministic scoring procedure, repeated evaluation of the same input can be identical. That narrow repeatability should not be confused with eliminating model uncertainty. Different seeds may produce materially different learning trajectories, and even a fixed model’s apparent competence depends on task-model decisions such as layer, probe, prompt, tokenization, and decoding. These sources are often controllable and estimable, but they are not absent.

The central asymmetry is thus structural. A child’s discrimination score is observed after several embodied response stages, whereas an ABX or cosine score may be calculated directly from a selected representation. Production moves model evaluation closer to an overt response, but child articulation and transcription still have no simple analogue in model decoding. Adding arbitrary noise to model outputs does not solve this problem: it can mimic the width of a behavioral distribution without reproducing its causal sources. Likewise, collecting more infant trials reduces sampling error but cannot by itself separate unstable knowledge from state-dependent behavior.

Calibrated analysis should instead propagate uncertainty on both sides. Human estimates should report exclusions, trial counts, coding reliability, individual and laboratory effects, and uncertainty around age-specific means. Model estimates should report independent training runs, checkpoint selection, task-model choices, and item-level confidence intervals. Where possible, the same hierarchical model should estimate stimulus-level and learner-level effects, while preserving the distinction between biological participants and random model instances. This does not make the noise structures identical; it prevents differences in precision from being mistaken for differences in competence.

2.3 Input alignment

Outcome alignment determines whether the two learners are measured comparably; input alignment determines whether they have been asked to solve comparable learning problems. “Child-scale” is not a property of token count alone. Children encounter spoken language in recurring activities, accompanied by visible events, interlocutors, and feedback. The distribution changes with age and with the child’s own behavior. A corpus can match one part of this experience while diverging sharply on others. For that reason, input claims should be decomposed into register, modality, interaction, quantity, and temporal organization rather than summarized

by a single realism label.

2.3.1 Quality, register, modality, and interaction

Language-only text. Web corpora such as Common Crawl, C4, and Wikipedia provide extensive coverage of lexical, syntactic, and factual patterns [Buck, 2014; Raffel, 2020; Kaplan, 2020]. They are nevertheless dominated by written, adult-directed genres: news, reference material, online discussion, and technical prose. Compared with child-directed speech, these sources differ in vocabulary, utterance length, discourse function, repetition, and topic distribution [Huebner, 2021a]. Shuffling documents also removes the contingent sequence in which one conversational turn responds to another. A model trained on this material may answer whether a pattern is learnable from broad written evidence, but not whether it is learnable from the experience available to a young child.

The BabyLM Challenge constrains training to 10 or 100 million words and combines child-directed transcripts with educational and simplified materials [Warstadt, 2023]. It therefore improves control over both scale and register and makes comparison across architectures possible within a common budget. Its input remains largely textual, however. Transcribed speech omits prosody, voice, timing, overlap, and much of the situational context that makes an utterance interpretable. A developmentally bounded text corpus is thus a valuable experimental resource, but it is not a complete reconstruction of infant input.

Naturalistic child-directed transcripts, especially CHILDES, more closely represent the vocabulary and conversational structure addressed to children [MacWhinney, 2000b]. Child-directed speech tends to contain shorter utterances, concrete and frequent vocabulary, repetition, reformulation, and prosodic adaptation; its composition predicts developmental outcomes [Weisleder, 2013]. The main limitation is not only corpus size. Recordings sample particular families and activities, transcription maps a continuous multimodal exchange into discrete words, and the resulting database does not preserve everything that was perceptually available. “Training on CHILDES” should consequently be accompanied by the languages, participants, age ranges, speaker roles, and preprocessing decisions used.

Synthetic augmentation attempts to expand such corpora without returning to adult web register. Paraphrases and variation sets generated from child-directed utterances can increase lexical or syntactic diversity and improve some formal evaluations [Haga, 2024a]. But register imitation is not equivalent to interaction. Models generating caregiver-like dialogue can reproduce local lexical and utterance statistics while overstating alignment and failing to reproduce turn-by-turn contingency [Liu, 2024]. Synthetic data are best described by the property they add—paraphrastic variety, simplified syntax, or a target vocabulary—rather than as globally child-like.

Language-only audio. Speech supplies information removed by transcription: segmental variation, prosody, speaker identity, timing, and cues to turn and utterance boundaries. These signals are central to phonetic category formation and segmentation [Kuhl, 2006; Saffran, 1996]. Read-speech and audiobook corpora offer clean acoustic conditions and relatively reliable annotation, but their adult-directed register and performance style differ from home interaction.

Spontaneous adult speech improves discourse naturalness without necessarily matching what is addressed to a child.

Long-form child-centered recordings preserve more of the natural acoustic ecology, including background sound, overlapping voices, changing distance from speakers, and extended daily routines [Cristia, 2020 ; Bergelson, 2023]. That realism creates a difficult learning signal: target speech is sparse, noisy, and only partly directed to the child. Systems evaluated on clean audiobooks and on long-form recordings are therefore changing acoustic quality and register together [Lavechin, 2024]. This trade-off is scientifically useful if it is reported explicitly; otherwise, a performance gap can be misattributed to the amount of data or to architecture.

Language with vision. Visual context makes referential learning possible in a way that text or audio alone does not. Large image–caption collections such as LAION and COCO provide millions of paired examples [Schuhmann, 2022 ; Lin, 2014]. Their captions describe images for adult readers rather than participating in a child’s unfolding activity. The pairing is curated and static, often identifying a salient object more cleanly than a natural scene would. These resources establish what vision–language architectures can learn from large aligned datasets, but they provide limited evidence about early cross-situational learning.

Synthetic child-directed captions occupy an intermediate position. Developmental vision–language work can pair child-perspective images with simplified descriptions and evaluate grounded combinations on child-oriented tests [Wang, 2025]. Such data improve register correspondence while retaining a clean supervised pairing. They do not recreate the ambiguity of a caregiver utterance whose referent depends on joint attention, current action, and conversational history.

Egocentric video more directly samples the visual stream available to individual children. SAYCam follows a small number of infants longitudinally and has supported demonstrations of word–referent learning from a single child’s visual and linguistic experience [Sullivan, 2021 ; Vong, 2024]. BabyView broadens the recording effort and improves visual quality and participant diversity [Long, 2024]. Yet a head-mounted camera is not an infant’s perceptual system: camera field of view differs from gaze and attention, and usable paired speech is far scarcer than web-scale image–text examples. High ecological validity therefore comes with limited coverage and measurement uncertainty of its own.

Language, vision, and social interaction. Children learn in exchanges structured by turn taking, joint attention, communicative intention, and contingent caregiver response [Tomasello, 1995 ; Csibra, 2009]. Dyadic corpora retain some of this structure. Shared book-reading recordings combine caregiver and infant perspectives in a constrained but genuinely interactive setting [Ahamat, 2025] ; ECOLANG annotates speech, gesture, and gaze in interactions with older children [Gu, 2025] ; and game-based corpora extend multimodal dialogue to middle childhood [Bodur, 2021 ; Goumri, 2024]. Their value lies in temporal coordination across channels, not merely in the coexistence of audio and video.

These corpora remain too small and specialized to serve as drop-in replacements for large

language-model training sets. They can instead support focused learning experiments, evaluation of contingent behavior, or estimation of the distributions that a larger simulation should preserve. Treating every audiovisual recording as “social” would erase the key distinction: co-occurrence presents signals together, whereas interaction makes one participant’s behavior conditional on the other’s.

TABLE 2.4 – Representative input resources organized by modality, register, interactional structure, and principal limitation. The entries characterize resource types and do not imply that every study uses identical preprocessing.

Resource	Modality and register	Interactional structure	Developmental value	Principal limitation
Common Crawl [Buck, 2014]	Written, mostly adult-directed web text	Documents rather than interaction	Broad lexical and domain coverage	Register, scale, and temporal mismatch
C4 [Raffel, 2020]	Filtered written web text	Documents rather than interaction	Reproducible large-scale LM training	Adult-written genres and weak grounding
BabyLM [Warstadt, 2023]	Mixed text with child-directed and simplified components	Some dialogue after transcription	Controlled 10M/100M-word budgets	Predominantly textual; heterogeneous sources
CHILDES [MacWhinney, 2000b]	Transcribed child-directed and child speech	Natural conversational turns	High register correspondence across languages	Sparse sampling and loss of acoustic/visual context
Synthetic CDS variation sets [Haga, 2024a; Liu, 2024]	Generated child-directed-style text	Simulated turns or paraphrases	Scalable control of surface register	Limited contingency and inherited generator bias
Read-speech collections [Lavechin, 2023]	Clean adult-directed audio	Narration rather than dialogue	Acoustic and prosodic learning at scale	Low interactional and register correspondence
Child-centered long-form audio [Cristia, 2020; Bergelson, 2023]	Naturalistic home audio	Extended daily interaction	Realistic acoustic ecology and exposure estimates	Noise, overlap, privacy, and sparse annotation
LAION [Schuhmann, 2022]	Web image–text pairs	No contingent interaction	Large-scale visual grounding	Adult web captions and curated pairing
COCO [Lin, 2014]	Curated images and descriptive captions	No contingent interaction	Controlled object and scene descriptions	Captioning task unlike child-directed talk
Developmental synthetic captions [Wang, 2025]	Child-oriented text with visual scenes	Static pairing	Register-controlled grounded evaluation	Generated language and absent joint attention
SAYCam [Sullivan, 2021; Vong, 2024]	Infant-perspective video with speech	Naturalistic home activity	Longitudinal, child-specific grounding	Few children and camera–attention mismatch
BabyView [Long, 2024]	Egocentric developmental video and audio	Naturalistic activity	Broader and higher-quality child-perspective sampling	Modest scale relative to web corpora
Shared book reading [Ahamat, 2025]	Dyadic audiovisual child-directed speech	Turn taking and joint attention	Aligned caregiver and infant perspectives	Narrow activity and small sample
ECOLANG [Gu, 2025]	Speech, gesture, and gaze	Naturalistic dyadic interaction	Explicit cross-channel coordination	Older age group and limited training scale

Continued on next page

Table 2.4 continued

Resource	Modality and register	Interactional structure	Developmental value	Principal limitation
CHICO/CHICA [Bodur, 2021 ; Goumri, 2024]	Multimodal task dialogue	Collaborative word games	Naturalistic communicative adaptation	Task- and age-specific interaction

2.3.2 Quantity and temporal organization

Claims about data efficiency require a defined denominator. Studies variously report words, subword tokens, utterances, audio hours, image-text pairs, gradient updates, or epochs. These units do not convert transparently. One hour of long-form audio contains silence, distant speech, overlapping speakers, and utterances not directed to the child; one hour of an audiobook is densely linguistic. Likewise, a word processed during a hundredth epoch is not equivalent to a newly encountered word in a new situation. Reporting a nominal corpus size without total presentations and sampling procedure can therefore make a heavily recycled training regime appear child-scale.

Estimates of children’s cumulative linguistic exposure vary with recording method, socioeconomic and cultural context, the definition of speech addressed to or merely audible to the child, and extrapolation from sampled days [Hart, 1992 ; Weisleder, 2013 ; Frank, 2023a]. A defensible comparison should state a range and sensitivity analysis rather than a single universal number. It should also distinguish raw exposure from *effective exposure*: what the learner can encode given signal quality, attention, vocabulary, and multimodal context.

Fixed-corpus trajectories. In a fixed-corpus design, one dataset is repeated while the model is evaluated across checkpoints [Chang, 2022 ; Evanson, 2023]. Holding composition constant helps isolate learning dynamics and makes frequent measurement inexpensive. It is useful for studying item acquisition order and the emergence of particular generalizations. But the developmental analogy is limited. Children do not receive the same closed corpus in a new random order on every pass; their vocabulary and activities expand, interlocutors respond to them, and old events cannot be replayed exactly.

Repeated epochs also change the effective quantity. A model trained for 100 epochs over ten million words has processed one billion word presentations, even though its unique corpus is described as ten million words. Memorization, overtraining, and declining lexical novelty may shape the resulting trajectory. At minimum, studies should report unique tokens, total token presentations, number of epochs, sampling with or without replacement, and the exposure count at every evaluated checkpoint.

Successive scaling. Scaling designs train otherwise comparable models on increasingly large samples and ask how much input is needed for a target behavior [Warstadt, 2020b]. BabyLM makes this comparison explicit within bounded text budgets [Warstadt, 2023 ; Hu, 2024] ; STELA and BabySLM extend related logic to quantities and registers of speech [Lavechin, 2023 ; Lavechin,

2024]. The input level at which a model crosses a milestone can be compared with a plausible exposure range for children.

This design better represents cumulative amount, but independent models do not form a longitudinal learner. Each scale may begin from a new initialization, and the larger training set is not necessarily an ordered extension of the smaller one. Apparent movement along a learning curve can therefore combine data quantity, sample composition, and between-run variation. Nested corpora, multiple seeds per scale, and identical evaluation make the interpretation stronger.

Curriculum and continual learning. Children’s input is incrementally arriving and contextually clustered. A caregiver repeats a topic within an exchange, adapts utterances to a response, and introduces new vocabulary through changing activities. Curriculum experiments attempt to preserve selected parts of that structure. Globally ordering child-directed speech by the child’s age has not consistently improved model outcomes, while keeping local conversational context can matter [Feng, 2024a]. The relevant curriculum may therefore lie in coherent interaction rather than a simple global ordering from short to long sentences.

Continual learning offers the closest formal analogy to an accumulating stream: the model updates on new data while retaining what it learned earlier. Standard neural systems are vulnerable to catastrophic forgetting, in which new learning disrupts earlier representations [McCloskey, 1989]. That failure is not merely an inconvenience to be hidden by replay. It identifies a property—stable integration of new evidence with long-term knowledge—that a developmental model must explain. Replay buffers, retrieval, and external memory can be tested as hypotheses, but their stored amount should count toward the model’s effective exposure.

No current design aligns quantity, lexical novelty, temporal order, interaction, and learning continuity simultaneously. Fixed corpora maximize experimental control but recycle a static distribution; successive scaling improves amount calibration but often substitutes different learners for developmental time; continual learning represents sequential arrival but introduces retention problems. The appropriate choice depends on the claim. Studies should state which dimension is controlled and which remains an abstraction.

2.4 Implications, reporting standards, and thesis research gaps

2.4.1 What a calibrated comparison licenses

Input and outcome alignment are distinct requirements. Strong outcome alignment with weak input alignment shows that a model can reproduce a behavioral contrast under a different learning regime. Strong input alignment with weak outcome alignment shows what representations or engineering scores arise from child-oriented data, but not whether child behavior has been reproduced. When both are reasonably aligned, model success supports a bounded sufficiency claim: the specified learner can acquire the specified outcome from the specified experience. Even then, success does not establish that children use the same internal mechanism.

This framework separates two legitimate research orientations. Developmentally oriented modeling uses a model to test hypotheses about learning and therefore requires calibrated input

and outcomes. Engineering-oriented evaluation uses human performance to diagnose or improve artificial systems; it can be scientifically valuable without recreating development, but its results should not be presented as evidence about how children learn. Problems arise when the orientation changes between method and conclusion—for example, when a web-trained model’s benchmark success is interpreted as a developmental proof, or when failure under a mismatched task is treated as evidence for an innate human constraint.

Model failure becomes informative through controlled contrasts. If input and evaluation are fixed while memory architecture changes, a performance difference can bear on the role of memory. If architecture and task are fixed while register changes, the comparison can isolate properties of the input distribution. If all dimensions vary at once, attribution is impossible. Calibration is therefore not a demand that every model imitate a child in every respect; it is a discipline for identifying which difference is the experimental variable.

2.4.2 Minimum reporting for interpretable evidence

Transparent reporting should make the inferential chain reconstructible. At a minimum, a developmental model comparison should specify:

1. **The orientation and claim.** State whether the study tests developmental sufficiency, mechanism, prediction of human behavior, or engineering performance, and formulate the conclusion at the same level.
2. **The input population.** Report language, register, speaker roles, child ages, modality, activity or genre, collection procedure, filtering, deduplication, and synthetic-data provenance.
3. **The exposure budget.** Give unique and total tokens or hours, epochs, sampling and curriculum, amount processed at each checkpoint, and a justified range for the human comparison.
4. **The outcome and task model.** Identify the construct, stimuli, response alternatives, prompt or probe, selected layer, scoring rule, threshold, and the linking assumption between child and model measures.
5. **The developmental comparison.** Predefine the human age range, checkpoints, acquisition criterion, and whether evidence concerns endpoint, order, curve shape, or item-level errors.
6. **Uncertainty and robustness.** Report participant exclusions and measurement reliability for humans; seeds, runs, numerical settings, checkpoint selection, and decoding for models; provide item- and learner-level uncertainty where possible.
7. **Residual mismatches.** Name the missing modalities, interaction, cognitive resources, or response stages and state how they limit the conclusion.

The checklist does not require every study to solve the entire calibration problem. It requires enough detail to know which problem was solved. Standardized benchmarks remain useful, but their scores should be accompanied by developmental task analyses. Naturalistic corpora remain useful despite incomplete coverage, but their sampling and preprocessing must be visible.

Cumulative evidence emerges when later work can reproduce both the learner and the comparison, not only the final score.

2.4.3 Research gaps addressed in this thesis

The review identifies three connected gaps that motivate the empirical chapters that follow. First, lexical development lacks a production-oriented measure that aligns model generation with the cumulative growth recorded for children while making input budgets explicit. Chapter 3 addresses this gap with CORPUS-CDI, comparing the expressive vocabularies of children and language models across developmental stages and controlled exposure. This operationalizes the present chapter’s argument that a vocabulary count is meaningful only when observation opportunities, lexical units, and developmental time are defined jointly.

Second, differences between child and model syntax may reflect memory and lexical exposure rather than grammatical representation alone. Chapter 4 tests whether episodic retrieval changes language models’ sensitivity to syntactic minimal pairs and whether it reduces a lexical-frequency gap. Retrieval is thereby treated as a controlled architectural hypothesis under a common outcome measure, rather than invoked after the fact to explain a mismatch.

Third, existing child-oriented corpora preserve linguistic register more readily than interactional contingency. Chapter 5 evaluates how well large language models reproduce child–caregiver dialogue and then isolates which forms of caregiver feedback support grammar learning. The benchmark distinguishes surface resemblance from turn-level behavior, while the learning experiment treats feedback as an input variable. Together, these studies address the interactional component of input alignment that static corpora leave unresolved.

These contributions do not attempt to build a globally child-equivalent learner. They make narrower comparisons in which the inputs, outcomes, or hypothesized mechanisms can be varied and measured. Their conclusions should therefore be read as evidence about lexical growth, episodic retrieval, and caregiver feedback under the stated conditions, not as a general equivalence between children and contemporary language models.

2.4.4 Limits of the framework

Calibration is necessary for developmental interpretation but not sufficient for mechanistic identity. Two systems can receive similar observable input and produce similar responses while relying on different internal representations. Stronger mechanistic evidence requires intervention, process-level prediction, and generalization to new tasks, not only behavioral correlation [Marr, 1982; Perfors, 2011].

The framework also abstracts over individual and cultural variation. Children’s environments differ in quantity, interlocutors, activity structure, multilingual exposure, and the social organization of learning [Weisleder, 2013; Bergelson, 2023]. Average developmental curves can conceal distinct routes and can make one narrowly sampled corpus appear universal. Model ensembles are not replicas of this variation because their differences arise from initialization and training rather than lived environments. Future comparisons should link individual input histories to

individual outcomes where the data permit.

Finally, the evidence base is uneven. Phonological and lexical perception have established infant paradigms, syntactic model evaluation has extensive minimal-pair resources, and word–referent matching provides a promising shared task. Naturalistic production, multimodal sentence meaning, and socially grounded pragmatics remain less well aligned. The selective study maps in this chapter represent that imbalance rather than a complete quantitative review, and claims about gaps should be updated as new corpora and matched paradigms become available.

2.5 Chapter summary

Computational models can inform language acquisition when the comparison is calibrated to the inference being made. Outcome alignment requires more than a shared linguistic label: child and model tasks must target the same construct through stimuli, response alternatives, and measures whose relationship is explicit. Thresholded tests, graded scores, developmental trajectories, and uncertainty each reveal different aspects of learning. They should be combined where possible rather than treated as interchangeable summaries.

Input alignment is similarly multidimensional. A child-oriented token budget does not compensate for adult-written register, and naturalistic child-directed language does not automatically preserve perceptual modality or social contingency. Repeated epochs, shuffled corpora, successive scaling, and continual learning embody different assumptions about developmental time. Consequently, every success and failure must be indexed to the quantity, quality, ordering, and interactional structure of the experience that produced it.

The strongest conclusion licensed by aligned success is a conditional sufficiency claim: a specified learner can acquire a specified outcome from specified input. Mechanistic conclusions require controlled variation and process-level evidence. This logic guides the following chapters on lexical growth under defined exposure, episodic retrieval with shared syntactic outcomes, and caregiver interaction as a component of the learning environment.

Chapitre 3

CORPUS-CDI: comparing expressive vocabulary growth in children and language models

Objectifs

Introduce CORPUS-CDI, a count-based lexical benchmark aligning language model generation with child word production across developmental stages, and use it to compare how children and character-level LMs acquire vocabulary under matched input budgets.

3.1 Introduction

Large language models (LLMs) have shown that a general-purpose architecture trained on large text corpora can acquire rich syntactic and semantic knowledge [Brown, 2020; Devlin, 2019; Touvron, 2023]. Beyond their impact on NLP applications [Bommasani, 2021], LLMs provide scalable tools for testing theories of early language acquisition [Dupoux, 2018; Evanson, 2023; Lavechin, 2022; Apidianaki, 2024; Hu, 2025]. In particular, they offer a controlled way to investigate the long-standing hypothesis that children bootstrap into language learning by accumulating statistics from linguistic input [Saffran, 1996].

A central challenge when applying LLMs as models of the early language learner is establishing valid *quantitative* comparisons between the learning trajectories of real children and those of LLMs fed with comparable data. For instance, it has been claimed that LLMs require significantly more input data than children to reach the same level of performance [Frank, 2023b]. But measuring this 'data gap' depends critically on developing sound metrics that can be applied consistently to both human learners and computational models.

We focus on measures of vocabulary, as there is a wealth of developmental data that provides human side of the comparison. Previous work has primarily explored two approaches for evaluating vocabulary in models: (i) **familiarity metrics**, which exploit LLMs' probabilistic outputs (e.g., surprisal or word-pseudoword preference) to probe lexical knowledge [Chang, 2022; Ma, 2025; Lavechin, 2025]; and (ii) **cross-modal matching**, which assesses word-referent grounding using multimodal training and behavioral benchmarks [Vong, 2024; Tan, 2024; Nikolaus, 2021c], analogous to looking-while-listening paradigms in children [Swingley, 2011]. While informative, familiarity metrics often rely on limited behavioral data, and cross-modal methods require multimodal input unavailable to unimodal language models. Also, neither of them has focused on expressive vocabulary.

To overcome these limitations and enable large-scale, quantitative comparison with child vocabulary, we introduce CORPUS-CDI, a benchmark that aligns model generations with children's expressive vocabulary. CORPUS-CDI calibrates three key dimensions: (1) training exposure, scaled from 6 to 36 months based on daylong recordings [Cristia, 2019]; (2) cumulative generated word count, matched to children's monthly word production rates; and (3) evaluation vocabulary, matched in part-of-speech categories and frequency distribution to the CDI word list [Frank, 2017].

Using CORPUS-CDI, we evaluate character-level LSTMs and Transformers trained on child-directed speech (CDS) and audiobook transcripts. We assess vocabulary acquisition across three complementary dimensions: lexical diversity (type-token ratio), word form accuracy (non-word rate), and vocabulary size (CORPUS-CDI score). Our analyses reveal that language models diverge from children in critical ways: models show excessive lexical diversity, minimal improvement in word form accuracy, and slower vocabulary growth. Frequency analyses further indicate that these gaps stem from long-tail truncation: low-frequency words are severely underrepresented in model acquisition, while high-frequency function words are overrepresented relative to children's productions.

3.2 Related Work

3.2.1 Developmental Trajectories

Prior work has examined the alignment between language model learning and child language development across multiple linguistic levels. A key challenge is mapping model training onto corresponding developmental milestones. Early studies used training checkpoints as proxies for age, tracking when models acquire words [Chang, 2022] or syntactic structures [Evanson, 2023]. However, these approaches repeatedly expose models to identical input batches, unlike children who encounter increasingly varied linguistic input. To address this, LAVECHIN et al. [Lavechin, 2023] varied input size to model developmental trajectories, improving zero-shot lexical accuracy with increased exposure. Beyond vocabulary, studies have investigated phonological development via word segmentation [Goriely, 2024], morphological acquisition through overregularization patterns [Haga, 2024b], and syntactic learning [Evanson, 2023]. Most prior work, however, calibrates only input quantity, leaving output production and evaluation frequency uncontrolled. In contrast, our framework aligns training exposure, generation quantity, and evaluation timing to enable direct comparison between models’ and children’s expressive vocabulary.

3.2.2 Vocabulary Evaluation

Evaluating vocabulary acquisition in models has followed two main strategies: developing developmentally inspired metrics or applying existing metrics to child-relevant tasks [Kosoy, 2023; Nikolaus, 2021a; Jiang, 2023]. Notably, CHANG et al. [Chang, 2022] first introduced the MacArthur-Bates Communicative Development Inventory (CDI) to evaluate when models acquire individual words during training. Their findings highlight that models rely disproportionately on input frequency, with more variance in acquisition timing than child age-of-acquisition norms. While children also benefit from Zipfian distributions, the challenge for models is primarily learning words in the *long tail* of the distribution [Wolters, 2024; Lavi-Rotbain, 2023; Lavi-Rotbain, 2022]. This partially explains the gap between model and human acquisition patterns, particularly for low-frequency words.

Cross-linguistic analyses show that contextual surprisal predicts age of acquisition for function words and predicates, but not nouns [Portelance, 2023], suggesting distributional learning mechanisms affect lexical categories differently. To address this, our evaluation sets are frequency-matched *and* controlled for part-of-speech categories to ensure that differences are not confounded by lexical class or frequency effects.

3.2.3 Calibrated Evaluation Frameworks

Direct comparisons between models and children require alignment across training data, learning objectives, and evaluation tasks [Warstadt, 2022]. The BabyLM Challenge [Warstadt, 2023; Hu, 2024; Charpentier, 2025] established shared tasks by restricting model training to 10M or 100M words to approximate child exposure at ages 2 and 13. Despite these constraints, findings remain mixed: some studies report benefits from child-directed speech [Feng, 2024a], whereas

others observe that Wikipedia training outperforms CHILDES-based input [Padovani, 2025b].

Input representation is another consideration: most models use BPE tokens [Huebner, 2021a; Evanson, 2023], whereas infants perceive continuous acoustic speech [Dupoux, 2018]. We use character-level inputs, forcing models to learn word boundaries unsupervised from orthographic sequences [Bunzeck, 2024; Goriely, 2024]. This provides an upper bound on performance compared to phoneme-based approaches.

Finally, production metrics such as lexical diversity often use the type-token ratio (TTR), which is sample-size dependent [Richards, 1987] and requires fixed-size windows for comparability [McKee, 2022]. Sampling temperature introduces tradeoffs between diversity and quality, with no single setting optimizing both [Verine, 2025].

Our framework addresses these challenges through three calibrations: (1) age-appropriate training exposure derived from daylong recordings [Cristia, 2019], (2) production quantity matched to children’s monthly output, and (3) frequency- and POS-matched evaluation sets reflecting child-directed speech patterns. Together, these enable a developmentally grounded evaluation of expressive vocabulary.

3.3 Methods

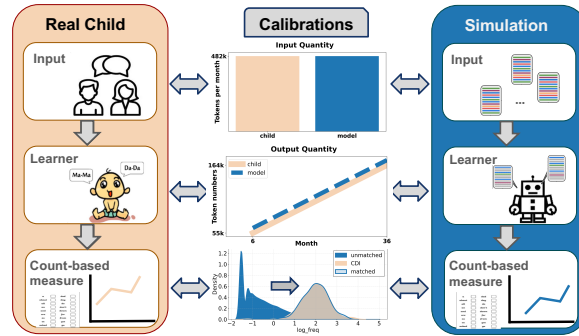


FIGURE 3.1 – **Overview of the corpus-CDI benchmark framework.** (a) Training and generation sizes are aligned with age-appropriate estimates of linguistic exposure and production from 6–36 months [Cristia, 2019]. (b) Evaluation sets match the frequency distribution of MacArthur–Bates CDI words in child-directed speech (English-CHILDES).

3.3.1 corpus-CDI Benchmark Overview

The CORPUS-CDI benchmark enables direct comparison between language model generation and child production through three methodological calibrations to address common mismatches in prior work (Figure 3.1):

(1) Age-appropriate training exposure: Training sets are scaled to match cumulative linguistic input from 6–36 months of age estimated from daylong recordings [Cristia, 2019].

(2) Production quantity matching: Generation sizes are calibrated to children’s monthly word production rates estimated from longitudinal corpora [Cristia, 2019] to ensure comparable sample sizes for vocabulary assessment.

(3) Frequency-matched evaluation words: The evaluation vocabulary is drawn from the MacArthur–Bates Communicative Development Inventory [Fenson, 2007] and frequency matched to words in English CHILDES corpora.

We evaluate models on three dimensions: *lexical diversity* (type–token ratio), *word form accuracy* (non-word rate), and *expressive vocabulary size* (CORPUS-CDI score).

3.3.2 Training Data

We construct two training corpora representing different linguistic environments:

Audiobook corpus: English audiobook transcripts from LibriVox [Kearns, 2014], representing narrative-rich, syntactically complex adult-directed language. **Child-directed speech (CDS) corpus:** English CHILDES transcripts [MacWhinney, 2000a], representing the naturalistic register of early child input.

Age-appropriate scaling Following CRISTIA et al. [Cristia, 2019], we estimate exposure from 250 hours at 6 months to 1,500 hours at 36 months of age. This corresponds to approximately 500 hours of speech input per year, representing a mid-range estimate of children’s linguistic environments. Converting speech hours to words using a standard rate of 3 words per second yields training sets ranging from 2.89M words (6 months) to 17.35M words (36 months). Developmental datasets are constructed every 6 months by randomly sampling the target quantity from each corpus (see Table 3.1). Given English CHILDES contains a total of 15.6M words of caregiver inputs, the available data is equivalent to those at the 30-month level, which thus serves as the maximum training input. We emphasize that this scaling controls for *total input quantity* rather than assuming uniform structure of children’s linguistic environments across months.

Preprocessing All text is lowercased, with spaces and basic punctuation (periods, commas, apostrophes) preserved as word boundary markers. No tokenization or subword segmentation is applied, forcing models to infer word boundaries directly from raw character sequences.

3.3.3 Model Architectures and Training

We train character-level autoregressive models under two architectures: **LSTM** Three layers with 200-dimensional character embeddings, 1024-dimensional hidden states, and 200-dimensional output projections, following LAVECHIN et al. [Lavechin, 2023]. **Transformer** GPT-2-style decoder-only architecture with 12 transformer layers, 512-dimensional hidden states, 12 attention heads per layer, following GORIELY et al. [Goriely, 2024].

Training procedure Models are trained to minimize cross-entropy loss using the Adam optimizer (learning rate 0.0001, $\beta_1 = 0.9$, $\beta_2 = 0.98$) with inverse square root learning rate decay and 10,000 warmup steps. We train until validation perplexity converges. For each age-corpus

pair, two models are trained with same random seeds and non-overlapping data subsets. Training details are provided in Table 3.2.

Generation procedure At inference time, we generate text using temperature sampling with $T \in \{0.3, 0.6, 1.0, 1.5\}$. Generation begins with a beginning-of-sequence (BOS) token and continues until reaching the target word count corresponding to children’s estimated monthly production at that developmental stage (see Table 3.3 in Appendix). Character sequences are segmented into words using space and punctuation boundaries. For months 31–36, outputs are generated from the same model checkpoints. Because the CDS corpus contains a total of 15.6M words of caregiver input, this limits the age-appropriate training to the 30-month equivalent. For months 31–36, we generate outputs using the same 30-month checkpoint, scaling generation lengths according to children’s estimated production, while noting that the training exposure does not increase beyond 30 months. This ensures a consistent comparison across developmental stages while respecting the corpus limitations.

3.3.4 Evaluation Set Construction

Base vocabulary and POS filtering We start with the MacArthur-Bates Communicative Development Inventory (CDI) [Fenson, 2007], a standardized checklist of 680 English words with normative age-of-production data. To control for lexical class effects, we apply POS tagging using a RoBERTa-based SpaCy pipeline. For each word, POS labels are aggregated across contexts and majority voting is applied separately in each corpus. Function words (articles, pronouns, prepositions, auxiliaries) are excluded, retaining nouns, verbs, adjectives, and adverbs.

Frequency stratification Filtered CDI words are divided into six frequency bands based on their occurrence in child-directed speech (adult input in English CHILDES), with an equal number of words per band.

Audiobook subset selection For a comparable evaluation, we construct a matched audiobook subset using the same POS filtering and frequency banding. Words are selected such that their frequency distribution mirrors the filtered CDI evaluation set. This process yields two aligned datasets: i.) Filtered CDI evaluation set with 420 words in six frequency bands; and ii.) frequency-matched audiobook subset with the same size per frequency band. This enables a developmentally grounded comparison between model generations and children’s expressive vocabulary.

3.3.5 Evaluation Metrics

We assess model performance using three complementary metrics capturing lexical diversity, form accuracy, and expressive vocabulary.

Type-Token Ratio (TTR) Measures lexical diversity as the ratio of unique word types to total word tokens. Following RÄSÄNEN et al. [Räsänen, 2024], we compute TTR over fixed-size chunks of 3,500 words to control for sample size effects, then average across all chunks.

Non-word Rate Quantifies word form accuracy as the proportion of generated word types not found in comprehensive English dictionaries: CELEX [Van Heuven, 2014], the Enchant library,¹ and Wiktionary.² A word is classified as valid if it appears in any of these three resources. This metric captures models’ ability to produce orthographically valid English words.

corpus-CDI Score Adapts the human CDI vocabulary assessment to model generations through a count-based criterion. We mark a word as "known" if it appears at least τ times in the generated sample. To determine τ , we analyze CHILDES child productions and fit the resulting vocabulary growth curve to the empirical CDI trajectory. We select $\tau = 60$, which yields slopes most consistent with observed developmental patterns. The CORPUS-CDI score is computed as:

$$\text{CDI} = \frac{|\{w \in \text{eval set} : \text{count}(w) \geq \tau\}|}{|\text{eval set}|}$$

where the evaluation set includes 420 words. This value represents the proportion of age-appropriate vocabulary that the model generates with sufficient frequency to be considered "known." Sensitivity analyses for alternative thresholds ($\tau \in \{1, 500\}$) are shown in Figure 3.6.

3.4 Results

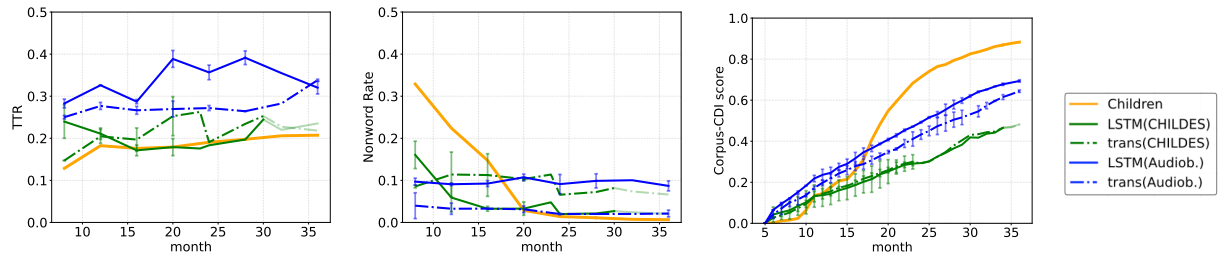


FIGURE 3.2 – **Core metrics comparing child and model vocabulary development across age groups.** Left: Type-token ratio (TTR) measuring lexical diversity. Middle: Non-word type rate indicating the proportion of invalid word forms. Right: CORPUS-CDI scores tracking vocabulary acquisition (words counted as known if produced >60 times). Results are shown for LSTM and Transformer models trained on audiobooks and child-directed speech (CDS), compared with CHILDES child production data. Error bars denote 95% confidence intervals across disjoint training subsets. For the final months (32–36), outputs are generated from the same model checkpoint and shown with lighter transparency.

3.4.1 Developmental Trajectories

Figure 3.2 presents model performance across three complementary metrics as a function of cumulative training exposure (6–36 months), compared against children’s developmental trajectories from CHILDES production data. All results use temperature $T = 1.0$ as the default setting.

1. <https://pypi.org/project/pyenchant/>

2. <https://www.wiktionary.org>

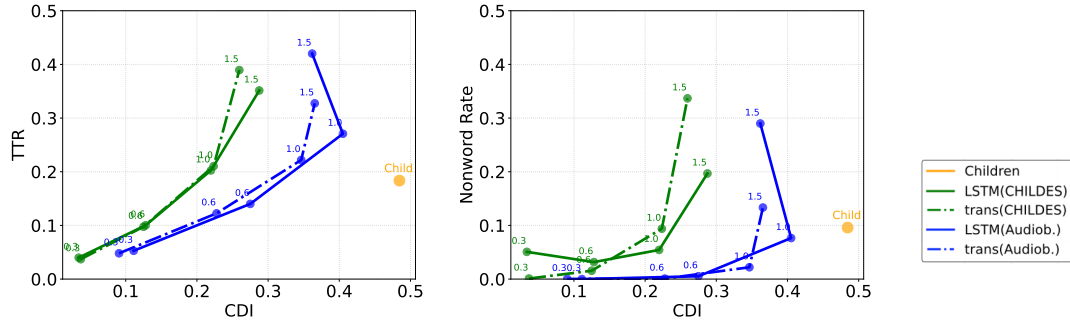


FIGURE 3.3 – **Temperature trade-off across different metrics; temperature settings (0.3-1.5)** Left: Trade-off between type-token ratio and CDI score. Right: Relationship between non-word rate and CDI score. Points show mean scores across models, with numbers illustrating temperature values.

Lexical diversity Both LSTM and Transformer models exhibit higher type-token ratios (TTR) than children across developmental stages. Models trained on audiobooks show slightly higher TTR than those trained on child-directed speech, reflecting broader lexical variety in narrative text. Unlike children, whose TTR gradually increases with age, model TTR remains relatively stable. This suggests models generate more lexically diverse outputs from early on, but does not increase substantially with additional training data.

Word form accuracy Models and children show different developmental patterns in non-word production rates. Children begin with high non-word rates that decrease sharply by the 24th month, reflecting the gradual development of articulatory control and phonological mastery. In contrast, both model architectures maintain relatively stable non-word rates across all training quantities, showing minimal improvement with increased data exposure. Transformers produce slightly fewer non-words than LSTMs, but neither captures the steep improvement observed in children. This divergence likely reflects a critical limitation of text-based models: they cannot capture the development of articulatory-motor control that drives children’s decreasing error rates.

Vocabulary growth CORPUS-CDI scores show that children’s vocabulary grows sigmoidally, with rapid acceleration during the vocabulary spurt. Models, by contrast, display roughly linear growth, with Transformers slightly outperforming LSTMs and audiobook-trained models outperforming CDS-trained ones. Critically, neither architecture reproduces the accelerating trajectory characteristic of human vocabulary development even as training data accumulates as the estimated quantity.

Across all three metrics, models exhibit qualitatively different developmental patterns from children. They show excessive lexical diversity, minimal improvement in word form accuracy, and slower vocabulary growth that lacks the characteristic acceleration of human development. These differences persist across both architectures and training corpora, suggesting they reflect fundamental limitations in how current language models learn from sequential exposure rather than artifacts of specific model choices. In the following sections, we investigate two potential

explanations for these gaps: whether alternative temperature settings during decoding can better align model behavior with children (Section 3.4.2), and whether the observed differences relate to models’ frequency-dependent learning patterns (Section 3.4.3).

3.4.2 Temperature Tradeoff

The developmental gaps observed in Section 3.4.1 raise an important question: could these differences arise from suboptimal generation parameters rather than inherent learning limitations? Temperature sampling controls generation stochasticity, with lower temperatures favoring high-probability outputs and higher temperatures increasing diversity. We evaluate whether any temperature setting can align models with children across all three metrics.

Figure 3.3 shows the relationships between metrics as temperature varies from 0.3 to 1.5. Each point represents the mean performance across all developmental stages for a given architecture, corpus, and temperature combination.

TTR-CDI trade-off Models exhibit a clear trade-off: increasing temperature simultaneously raises type-token ratio (TTR) and CORPUS-CDI scores. Low temperatures produce TTR values closer to children’s range but severely underrepresent vocabulary size, whereas high temperatures improve vocabulary coverage at the cost of excessive lexical diversity. No temperature setting positions models within the child-like range for both metrics. Audiobook-trained models consistently overshoot children’s TTR, while CDS-trained models come closer but still fall short in vocabulary size. This indicates that the repetitive, high-frequency vocabulary characteristic of early child speech cannot be recovered through decoding adjustments alone.

Non-word-CDI trade-off Higher temperatures increase non-word rates for both architectures. While some CDS-trained models at intermediate temperatures approximate children’s average non-word rates, this alignment is misleading: children’s non-word rates decrease sharply over development, whereas models remain largely static. Moreover, achieving this partial alignment comes at the expense of TTR matching, suggesting that it is still implausible to simultaneously satisfy multiple developmental constraints.

These results demonstrate that temperature tuning cannot resolve the developmental misalignment between models and children. The gaps reflect fundamental differences in how models and children balance vocabulary size, lexical diversity, and production accuracy. Post-hoc decoding adjustments cannot substitute for differences in underlying learning and production mechanisms. In the next section, we examine whether these differences relate to frequency-dependent learning patterns.

3.4.3 Frequency effect

The analyses so far show that models lag far behind children in vocabulary acquisition, and this gap cannot be closed by tuning generation parameters. We now ask whether this delay is uniform across input frequency. To test this, we divided the vocabulary into six frequency bands based on child-directed speech and tracked the proportion of words acquired in each band, using

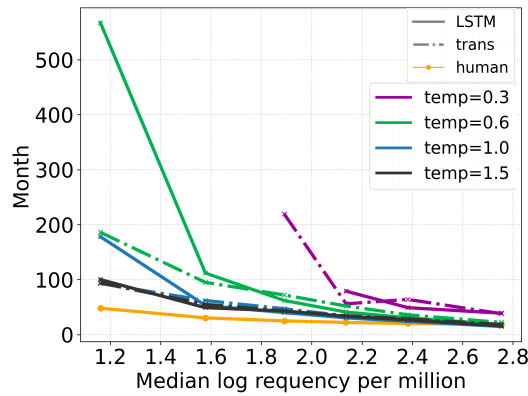


FIGURE 3.4 – **Extrapolated months across different datasets for models trained on audiobooks.** x-axis: log frequency per million in training data; y-axis: estimated developmental month at which models reach 80% mastery for each frequency band. Missing points of the lower frequency bands of temperature =0.3 come from 0

the same sigmoid growth curve fitting as above. For each band, we extracted the *extrapolated age of acquisition*, defined as the cumulative training exposure at which the model reaches 80% mastery, matching the maximum rate observed in CDI norms. This “months-to-mastery” measure quantifies how quickly different frequency bands are learned as a function of cumulative training exposure.

Divergent frequency sensitivity Figure 3.4 shows that children show much weaker frequency dependence than models, with only modest variation between high- and low-frequency vocabulary. By contrast, both models display considerable frequency dependence: high-frequency words are acquired relatively quickly, but acquisition times for low-frequency words increase by orders of magnitude. This indicates that models rely heavily on repeated exposure, whereas children generalize more effectively from sparse input.

Temperature modulation Temperature settings systematically modulate frequency effects. At low temperatures ($T = 0.3$), models fail to acquire rare words, as deterministic decoding prevents them from being generated sufficient times to count as “known”. Higher temperatures ($T = 1.0$ – 1.5) alleviate this issue by surfacing rare words more often, but the improvement remains limited: even under the most favorable conditions, models require many times more exposure than children to acquire low-frequency vocabulary. Thus, temperature affects severity of the bias but not the underlying cause.

Long-tail truncation These results point to long-tail truncation as the primary mechanism underlying models’ vocabulary acquisition gap. Language models are trained to minimize cross-entropy loss, which optimizes for predicting the most probable next character given context. This objective inherently prioritizes frequent patterns: prediction errors on common words contribute more to loss than on rare words, leading models to allocate representational capacity proportionally to frequency.

Children, by contrast, do not optimize for predictive accuracy. Their learning mechanisms such as explicit memory for individual word-learning episodes [Markson, 1997], cross-situational inference [Smith, 2008], and social-pragmatic cues [Tomasello, 2000] allow them to acquire words from sparse observations without requiring proportional exposure. This qualitative difference in learning objective, rather than mere data inefficiency, explains why children succeed in the long tail where models consistently fail.

3.4.4 Production of Content Words

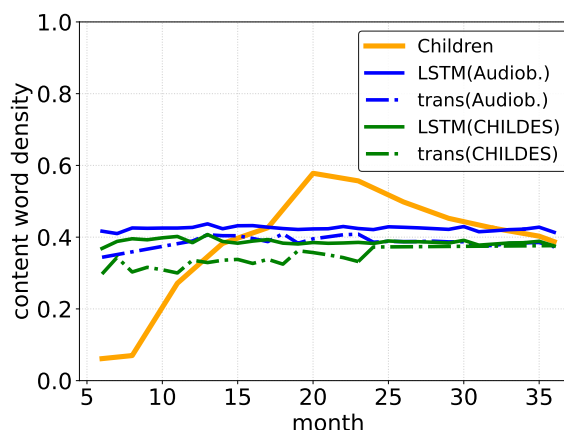


FIGURE 3.5 – **Proportion of content words in generated text across development.** Children show an inverse U-shaped trajectory, while all model configurations maintain stable proportions around 40% across training stages.

The frequency-dependent learning patterns in Section 3.4.3 may partly stem from differences in functional versus content word production. Function words dominate the highest-frequency bands [Zipf, 2016], whereas our CORPUS-CDI evaluation targets content words (Section 3.3.4). Over-reliance on function words could explain why models achieve lower CDI scores and limited lexical diversity: frequent repetition of a small set of functional items increases token counts without expanding the number of unique word types.

Children’s inverse U-shaped trajectory Figure 3.5 shows that children exhibit a pronounced inverse U-shaped trajectory in content word production. Early productions contain few content words, followed by a sharp rise during the vocabulary spurt, and then a decline as grammatical function words become integrated [Bates, 1994; Brown, 1973]. Critically, this declining phase coincides with children’s steepest CORPUS-CDI growth (Figure 3.2), showing that children’s vocabulary growth is not simply driven by frequency exposure but by a qualitative reorganization of their lexical production.

Models’ invariant balance In contrast, all model configurations maintain stable content-word proportions around 40%, which closely matches adult English distributions [Biber, 2021]. This invariance holds across architectures and training corpora: both CDS- and audiobook-trained

models converge almost immediately to corpus-level statistics and remain fixed throughout training.

This invariance helps explain several earlier findings: (1) Frequency truncation (Section 3.4.3) disproportionately harms content words, which spread more evenly across frequency bands. (2) CDI underperformance reflects systematic undergeneration of content words, on which CDI norms are based. (3) Elevated TTR may paradoxically arise from over-allocating tokens to a small set of high-frequency function words, forcing broader but shallower sampling of content vocabulary.

Taken together, these results reveal a fundamental mismatch: cross-entropy training aligns models with corpus-level statistics but not with children’s developmental trajectory. While children dynamically restructure their lexicon and functional–content balance, models remain tied to static frequency distributions.

3.5 Discussion

We introduced CORPUS-CDI, a benchmark that enables direct comparison between children’s expressive vocabulary and language model generations by aligning training exposure, production quantity, and frequency distributions. Our analyses yield three main findings that highlight key differences between human and model vocabulary acquisition.

First, while both character-level LSTMs and Transformers expand their vocabularies with increased training data, their developmental trajectories diverge largely from those of children. Models consistently produce far greater lexical diversity, maintain stable non-word rates across training, and exhibit slower vocabulary growth without the characteristic acceleration observed in children. These patterns generalize across architectures (LSTM vs. Transformer) and training corpora (audiobooks vs. child-directed speech), suggesting that the underlying learning mechanisms driving word acquisition differ fundamentally between children and current language models. In other words, the mismatch is not tied to the choice of corpus or model architecture, but reflects mechanistic differences in how children versus models acquire and prioritize vocabulary.

Second, these divergences cannot be reconciled through parameter tuning at generation time. Temperature variation experiments show that no single setting simultaneously aligns models with children across TTR, non-word rate, and vocabulary size. The gap thus arises from fundamental properties of the learning and production mechanisms rather than from suboptimal decoding choices.

Third, frequency analyses reveal the mechanistic source of these differences. Children acquire words across frequency bands within a narrow age window (around 35–40 months), whereas models require exponentially more exposure for low-frequency words (equivalent to 300–600 months of input for words around log frequency 1.2–1.4 per million). This extreme frequency sensitivity follows directly from cross-entropy optimization, which down-weights rare words during training. Importantly, the interaction with word categories amplifies the divergence: models preserve adult-like content/function ratios (around 40% content words) throughout training, while children show a U-shaped trajectory driven by the integration of function words. Since

function words cluster in high-frequency bands whereas content words (the CDI evaluation focus) distribute more evenly, models’ frequency-proportional learning underproduces the vocabulary class children prioritize.

These findings point to limitations of predictive accuracy as the sole training objective for modeling acquisition. Children do not rely solely on distributional statistics; they integrate referential grounding, social-pragmatic inference, and episodic memory of word-learning events to acquire vocabulary from sparse observations. Progress in cognitively grounded language modeling may therefore require learning objectives beyond predictive accuracy, such as memory-augmented architectures for rare word retention, contrastive objectives that emphasize lexical representations, or interactive learning paradigms where production influences subsequent input.

3.6 Limitations

Our evaluation framework has several limitations. First, we evaluate character-level models rather than speech-based systems. While character input provides an upper bound by preserving invariant word forms and explicit boundaries, it bypasses the acoustic variability and segmentation challenges that infants solve [Dupoux, 2018]. Speech-based language models [Lakhotia, 2021; Nguyen, 2024] must additionally contend with phonetic variability and ambiguous word boundaries. As such, our reported gaps likely underestimate the difficulty of vocabulary acquisition in more realistic, speech-based models. Extending CORPUS-CDI to speech therefore requires addressing the non-trivial task of mapping continuous acoustic output into discrete word forms suitable for CDI scoring and we would expect even larger divergences from children’s developmental trajectories. Moreover, our models lack multimodal grounding—the word-object associations and interactive feedback that children exploit. Such grounding likely supports children’s frequency-independent learning by providing alternative cues beyond distributional statistics.

Second, the CHILDES transcripts used as reference data introduce possible biases. Transcribers often normalize children’s productions by correcting errors or completing fragments, which may underestimate non-word rates and inflate apparent model–child gaps. Conversely, our character-level models inherently generate orthographically valid forms, potentially overstating their production accuracy relative to children’s speech.

Third, our comparison of parental CDI reports with model outputs involves methodological misalignment. CDI scores reflect whether parents judge a child to “say” a word, capturing semantic knowledge and contextual usage in addition to production capability. In contrast, our CORPUS-CDI metric counts raw production frequency in unconstrained model text. A child uttering dog once demonstrates robust lexical knowledge, whereas a model generating dog 60 times may not. While our threshold ($\tau = 60$) calibrates vocabulary growth slopes, the construct validity gap between CDI reports and model generation remains.

Finally, our study focuses on English and a single vocabulary assessment tool (CDI). Cross-linguistic validation is essential for establishing universality: languages with richer morphology, less transparent orthography, or different child-directed registers may yield different alignments

between model and child learning. Character-level models in particular may be sensitive to orthographic properties, limiting generalizability beyond English.

3.7 Ethics Statement

All data used in this study are publicly available through the CHILDES database. Our work focuses exclusively on model benchmarking and internal simulations, with the goal of advancing fundamental research on language acquisition. We emphasize that the evaluations conducted here are not sufficient for deployment in real-world applications. Any future use involving human data should undergo rigorous validation and appropriate de-identification procedures to ensure privacy and ethical compliance.

3.8 Appendix

3.8.1 Training and generation details

Table 3.1 summarizes the age-appropriate scaling of training data across three exposure scenarios. Linguistic exposure estimates are based on daylong recording studies [Cristia, 2019], which document substantial variability in the amount of speech heard by children across environments. Annual exposure ranges from approximately 100 hours in low-input households to 1000 hours in high-input ones, with 500 hours per year representing a mid-range estimate.

Cumulative exposure thus increases from 50–500 hours at 6 months to 300–3000 hours by 36 months. Assuming a standard speech rate of three words per second, this corresponds to training sets ranging from 0.58M words (6 months, low exposure) to 34.71M words (36 months, high exposure).

Age	Hours	Words	Characters	Utterances
<i>Low exposure scenario (100 hours/year)</i>				
6	50	0.58	2.95	58
12	100	1.16	5.89	116
18	150	1.73	8.84	173
24	200	2.31	11.78	231
30	250	2.89	14.73	289
36	300	3.47	17.67	347
<i>Medium exposure scenario (500 hours/year)</i>				
6	250	2.89	14.73	289
12	500	5.78	29.45	578
18	750	8.68	44.18	868
24	1000	11.57	58.90	1157
30	1250	14.46	73.63	1446
36	1500	17.35	88.35	1735
<i>High exposure scenario (1000 hours/year)</i>				
6	500	5.78	29.45	578
12	1000	11.57	58.90	1157
18	1500	17.35	88.35	1735
24	2000	23.14	117.80	2314
30	2500	28.92	147.25	2892
36	3000	34.71	176.70	3471

TABLE 3.1 – **Age-appropriate scaling of training datasets across three exposure scenarios.** Estimates are derived from three exposure scenarios (100h, 500h, 1000h per year), reflecting observed variability in children’s linguistic environments [Cristia, 2019]. Word counts assume 3 words/second; character counts assume 5.1 characters/word; utterance counts assume 10 words/utterance.

Model training employed carefully tuned hyperparameters (Table 3.2), selected through systematic validation experiments to ensure training stability and computational efficiency across different data scales.

The examples in Table 3.4 highlight distinct generation behaviors across datasets and model

Hyperparameter	Value
Max sequence length	1024
Batch size	64
Learning rate	0.0001
Learning rate scheduler	inverse sqrt
Warmup steps	10000
Optimizer	Adam
Adam-beta1	0.9
Adam-beta2	0.98
Dropout	0.1
LSTM hyperparameter	Value
decoder layers	3
hidden size	1024
embedding dimension	200
Transformer hyperparameter	Value
Transformer layers	12
Intermediate hidden size	512
Attention heads	12
Attention dropout	0.05

TABLE 3.2 – **Language model training hyperparameters.** Configurations balance model capacity, stability, and efficiency across training scales.

architectures under a fixed sampling temperature ($T = 1.0$). Models trained on child-directed input (Child, CHILDES) tend to produce shorter, phonologically simple, and semantically grounded utterances, resembling early stages of language development. In contrast, audiobook-trained models generate longer, syntactically structured sequences but often lack pragmatic coherence. Comparing architectures, Transformers exhibit more globally consistent syntax and longer dependency spans, while LSTMs display stronger local predictability but higher rates of morphological irregularities. These qualitative patterns align with quantitative trends observed in our lexical and nonword diversity metrics, underscoring how data type and inductive bias jointly shape emergent linguistic structure.

3.8.2 Result details

To establish reliable vocabulary acquisition metrics, we conducted systematic analyses of different word count thresholds (Figure 3.6). This calibration process aimed to align our CORPUS-CDI scores with empirical observations of child vocabulary development.

Notably, the frequency effects observed in the Machine CDI are evaluated on a selected subset matched with the human-CDI set. It is unclear whether this effect is influenced by the randomness of sampling. Also, the selected words are typically in the median frequency bands, making it unknown whether such an effect stems from the limited generation size. To address this concern, we expanded our analysis of the model generations using the same and larger sampling sets. Figure 3.7 shows the word frequency distributions of the training and generated datasets by the LSTM model trained on 3.4M words, with different generation sizes. Expanding the

month	#sec/hour	estimation	production
6	61.25	55,125	782
7	65.30	58,769	1,323
8	69.35	62,412	1,278
9	73.40	66,056	3,964
10	77.44	69,699	627
11	81.49	73,343	1,148
12	85.54	76,986	1,361
13	89.59	80,630	568
14	93.64	84,273	2,720
15	97.69	87,917	13,765
16	101.73	91,560	3,620
17	105.78	95,204	7,589
18	109.83	98,847	15,197
19	113.88	102,491	17,090
20	117.93	106,134	19,912
21	121.98	109,778	33,458
22	126.02	113,421	33,880
23	130.07	117,065	63,738
24	134.12	120,708	191,062
25	138.17	124,352	130,525
26	142.22	127,995	136,195
27	146.27	131,639	148,741
28	150.31	135,282	158,910
29	154.36	138,926	190,263
30	158.41	142,569	195,549
31	162.46	146,213	196,677
32	166.51	149,856	184,024
33	170.56	153,500	185,534
34	174.60	157,143	185,691
35	178.65	160,787	153,515
36	182.70	164,430	299,537

TABLE 3.3 – **Monthly progression of speech metrics.** Estimated exposure and production rates over developmental time.

generation set mitigates some truncation effects, as more high-frequency words are generated, but low-frequency words remain underrepresented. This pattern reflects the model’s limited ability to reproduce realistic word distributions, especially for rare words, even with larger sampling sizes.

Although models trained on higher-exposure estimates (1000 h) achieve slightly higher CORPUS-CDI scores than those trained on smaller sets (100 h, 500 h), the overall developmental trajectories remain similar. This pattern indicates that scaling input exposure alone does not yield more human-like developmental dynamics; instead, the gap likely arises from structural differences in learning mechanisms rather than data quantity.

Setting	Generated Sequences
Child	dad dada mommy looked it missed its wheel i want my plate for it
Audiobook (LSTM)	wounded for woman says no i don't know what i fuller over his week scully a
Audiobook (Trans)	to o he had been and the work of the lord ascend mass of softly rounded head
CHILDES (LSTM)	can bump he said I was wondering what yeah detfive wroggy on drh aulf
CHILDES (Trans)	on helsewhere he doesn't like him does he hafta dragged on doh aunty quite

TABLE 3.4 – **Examples of generated sequences.** Samples are drawn from models trained on different datasets and architectures at a sampling temperature of 1.0. The boundary marker is replaced with blank space for readability.

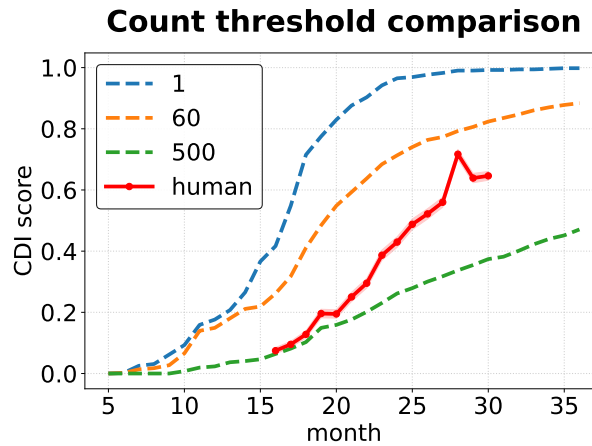


FIGURE 3.6 – **Count-based calibration** Comparison of different word frequency thresholds for determining word acquisition. The optimal threshold (60 occurrences) was selected based on alignment with parental vocabulary size reports (red line). Each threshold produces distinct vocabulary growth trajectories, with higher thresholds resulting in more conservative estimates.

Dataset (Model)	Temp	TTR (%)		NWType (%)		NWToken (%)		CDI(%)	
		Mean	Last	Mean	Last	Mean	Last	Mean	Last
CHILDES	-	18.3	20.7	9.6	0.7	4.7	0.2	48.6	88.
Audiobook (LSTM)	0.3	43.8 [41.3, 46.1]	44.2	13.5 [12.0, 14.8]	13.8	10.4 [9.2, 11.6]	10.6	62.5 [59.2, 65.8]	63.
	0.6	45.9 [43.4, 48.2]	46.3	14.8 [13.3, 16.1]	15.1	11.7 [10.5, 12.9]	11.9	65.2 [61.9, 68.5]	65.
	1.0	48.1 [45.6, 50.4]	48.5	16.4 [14.9, 17.7]	16.7	13.1 [11.9, 14.3]	13.3	68.8 [65.5, 72.1]	69.
	1.5	50.3 [47.8, 52.6]	50.7	18.0 [16.5, 19.3]	18.3	14.8 [13.6, 16.0]	15.0	71.9 [68.6, 75.2]	72.
Audiobook (Trans)	0.3	44.5 [42.0, 46.8]	44.9	13.8 [12.3, 15.1]	14.1	10.7 [9.5, 11.9]	10.9	63.2 [59.9, 66.5]	63.
	0.6	46.6 [44.1, 48.9]	47.0	15.1 [13.6, 16.4]	15.4	12.0 [10.8, 13.2]	12.2	65.9 [62.6, 69.2]	66.
	1.0	48.8 [46.3, 51.1]	49.2	16.7 [15.2, 18.0]	17.0	13.4 [12.2, 14.6]	13.6	69.5 [66.2, 72.8]	70.
	1.5	51.0 [48.5, 53.3]	51.4	18.3 [16.8, 19.6]	18.6	15.1 [13.9, 16.3]	15.3	72.6 [69.3, 75.9]	73.
CHILDES (LSTM)	0.3	43.8 [41.3, 46.1]	44.2	13.5 [12.0, 14.8]	13.8	10.4 [9.2, 11.6]	10.6	62.5 [59.2, 65.8]	63.
	0.6	45.9 [43.4, 48.2]	46.3	14.8 [13.3, 16.1]	15.1	11.7 [10.5, 12.9]	11.9	65.2 [61.9, 68.5]	65.
	1.0	48.1 [45.6, 50.4]	48.5	16.4 [14.9, 17.7]	16.7	13.1 [11.9, 14.3]	13.3	68.8 [65.5, 72.1]	69.
	1.5	50.3 [47.8, 52.6]	50.7	18.0 [16.5, 19.3]	18.3	14.8 [13.6, 16.0]	15.0	71.9 [68.6, 75.2]	72.
CHILDES (Trans)	0.3	44.5 [42.0, 46.8]	44.9	13.8 [12.3, 15.1]	14.1	10.7 [9.5, 11.9]	10.9	63.2 [59.9, 66.5]	63.
	0.6	46.6 [44.1, 48.9]	47.0	15.1 [13.6, 16.4]	15.4	12.0 [10.8, 13.2]	12.2	65.9 [62.6, 69.2]	66.
	1.0	48.8 [46.3, 51.1]	49.2	16.7 [15.2, 18.0]	17.0	13.4 [12.2, 14.6]	13.6	69.5 [66.2, 72.8]	70.
	1.5	51.0 [48.5, 53.3]	51.4	18.3 [16.8, 19.6]	18.6	15.1 [13.9, 16.3]	15.3	72.6 [69.3, 75.9]	73.

TABLE 3.5 – Metric scores across different model settings presented in percentages (%). Mean values are averaged across different months, followed by range across different models. We also reported the score of the 36th month. TTR: Type-token ratio, NWType: Non-word type rate, NWToken: Non-word token rate. CHILDES child production. Best scores for each setting are in **bold**.

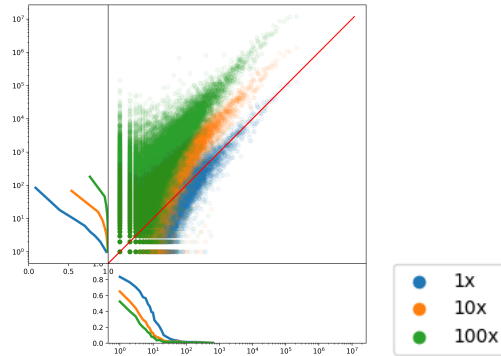


FIGURE 3.7 – **Word count distribution(temperature = 1.0)**. Word frequency relationships between training and generation sets under different sampling conditions (1x, 10x, 100x generation volume). x-axis: word frequency in the training set; y-axis: word frequency in the generation set. Side plot on the bottom show Missing rates as a function of word count in input corpus.

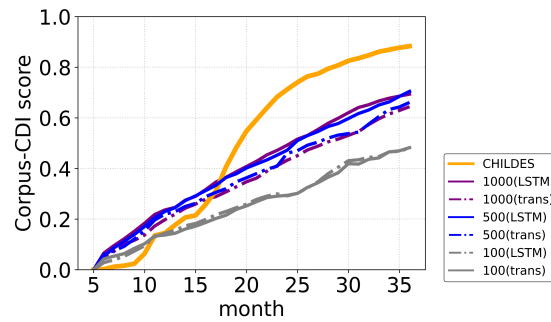


FIGURE 3.8 – **Model vocabulary growth across estimated annual exposure levels.** Comparison of CORPUS-CDI trajectories for models trained on 100-, 500-, and 1000-hour exposure estimates, alongside CHILDES child production data. Models show roughly linear vocabulary growth that scales modestly with total exposure but remains far below the human developmental curve.

Chapitre 4

Does episodic memory help close the lexical frequency gap in sensitivity to syntactic contrasts? A test using retrieval-augmented language models

Objectifs

Test whether retrieval-augmented language models (k -nearest-neighbor LMs) reduce the frequency gap that vanilla LMs show on syntactic minimal-pair contrasts, and identify which retrieval signals (structural vs. semantic) drive any gains.

4.1 Introduction

Language models are known to exhibit systematic lexical frequency biases: their performance on linguistic evaluations correlates with how often the lexical items in the test examples appear in training data, leading to low-frequency items posing a persistent challenge [Wei, 2021 ; Kim, 2022 ; Diehl Martinez, 2024]. While frequency does influence human performance in empirical tests of grammatical knowledge such as acceptability judgments [lau2017grammaticality ; Kempen, 2005 ; Bader, 2010, i.a.], recent work [Christensen, 2024] has shown evidence for robustness to lexical frequency in delineating the critical contrasts. What accounts for this gap between human robustness and model fragility?

One possible answer comes from Complementary Learning Systems (CLS) theory [McClelland, 1995 ; Kumaran, 2016], which proposes that the brain employs two distinct memory systems: a hippocampal system that rapidly encodes and retrieves specific experiences, and a neocortical system that gradually extracts generalizable statistical patterns. When parametric knowledge is weak, as it is for rare items, episodic memory can compensate by directly retrieving relevant past encounters [Davis, 2009]. This division of labor has also been proposed to be relevant to linguistic processing [Borensztajn, 2011 ; Duff, 2012], leading us to hypothesize that episodic retrieval may reflect the mechanism that underlies human robustness to lexical frequency in syntactic processing.

In this work, we computationally test this hypothesis by asking whether augmenting language models with an explicit retrieval mechanism helps close the lexical frequency gap in syntactic contrast evaluations. We adopt k -nearest-neighbor language models (k NN-LM; [Khandelwal, 2020]) as our proxy for episodic memory: the non-parametric datastore caches training instances for direct retrieval at inference time (analogous to hippocampal episodic traces), while the underlying parametric network consolidates statistical patterns through training (analogous to neocortical learning). The original k NN-LM has already been shown to be particularly helpful for rare patterns [Mallen, 2023 ; Kandpal, 2023], and recent work shows that the information available in the context can be used more flexibly than the same information stored parametrically [chan2022transformers ; Lampinen, 2025], both consistent with the CLS prediction. At the same time, a recent finding that k NN-LMs do *not* improve prediction of low-frequency tokens in standard language modeling evaluation [Nishida, 2025] makes it a non-trivial and empirically open question whether retrieval benefits extend to the domain of syntactic knowledge.

We investigate this question through syntactic contrast tests across three different phenomena (subject-verb agreement, wh-questions, relative clauses) using frequency-stratified test items, under both child-realistic and large-scale pretraining. We further analyze what drives retrieval effectiveness by contrasting full versus semantics-only retrieval, by examining the role of structural similarity in retrieved instances, and by varying the configurations of retrieval (retrieval granularity, context window size, and number of retrieved neighbors). Our results show that retrieval augmentation consistently narrows the lexical frequency gap across all phenomena and both model scales, with greater gains for more syntactically complex phenomena. Nevertheless, the gap remains not fully closed. To this end, we analyze the limitations of the current modeling

approach and posit preferential reweighting, improved representations and retrieval for structural information, and flexible configurations of storage and retrieval as promising future directions.

4.2 Related Work

Frequency effects on linguistic evaluation in language models. Language models exhibit systematic frequency biases, where performance on linguistic evaluation often degrades when the test items contain low-frequency lexical items [Wei, 2021 ; Kim, 2022]. In the context of syntactic contrasts, models perform worse on minimal pairs that contain low-frequency words, even when the underlying grammatical contrast is identical to high-frequency cases. DIEHL MARTINEZ et al. [Diehl Martinez, 2024] directly quantify this frequency bias in BLiMP (one of our syntactic test datasets) and propose a training-time intervention (Syntactic Smoothing) to reduce it, finding that substantial gaps persist even after mitigation.

WETTIG et al. [Wettig, 2023] examine how the masking rate during training determines which co-occurring token pairs (measured via pointwise mutual information) are exposed to the model, shaping both convergence speed and final performance. Since pointwise mutual information is itself derived from corpus-level token frequencies, these findings are consistent with frequency distributions in pretraining data influencing model behavior, though this connection is inferential. More directly, KANDPAL et al. [Kandpal, 2023] show that language models struggle to acquire long-tail knowledge whose supporting documents are rare in the pretraining corpus, establishing a clear link between corpus frequency and what is learned.

Together, these findings establish that (lexical) frequency creates differential strength in parametric representations, with low-frequency items learned far less robustly. Our work asks a related question from a different methodological angle: rather than reducing frequency bias through better training, can it be compensated for at inference time through retrieval?

Complementary Learning Systems theory. Complementary Learning Systems (CLS) theory [McClelland, 1995 ; Kumaran, 2016] proposes that the human brain employs two specialized memory systems: a hippocampal system that rapidly encodes specific episodes, and a neocortical system that gradually extracts statistical regularities across experiences. This allows new information to be acquired quickly without disrupting slower cortical learning, and computational implementations of CLS show that it enables both rapid memorization and gradual generalization [McClelland, 1995]. Critically for our purposes, EICHENBAUM [Eichenbaum, 2001] argues that the hippocampus does not merely store episodes but actively supports flexible use of past experiences through retrieval, enabling inferences that parametric learning alone may fail to achieve. Applied to language models, this predicts that when parametric representations are weak (as for low-frequency items), episodic retrieval should provide compensatory benefits by making specific relevant experiences directly accessible. LAMPINEN et al. [Lampinen, 2025] in particular demonstrates that language models fail to generalize certain types of knowledge learned parametrically (e.g., relation reversals) but succeed when the same information is available in context, suggesting that episodic retrieval enables more flexible use of past experiences than parametric consolidation. Similarly, studies of in-context learning show that Transformers can apply

procedures to information presented in context that they cannot apply to parametrically-stored knowledge [chan2022transformers]. Our work extends these insights to syntactic processing by testing whether retrieval enables more robust generalization across frequency bands than parametric-only inferences.

Retrieval-augmented language models. Retrieval-augmented methods enhance language models by supplementing parametric knowledge with information drawn from an explicit memory at inference time. k NN-LMs [Khandelwal, 2020], which we adopt in this work, interpolate the base model’s next-token distribution with a distribution computed from nearest-neighbor contexts retrieved from a datastore of training instances. While prior work shows benefits for rare patterns and knowledge-intensive tasks [Borgeaud, 2022; Kandpal, 2023; Mallen, 2023], recent work finds that standard k NN-LMs do not improve prediction performance for low-frequency tokens in language modeling evaluation [Nishida, 2025], leaving open questions about retrieval benefits for linguistic processing.

4.3 Method

We first show that frequency gaps in tests of grammatical knowledge exist in vanilla language models (“base models”), and then test whether k NN-LMs help close these gaps. To this end, we use targeted syntactic contrast datasets where we divide test instances based on the frequency of the content words they contain. We describe the details of the models, datasets, evaluation tasks and the retrieval procedure below.

4.3.1 Models and Training

Base models. We evaluate two decoder-only Transformer models that vary in parameter size and training data scale: (1) GPT-2 XL (1.5B parameters), pretrained on large-scale web text [Radford, 2019], and (2) GPT-2 Small (124M parameters) pretrained on child-directed speech corpora with 50M words selected from the BabyLM Challenge [Hu, 2024]. Both use a standard autoregressive objective and serve as parametric-only baselines. Further details are provided in Appendix 4.8.1.

k NN language models. Following KHANDELWAL et al. [Khandelwal, 2020], we augment base models with a datastore $\mathcal{D}$ containing context–target pairs. For each token x_t , we store the key-value pair: $(\mathbf{f}_\theta(\mathbf{x}_{<t}), x_t)$, where $\mathbf{f}_\theta(\mathbf{x}_{<t})$ is the layer-normalized contextual representation fed as input to the feed-forward sublayer of the final Transformer block. For GPT-2, this corresponds to the output of the final block’s second layer-normalization module and the input to its MLP. At inference time, we:

1. Compute the query: $\mathbf{q}_t = \mathbf{f}_\theta(\mathbf{x}_{<t})$
2. Retrieve k nearest neighbors via squared L2 distance
3. Compute the k NN probability: $p_{kNN}(x_t|\mathbf{x}_{<t}) \propto \sum_{(\mathbf{k}, v) \in \mathcal{N}} \mathbb{1}_{x_t=v} \exp(-d(\mathbf{k}, \mathbf{q}_t)/\tau)$
4. Interpolate: $p(x_t|\mathbf{x}_{<t}) = \lambda p_{kNN} + (1 - \lambda)p_{LM}$

We fix $\lambda = 0.25$ following KHANDELWAL et al. [Khandelwal, 2020] and vary $k \in \{1, 16, 1024\}$, using FAISS [Johnson, 2019] for retrieval.

4.3.2 Evaluation Task

Syntactic contrasts. We evaluate sensitivity to three syntactic phenomena: Subject-Verb agreement (SV), Wh-questions (Wh), and Relative Clauses (RC). We collect examples in these categories from BLiMP [Warstadt, 2020a], Zorro [Huebner, 2021a], and BIG-bench [Srivastava, 2023], filtering examples containing out-of-vocabulary items for each model. Each test item is a minimal pair with one grammatical and one ungrammatical sentence, where models must assign a higher probability to the grammatical sentence.

Frequency stratification. To test our hypothesis regarding lexical frequency, we further divided the evaluation data based on mean frequency across all the content words in the sentences (*high-frequency* ($f > 10^4$) versus *low-frequency* ($f < 10^3$); everything in-between *mid-frequency*). Our main hypothesis concerns the contrast between the high- and low-frequency bands; we report the mid-frequency band throughout to give a complete picture of how retrieval benefits scale with frequency.

We emphasize that all frequency gap claims are scoped *within* a single model/corpus: each gap and its narrowing under retrieval is measured against the same model’s own baseline. Because the two pretraining corpora differ substantially in size and composition, the absolute frequency thresholds do not render items in the same band directly comparable across pretraining settings, and we therefore do not compare numerical trends across the two settings.

4.3.3 Retrieval Procedure

We construct three datastore variants to disentangle the contributions of structural and semantic information: (1) *Full Retrieval*: standard contextualized embeddings encoding both semantic and syntactic information; (2) *Semantics-only (averaged)*: position-averaged contextualized embeddings from (1);¹ and (3) *Semantics-only (uncontextualized)*: summed GloVe [Pennington, 2014] and fastText [Bojanowski, 2017] embeddings, capturing lexical semantics without structural information. We additionally vary three retrieval configuration parameters: granularity of retrieved instances (token, phrase, or sequence), context window size (τ), and number of retrieved neighbors (k), to characterize which retrieval configurations are most effective across syntactic phenomena.

TABLE 4.1 – Syntactic contrast accuracy and perplexity (PPL) for baseline and kNN-LM models ($k = 16$, $\tau = 3$, sequence-level retrieval). kNN augmentation improves accuracy across all phenomena and both pretraining scales, with relative clauses showing the largest gains. Perplexity reductions are larger for the large-scale model. Bold indicates improvement over baseline.

	Child-realistic		Large-scale	
	baseline	kNN	baseline	kNN
PPL. ↓	14.88	14.30	13.68	11.60
SV ↑	0.76	0.82	0.84	0.89
Wh ↑	0.77	0.81	0.81	0.86
RC ↑	0.68	0.78	0.65	0.84

1. In practice, averaging contextualized embeddings may not fully scrub syntactic information; we return to this issue later.

4.4 Results

4.4.1 Retrieval Benefits Syntactic Contrasts

We begin by investigating the general effectiveness of retrieval augmentation on syntactic contrasts. Table 4.1² shows that k NN augmentation improves performance across all three syntactic phenomena and both pretraining scales, as well as replicating previously reported perplexity improvements [Khandelwal, 2020]. Relative clauses benefit the most from retrieval augmentation across both model scales. This pattern holds consistently regardless of pretraining scale, though perplexity reductions are larger for the large-scale model. The disproportionate benefit for relative clauses suggests that retrieval is most helpful where parametric representations are weakest, which motivates us to analyze the performance gains stratified by frequency.

4.4.2 Frequency Gap is Narrowed but not Fully Closed

Next, we examine whether retrieval helps close the lexical frequency gap. Table 4.2 confirms the existence of frequency gaps in both baseline models across three syntactic phenomena, with performance on low-frequency items consistently lower than on high-frequency items, and mid-frequency items falling in between. With retrieval augmentation, the high-low performance gap narrows across all phenomena and both model scales. In terms of error reduction ($\Delta\text{acc}/(1 - \text{acc}_{\text{base}})$), low-frequency items show substantial relative gains (e.g., 49.2% for RCs in the large-scale model), though we note that this metric can be misleading near ceiling, where small absolute changes in already-high accuracy

TABLE 4.2 – Frequency-stratified syntactic contrast performance ($k = 16$, $\tau = 3$; sequence-level retrieval). ER denotes error reduction, $\Delta\text{acc}/(1 - \text{acc}_{\text{base}})$. Bold indicates improvement over baseline.

		Child-realistic			Large-scale		
		base	kNN	ER (%)	base	kNN	ER (%)
SV	high	0.91	0.90	−11.1	0.96	0.95	−25.0
	mid	0.82	0.83	5.6	0.90	0.89	−10.0
	low	0.60	0.76	40.0	0.71	0.83	41.4
Wh	high	0.92	0.88	−50.0	0.93	0.93	0.0
	mid	0.83	0.86	17.6	0.90	0.91	10.0
	low	0.61	0.70	23.1	0.69	0.74	16.1
RC	high	0.88	0.89	8.3	0.89	0.95	54.5
	mid	0.73	0.81	29.6	0.82	0.89	38.9
	low	0.48	0.65	32.7	0.41	0.70	49.2

produce large negative percentages (e.g., child-realistic high-frequency Wh, $0.92 \rightarrow 0.88$, yields −50%). Raw accuracies should be used to adjust the interpretations in such cases.

Statistical analysis supports the interpretation that narrowing is driven primarily by gains on low-frequency items. Bootstrapped 95% confidence intervals on the per-frequency band performance delta (k NN − baseline) confirm a robust separation: for the child-realistic model, low-frequency items show $\Delta = 0.140$ (95% CI [0.115, 0.166]) versus $\Delta = -0.013$ (95% CI

2. Note for all tables in this section: Baseline models use parametric knowledge only; k NN models are retrieval-augmented. Abbreviations: SV (subject-verb agreement), Wh (wh-questions), RC (relative clauses), PPL (perplexity). We use $k = 16$, $\tau = 3$, and sequence-level retrieval (see Section 4.3.3) as the representative configuration, since this setting yields general gains in low-frequency conditions. However, as discussed in Section 4.5.2, there may exist alternative configurations that outperform this representative configuration for individual phenomena.

$[-0.030, 0.004]$) for high-frequency items, with non-overlapping intervals (large-scale model shows the same pattern; see Appendix 4.8.3).

A mixed-effects logistic regression using treatment coding with high-frequency items and the baseline model as reference levels ($\text{correct} \sim \text{Frequency} * \text{ModelType} + (1|\text{item}) + (1|\text{phenomenon})$) further shows that the retrieval effect is significantly larger for low-frequency than for high-frequency items (Frequency [low] $\times$ ModelType [kNN] interaction: $\beta = 0.52$, $p < 0.001$. The effect of retrieval within the high-frequency band is not significant (ModelType [kNN]: $\beta = -0.04$, $p = 0.62$; Appendix 4.8.3).

Nevertheless, this compensation is not sufficient to eliminate the frequency gap: low-frequency items still yield lower accuracy than high-frequency items across all tested phenomena and both model scales, even after retrieval augmentation.

4.5 Analysis

The analyses in this section fix the retrieval configuration to $k = 16$, $\tau = 3$ with sequence-level retrieval, which isolates the contribution of the factors studied from the effects of configuration.

4.5.1 Importance of Structural vs. Semantic Information

We proceed to investigate which aspects of the retrieved information drive the observed benefits by examining (1) the effect of semantics-only retrieval, and (2) the effect of retrieving structurally similar instances.

Semantics-only retrieval is mostly ineffective. Figure 4.1 shows that semantics-only retrieval (FastText, GloVe, G-Avg; see Section 4.3.3) performs at or substantially *below* the baseline without retrieval across most conditions. This shows that semantic similarity of the retrieved examples alone does not reliably improve syntactic judgments. One partial exception is GPT2-Averaged (G-Avg), where retrieval does lead to substantive gains over the baseline for low-frequency items for RCs, although performance on high-frequency items degrades by a large margin. FastText and GloVe also yield gains for low-frequency RCs in the large-scale setting, although relatively minor. There are two possible, non-mutually exclusive interpretations for this: (1) semantic information does sometimes yield benefits, as discussed in the next paragraph; or (2) that averaging of contextualized embeddings does not totally remove structural information.

Structural information is critical, and similarity-based example selection does not compensate for missing structure. Figure 4.2 examines retrieval benefits from two angles: whether the set of retrieved items contains structurally matching examples (*with* (w) vs. *without* (w/o)),³ and whether retrieval ranks them by embedding similarity or at random ($+r$). Across all phenomena and both model scales, the w bars consistently exceed w/o for low-frequency items (orange bars) with the strongest effects for relative clauses. This illustrates the importance of structure-matched examples in the retrieved set. For low-frequency items, both structural matches and similarity-based selection contribute. However, similarity does not compensate for missing

3. See Appendix 4.8.4 for our structural match metric.

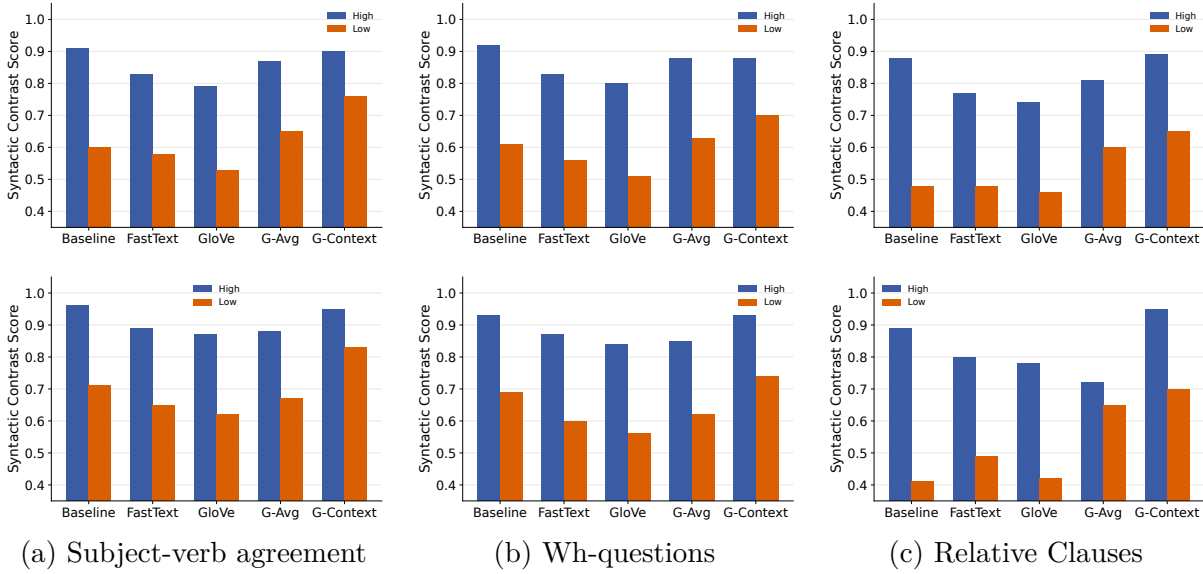


FIGURE 4.1 – Performance on syntactic contrasts using semantics-only retrieval strategies compared to full retrieval ($k = 16$, $\tau = 3$) of the child-realistic (top row) and large-scale (bottom row) models. Blue bars indicate high-frequency items and orange bars low-frequency items.

structural information. While it does add smaller, consistent gains overall, similarity-based selection without structural matches (w/o : middle orange bars with hatched gains) consistently fails to outperform random selection with structural matches ($w/+r$: right orange bars without hatched gains) for any model or phenomenon. Thus, structurally matched examples are the primary driver of frequency-selective retrieval benefits, with secondary contributions from similarity-based selection. Given the ineffectiveness of semantics-only retrieval discussed above, the embedding similarity gains likely reflect partial structural similarity between the retrieved examples and the target, in addition to semantic similarity.

4.5.2 Effects of Retrieval Configuration

We next analyze whether and how retrieval effectiveness is modulated by three configuration parameters: retrieval granularity (token, phrase, or sequence), number of neighbors (k), and context window size (τ).

Effects of retrieval granularity. Figure 4.3 shows that, at $k = 16$ and $\tau = 3$, sequence-level retrieval numerically provides the strongest or tied-strongest performance for low-frequency items across all phenomena and both model settings, but the effect of retrieval granularity overall is small.

Effects of context window size. Figure 4.4 shows that the effect of context window size is idiosyncratic. For RC, $\tau = 10$ yields the highest accuracy across both model scales and both frequency bands, whereas for SV, $\tau = 3$ yields the strongest performance for low-frequency. For Wh, the effect of τ is minimal, except $\tau = 3$ leading to degradation in high-frequency items in the child-realistic model. Thus, increasing context window size does not uniformly improve performance, and the preferred window size depends on the phenomenon, frequency band, and

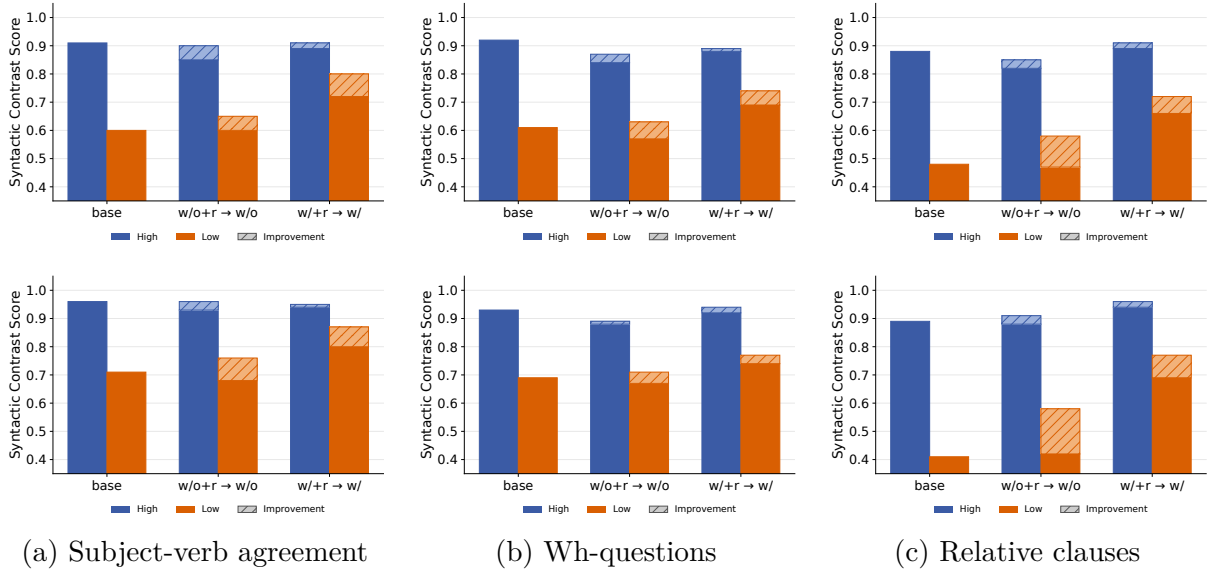


FIGURE 4.2 – Performance when retrieved sequences contain target syntactic structures (with: w/) versus when they do not (without: w/o). Top row: child-realistic model; bottom row: large-scale model ($k = 16$, $\tau = 3$, sequence-level). Blue and orange bars correspond to high- and low-frequency items, respectively. Within each panel, the x -axis shows three conditions: **base** (no retrieval), **w/o+r $\rightarrow$ w/o** (retrieved set does not contain structurally matched examples, random selection $\rightarrow$ embedding-similarity), and **w/+r $\rightarrow$ w/** (retrieved set contains structurally matched examples, random selection $\rightarrow$ embedding-similarity). Solid bars denote random selection (+r); hatched segments denote the additional gain from embedding-similarity ranking over random selection.

model.

Effects of number of retrieved instances. Figure 4.5 shows that the number of retrieved instances has a non-monotonic effect on performance in the low-frequency regime. Moving from $k = 1$ to a moderate number ($k = 16$) consistently improves performance across most conditions, indicating the benefit of aggregating evidence from multiple exemplars. However, further increasing the number of neighbors ($k = 1024$) leads to performance degradation, likely due to noise from distant and less relevant neighbors. This pattern holds across models, granularities, and syntactic phenomena, suggesting the optimal choice lies at intermediate values of k . The effect for high-frequency items is smaller and mixed.

No cross-phenomena optimal configuration. Across the three parameters (retrieval granularity, context window size, and number of retrieved instances), no single configuration consistently yields optimal performance across all syntactic phenomena, model, and frequency strata. While sequence-level retrieval, larger context windows, and a moderate number of neighbors ($k = 16$) provide the most robust overall performance (and therefore serves as our representative configuration), it does not uniformly dominate across all settings, particularly for simpler or high-frequency dependencies where more localized representations or smaller contexts can be competitive. These results suggest that effective retrieval is jointly modulated by dependency complexity, data frequency, and model capacity, and that optimal retrieval configurations are likely to be phenomenon-dependent rather than universally fixed.

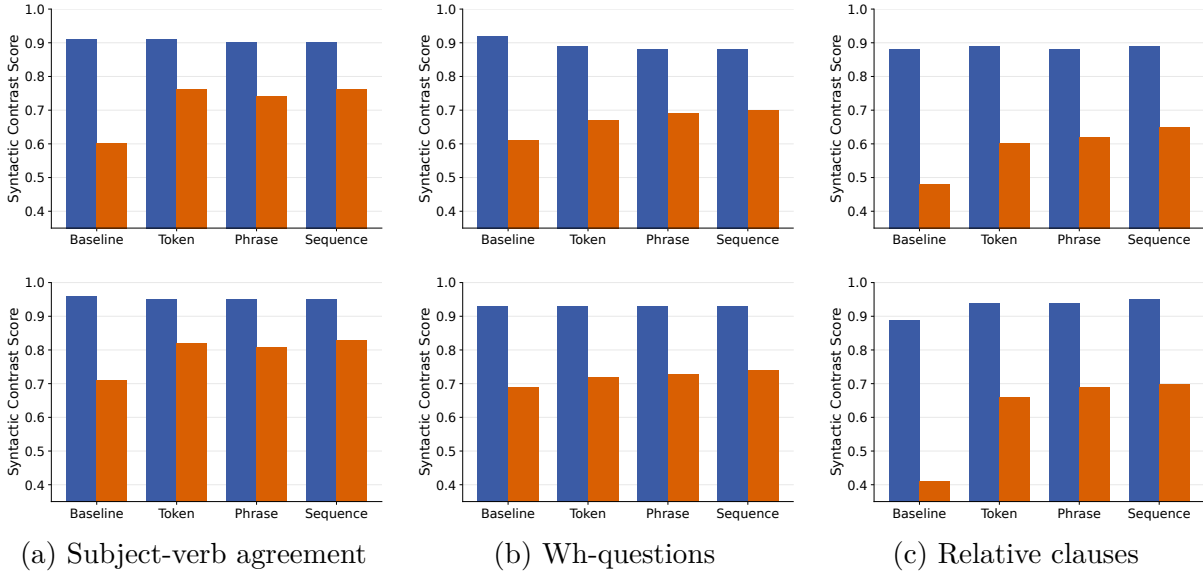


FIGURE 4.3 – Performance across retrieval granularities (token, phrase, sequence) in child-realistic (top row) and large-scale (bottom row) models ($k = 16$, $\tau = 3$). Blue bars indicate high-frequency items and orange bars low-frequency items. Sequence-level retrieval is numerically the overall strongest, but the effect of retrieval granularity is overall small.

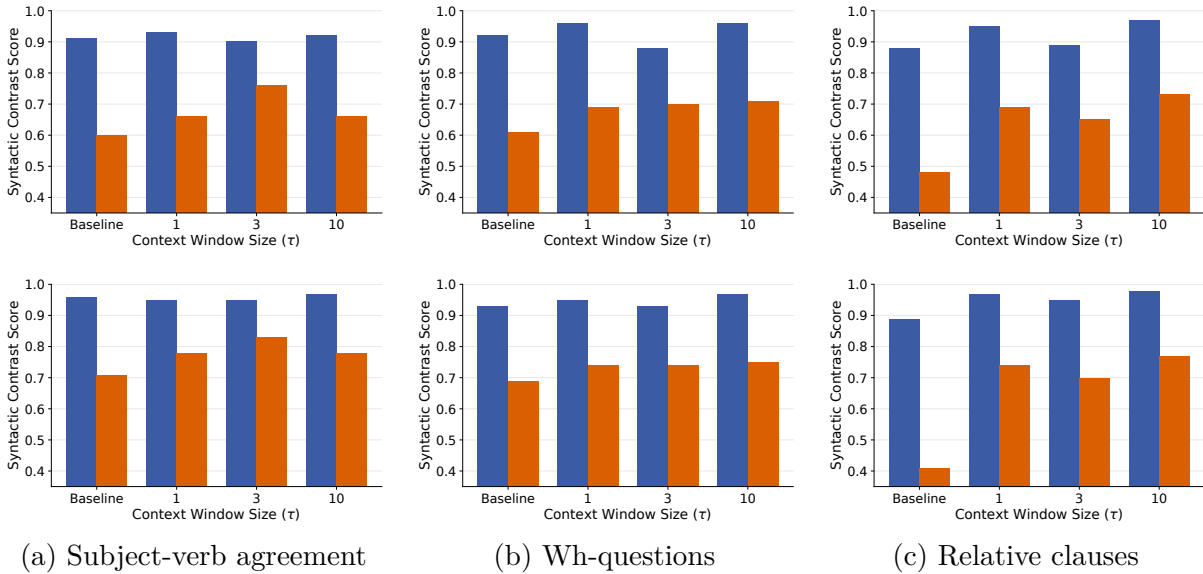


FIGURE 4.4 – Performance across context-window sizes for the child-realistic (top row) and large-scale (bottom row) settings, using sequence-level retrieval with $k = 16$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items. The effects of context size are non-monotonic and phenomenon-, frequency band- and model-dependent.

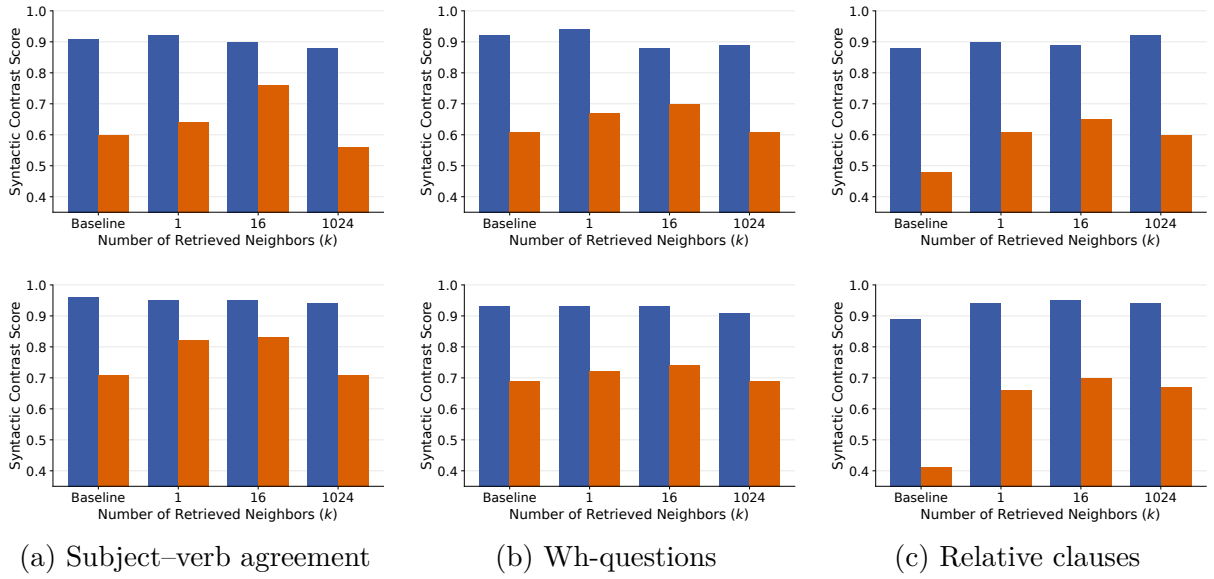


FIGURE 4.5 – Performance across numbers of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic (top row) and large-scale (bottom row) settings, using $\tau = 3$ and sequence-level retrieval. Blue bars indicate high-frequency items and orange bars indicate low-frequency items. For all six low-frequency conditions, $k = 16$ outperforms both $k = 1$ and $k = 1024$. Effects on high-frequency items are smaller and mixed.

4.6 Discussion

We tested whether an episodic retrieval mechanism as implemented via k NN-LMs can help close the lexical frequency gap that language models exhibit in syntactic contrast tests. Our findings reveal that retrieval augmentation provides substantial compensatory benefits for low-frequency items, and that both structural and semantic information contribute to the gains, although the role of structural information is more critical. However, the compensation is insufficient: the frequency gap is narrowed but not fully closed. In this section, we discuss which modeling considerations may help narrow this gap further in the future based on the limitations we observe in our analyses. Specifically, our analyses suggest that preferential reweighting of retrieved instances, improved representations and retrieval strategies for structural information, and an adaptive storage and retrieval mechanism that allow for flexible configurations of instances may be promising next steps.

Preferential reweighting. The hippocampus has been proposed to engage in preferential reweighting, where infrequent (but significant) instances are prioritized in storage as well as retrieval [Kumaran, 2016]. This kind of modulation is consistent with error-driven learning accounts [Chang, 2006; Rescorla, 1972], which predict that unexpected or surprising items receive stronger consolidation. Standard k NN-LM as implemented in this study retrieves instances by similarity alone, without modulating retrieval by the rarity or informativeness of stored instances. This leads to suboptimal retrieval of relevant examples especially for more difficult phenomena like RCs in the child-realistic regime (e.g., the low rate of % structurally similar instances retrieved for RCs: Table ??), indicating that improving strategies for retrieval and storage to identify better retrieval targets could lead to closing the frequency gap further. Recent computational work similarly finds benefits from selective data weighting [Toneva, 2019] and importance-based sampling [Xie, 2023], suggesting that incorporating reweighting strategies into the retrieval mechanism could amplify benefits in the low-frequency regime.

Better representations and retrieval strategies for structural information. Our analysis shows that structural information is critical for retrieval benefits we observe, which is in line with episodic accounts of syntactic processing that focus on the structural aspects of the retrieved content [Borensztajn, 2011]. In light of this observation, we note two potential issues that may contribute to the persistent frequency gap. First, as discussed above, retrieval based on representation similarity is not always effective in identifying structurally relevant instances, particularly in child-realistic models. Second, even when relevant instances are retrieved, the structural signals encoded in the representations may be too weak to fully bootstrap processing of low-frequency instances. Addressing the first issue may benefit from selective indexing strategies and constraint-based filtering strategies [Borgeaud, 2022; Izacard, 2022], and the second may require expanding the datastore to maintain the full residual stream representations or exploring alternative pooling strategies in the datastore.

Flexible configurations of storage and retrieval. The absence of a universal retrieval configuration across different syntactic phenomena suggests that maximally effective episodic retrieval requires adaptive mechanisms that modulate granularity, context scope, and neighbor

count. Indeed, human episodic memory has been proposed to support this kind of flexibility [Duff, 2012] and adaptive retrieval mechanisms grounded in metacognitive monitoring have been discussed in memory theory [Nelson, 1990]. Current k NN-LM uses a fixed configuration across all inputs. Developing retrieval strategies that adapt their configuration to the structural demands of the input represents a promising direction for closing the remaining frequency gap.

4.7 Conclusion

We tested whether an episodic retrieval mechanism can help close the lexical frequency gap that language models exhibit in syntactic contrast evaluations. Using k NN-LMs as a computational proxy for hippocampal episodic memory, we find consistent compensatory benefits for low-frequency items across syntactic phenomena and model/data scales, with structural information playing a critical role in driving these benefits. However, the frequency gap remains not fully closed—test items with low frequency lexical items still systematically lead to lower performance even after retrieval augmentation. The limitations of the current implementation of episodic memory identified through our analyses lead to three concrete future directions for improvement: implementation of a preferential reweighting mechanism, stronger structural representations and retrieval strategies, and adaptive retrieval configurations rather than fixed. More broadly, these results offer proof-of-concept support for the CLS hypothesis that episodic memory contributes to the robustness of syntactic processing to lexical frequency effects, while identifying areas for improvement for future modeling work.

4.8 Appendix

4.8.1 Pretrained models and datasets

Child-realistic pretraining data. We constructed a dataset to approximate the language input a child receives during early development, drawing on child-directed speech (CDS) from CHILDES [MacWhinney, 2000a], transcribed conversational speech from the Switchboard Dialog Act Corpus [stolcke2000dialogue; Godfrey, 1992], children’s stories from the Children’s Stories Text Corpus [bensaid2021cstc; bensaid2021fairytailor], and movie and television subtitles from OpenSubtitles [lison2016opensubtitles]. These four sources also appear in the 2023 BabyLM corpus release [Warstadt, 2023]. The complete dataset comprises 50.03M words, consistent with estimates of cumulative linguistic input by early childhood [Gilkerson, 2017; Frank, 2023b]. Table 4.3 shows the distribution of constituent sub-datasets.

Large-scale pretraining data. For the large-scale condition, we use GPT-2 XL (1.5B parameters) pretrained on the WebText corpus [Radford, 2019], which contains approximately 40GB of text scraped from internet sources.

Model architectures and training. Table 4.4 summarizes the architectural specifications for both models. The child-realistic model uses the GPT-2 Small architecture with 124M parameters, trained from scratch for 10 epochs, following the BabyLM Challenge setup [Hu, 2024]. The GPT-2

TABLE 4.3 – Composition of child-realistic pretraining corpus combining child-directed speech, conversational dialogue, and children’s books to approximate naturalistic language exposure during early childhood. CDS = child-directed speech; SWBD = Switchboard; CSTC = Children’s Stories Text Corpus; OSub = OpenSubtitles.

Dataset	Domain	# words	Prop. (%)
CHILDES	CDS	15M	30.0
SWBD	Dialogue	1.18M	2.4
CSTC	Books	3.22M	6.4
OSub	Dialogue	30.63M	61.2
Total		50.03M	100.0

XL model uses the publicly available pretrained checkpoint from RADFORD et al. [Radford, 2019]. The training details for the child-realistic model are provided in Table 4.5.

TABLE 4.4 – Model architecture specifications. GPT-2 Small was trained from scratch on child-realistic data (50M words); GPT-2 XL uses the publicly released pretrained checkpoint from RADFORD et al. [Radford, 2019] trained on WebText (~ 40 GB).

Parameter	GPT-2 Small	GPT-2 XL
# parameters	124M	1.5B
Transformer layers	12	48
Hidden size	768	1600
FFN size	3072	6400
Attention heads	12	25
Attention dropout	0.1	0.1
Embedding dimension	768	1600
Vocabulary size	50257	50257

4.8.2 Evaluation set construction

To ensure focused evaluation on syntactic phenomena relevant to our research questions, we selected subsets corresponding to our target phenomena from three evaluation benchmarks: BLiMP [Warstadt, 2020a], Zorro [Huebner, 2021a], and BIG-bench [Srivastava, 2023] as discussed below.

4.8.2.1 Source benchmarks

BLiMP. The Benchmark of Linguistic Minimal Pairs for English [Warstadt, 2020a] consists of 67 sub-datasets, each containing 1,000 minimal pairs (67,000 pairs total) isolating specific contrasts in syntax, morphology, or semantics. The data is automatically generated according to linguist-crafted grammar templates, with 96.4% aggregate human agreement with the labels. For subject–verb agreement, we selected the relational-noun distractor paradigm and four plural-agreement paradigms: regular and irregular plurals, each with a verb-number manipulation

TABLE 4.5 – Training hyperparameters for GPT-2 Small on the child-realistic corpus, trained for 10 epochs using a standard autoregressive language modeling objective with cross-entropy loss.

Hyperparameter	Value
Max sequence length	1024
Batch size	64
Training epochs	10
Learning rate	0.0001
LR scheduler	inverse sqrt
Warmup steps	1000
Optimizer	Adam
Adam- β_1	0.9
Adam- β_2	0.98
Dropout	0.1
Weight decay	0.01
Gradient clipping	1.0

holding the subject fixed (variant 1) and a subject-number manipulation holding the verb fixed (variant 2). For wh-questions, we selected the object-gap, subject-gap, and long-distance subject-gap wh-question paradigms, together with the four wh-versus-that paradigms crossing gap presence and dependency length. For relative clauses, we selected the relative-clause distractor agreement paradigm.

Zorro. Zorro [Huebner, 2021a] is a targeted grammar test suite designed for small-data, restricted-vocabulary evaluation. It comprises 23 paradigms spanning 13 grammatical phenomena, with 2,000 minimal pairs (4,000 sentences) per paradigm. Its lexical items were selected from whole-word entries in BabyBERTa’s custom vocabulary, making the suite suitable for evaluating models trained on comparatively limited, child-directed linguistic input. For subject-verb agreement, we selected agreement across a prepositional phrase, agreement in questions with an auxiliary, and agreement in simple questions. For wh-questions, we selected the object and subject wh-question paradigms. For relative clauses, we selected agreement across a relative clause.

BIG-bench. The Beyond the Imitation Game Benchmark [Srivastava, 2023] includes several linguistic evaluation tasks. For subject-verb agreement, we selected the simple-English, single-adverb, double-adverb, adverb-conjunction, name-plus-prepositional-phrase, noun-plus-prepositional-phrase, noun-plus-prepositional-phrase-and-adverb, Gulordava-nonce, and frequency-disbalance-nonce subtasks. For relative clauses, we selected the long-nested-inner and long-nested-outer subtasks. No BIG-bench items were used for the wh-question evaluation.

4.8.3 Statistical analyses

Bootstrapped confidence intervals on per-frequency band deltas. For each frequency band, we computed the performance delta (kNN – baseline) and estimated 95% confidence intervals via bootstrap resampling over test items (10,000 resamples). As seen in Table 4.6, in both settings, the intervals for low- and high-frequency items do not overlap, indicating a robust separation in retrieval gains.

TABLE 4.6 – Bootstrapped 95% confidence intervals on the per-frequency band accuracy delta (kNN – baseline), 10,000 resamples over test items ($k = 16$, $\tau = 3$, sequence-level).

Pretraining	Freq	Δ	95% CI
Child-realistic	high	−0.013	[−0.030, 0.004]
	mid	0.040	[0.022, 0.059]
	low	0.140	[0.115, 0.166]
Large-scale	high	0.017	[0.002, 0.033]
	mid	0.023	[0.008, 0.039]
	low	0.153	[0.125, 0.182]

Mixed-effects logistic regression. To test whether the effect of retrieval is larger for low-frequency items, we fit a mixed-effects logistic regression predicting correctness from Frequency (high/low) and Model Type (baseline/kNN), with random intercepts for item and for phenomenon. We used treatment coding, with high-frequency items and the baseline model as the reference levels:

$$\text{correct} \sim \text{Frequency} * \text{ModelType} + (1|\text{item}) + (1|\text{phenomenon})$$

Coefficients are reported in Table 4.7. Under this coding, the Model Type coefficient estimates the effect of kNN augmentation within the high-frequency band. This coefficient was not statistically significant ($\beta = -0.04$, $p = 0.62$), indicating no detectable retrieval effect for high-frequency items. Crucially, the Frequency $\times$ Model Type interaction was positive and significant ($\beta = 0.52$, $p < 0.001$), indicating that the retrieval effect (in log odds) was larger for low-frequency than for high-frequency items. Together with the positive accuracy differences and bootstrap confidence intervals for low-frequency items in Table 4.6, this result supports the hypothesis that retrieval disproportionately benefits items for which parametric representations are weak, rather than providing a uniform improvement across frequency bands.

TABLE 4.7 – Mixed-effects logistic regression coefficients ($\text{correct} \sim \text{Frequency} * \text{ModelType} + (1|\text{item}) + (1|\text{phenomenon})$; $k = 16$, $\tau = 3$, sequence-level).

Term	β	SE	z	p
Frequency (low)	−1.52	0.11	−13.8	< 0.001
ModelType (kNN)	−0.04	0.08	−0.50	0.62
Frequency $\times$ ModelType	0.52	0.10	5.20	< 0.001

4.8.4 Structural match analysis

For each evaluation item and retrieved sequence, we defined a binary structural match indicator equal to 1 if the sequence contained the same target phenomenon—subject-verb agreement (SV), a wh-question (Wh), or a relative clause (RC)—and 0 otherwise. We identified these phenomena using spaCy [Honnibal, 2020] as specified below:

Subject-Verb detection. We applied spaCy’s dependency parser to identify subject-verb relationships. Instances were flagged if they contained `nsubj` (nominal subject) or `nsubjpass` (passive nominal subject) dependencies that link a noun phrase to a finite verb.

Wh-Sentence detection. We used part-of-speech (POS) tagging to identify wh-words (POS tags: WDT, WP, WP\$, WRB) at clause boundaries. Instances containing interrogative or relative wh-elements were marked for inclusion.

Relative Clause detection. We combined POS tagging with dependency parsing to identify relative clause structures. Specifically, we detected: (1) Relative pronouns (WDT, WP) or relativizers (*that*, *which*, *who*); (2) Clausal complements with `relcl` (relative clause modifier) dependency relations; and (3) Appropriate structural embedding within a noun phrase.

Table 4.8 reports the mean proportion of the 16 retrieved neighbors that contain the target syntactic structure, and Table 4.9 reports two-sample *t*-tests comparing the effect of retrieval with vs. without structural matches, each against the baseline and against each other.

TABLE 4.8 – Mean proportion of the 16 retrieved neighbors containing the target syntactic structure ($k = 16$, $\tau = 3$, sequence-level retrieval). Structural matches remain particularly sparse for relative clauses in the child-realistic setting.

	SV		Wh		RC	
	high	low	high	low	high	low
Child-realistic	0.74	0.70	0.49	0.62	0.09	0.15
Large-scale	0.95	0.89	0.67	0.76	0.74	0.70

TABLE 4.9 – *p*-values for structural-retrieval comparisons in the child-realistic and large-scale models ($k = 16$, $\tau = 3$, sequence-level retrieval).

Comparison	Freq	Child-realistic			Large-scale		
		SV	Wh	RC	SV	Wh	RC
Baseline vs. Without	low	0.16	0.38	0.022	0.071	0.49	< 0.001
	high	0.67	0.14	0.43	0.76	0.008	0.28
Baseline vs. With	low	0.006	0.021	< 0.001	< 0.001	0.004	< 0.001
	high	0.36	0.23	0.016	0.44	0.52	< 0.001
With vs. Without	low	0.003	0.046	< 0.001	< 0.001	0.031	< 0.001
	high	0.31	0.078	0.012	0.44	0.001	0.001

The tests support the claim that retrieval benefits are greater when the retrieved examples contain structurally matched examples, and reveal additional nuance: the structural-match advantage is more pronounced in low-frequency bands and for structurally complex phenomena (RCs), consistent with broader trends elsewhere in the manuscript.

4.8.5 Additional Retrieval Configuration Results

This section provides additional results for the retrieval configuration analyses in the main text. All results use the *Full Retrieval* datastore defined in Section 4.3.3; baseline results are shown for reference. Figures 4.6–4.9 vary context-window size ($\tau \in \{1, 3, 10\}$) at $k \in \{1, 1024\}$, whereas Figures 4.10–4.13 vary the number of retrieved neighbors ($k \in \{1, 16, 1024\}$) at $\tau \in \{1, 10\}$. Together with the main analyses, these figures cover the evaluated combinations of neighbor count, context-window size, and retrieval granularity for the displayed high- and low-frequency

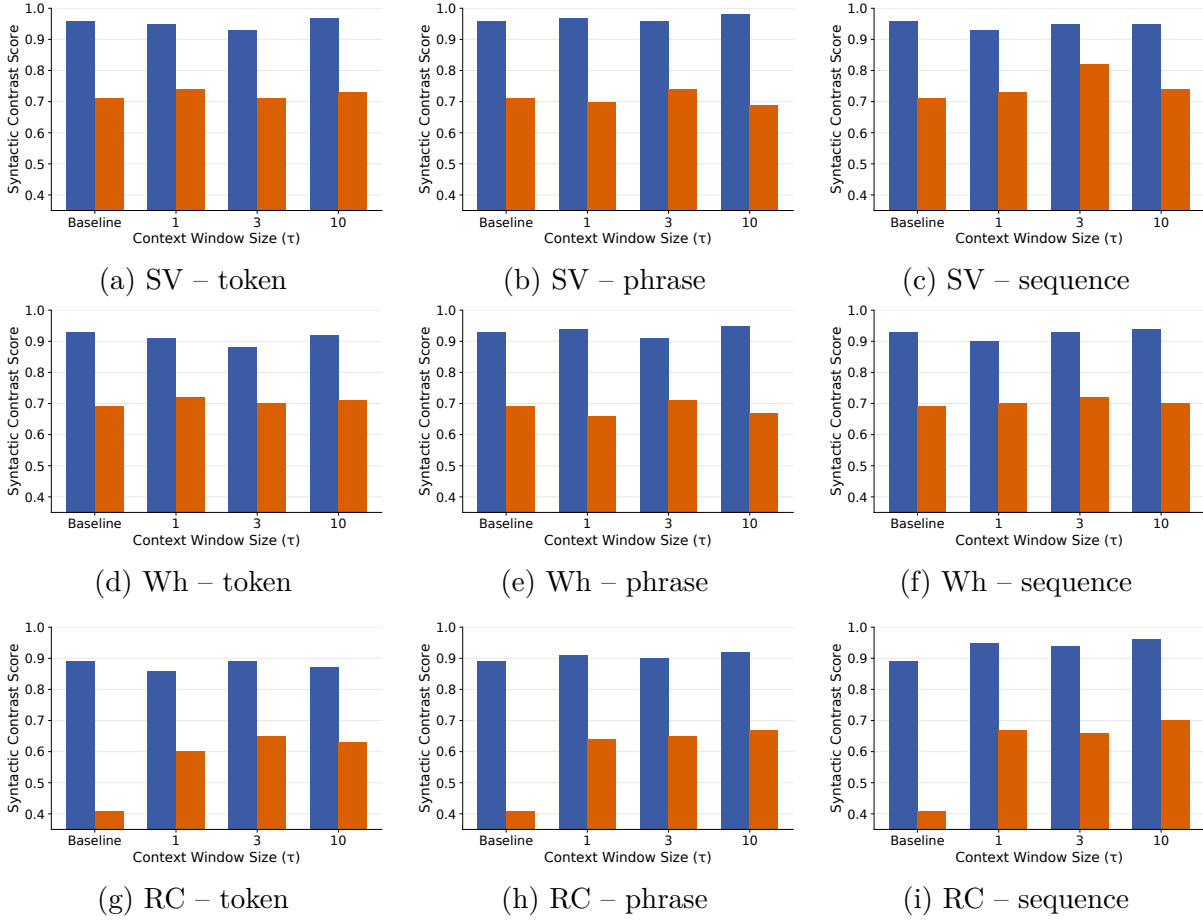


FIGURE 4.6 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for GPT-2 XL at $k = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

bands in both pretraining settings. Given many differences between individual configurations are small, we focus here on broad patterns rather than numerical rankings within individual panels. **Effects of context window size.** Figures 4.6–4.9 extend Figure 4.4 to $k \in \{1, 1024\}$ and to all retrieval granularities. As in the main analysis, accuracy is non-monotonic with context window size. The direction and magnitude of the effects vary by number of neighbors, retrieval granularity, phenomenon, and frequency band; and are often small. These additional results therefore do not identify a context window size that is uniformly preferred across conditions.

Effects of the number of retrieved neighbors. Figures 4.10–4.13 extend Figure 4.5 to $\tau \in \{1, 10\}$ and to all retrieval granularities. The results are broadly consistent with the pattern reported in the main text: for low-frequency items, the intermediate setting $k = 16$ generally yields the strongest performance among the tested values, whereas effects on high-frequency items are smaller and more mixed. The magnitude and direction of the effects vary across conditions, and the effects are often small. We therefore treat $k = 16$ as a useful representative setting rather than as a universally optimal neighbor count. Comparisons between $\tau = 1$ and $\tau = 10$ similarly provide no evidence for an interesting interaction effect with context window size.

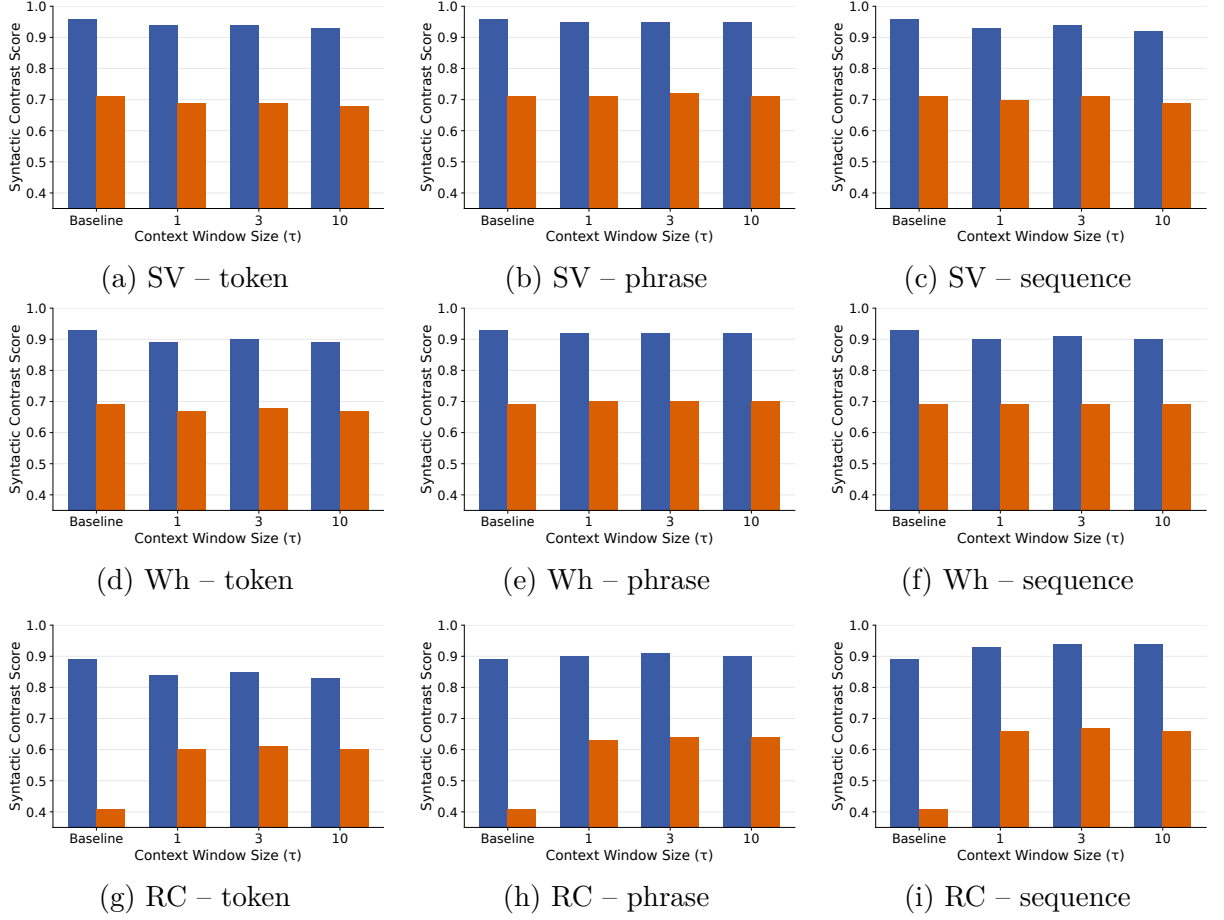


FIGURE 4.7 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for GPT-2 XL at $k = 1024$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

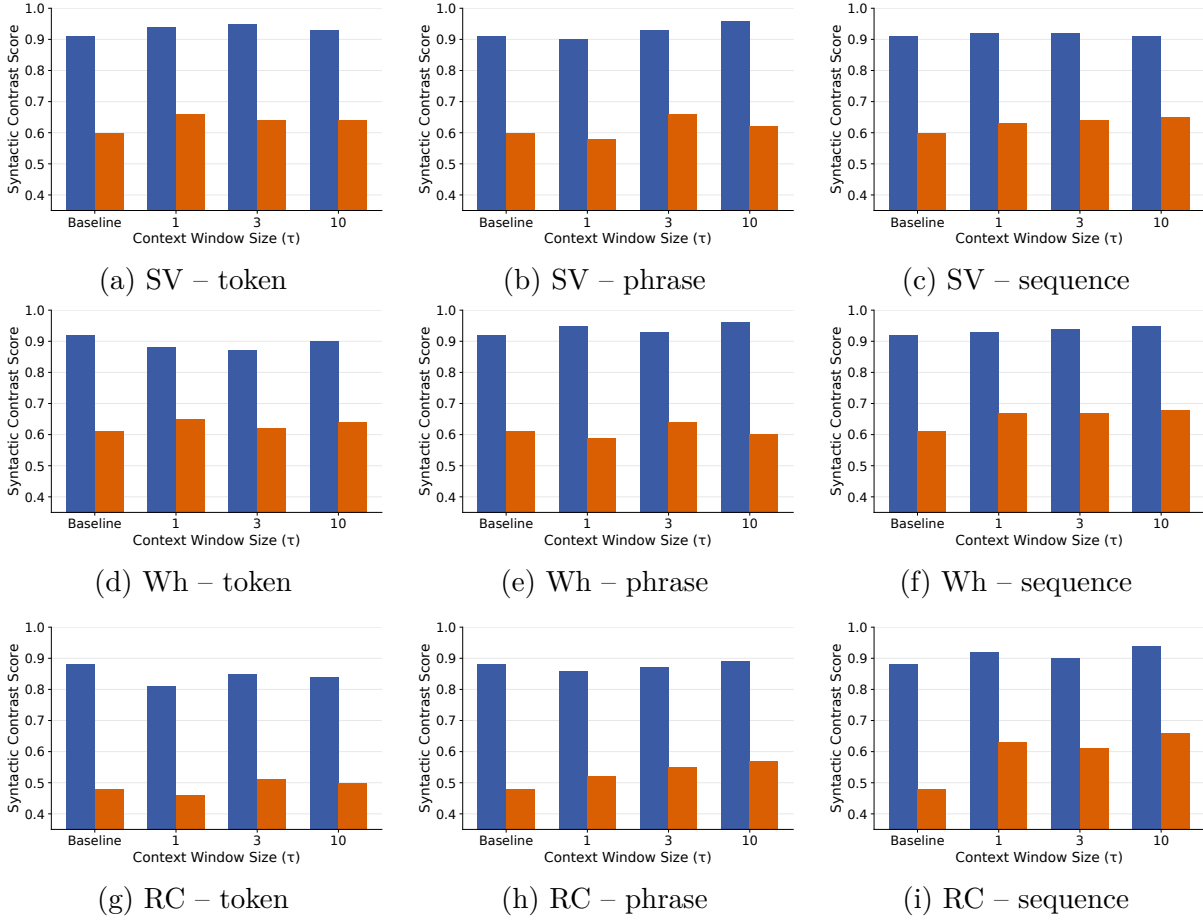


FIGURE 4.8 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for the child-realistic model at $k = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

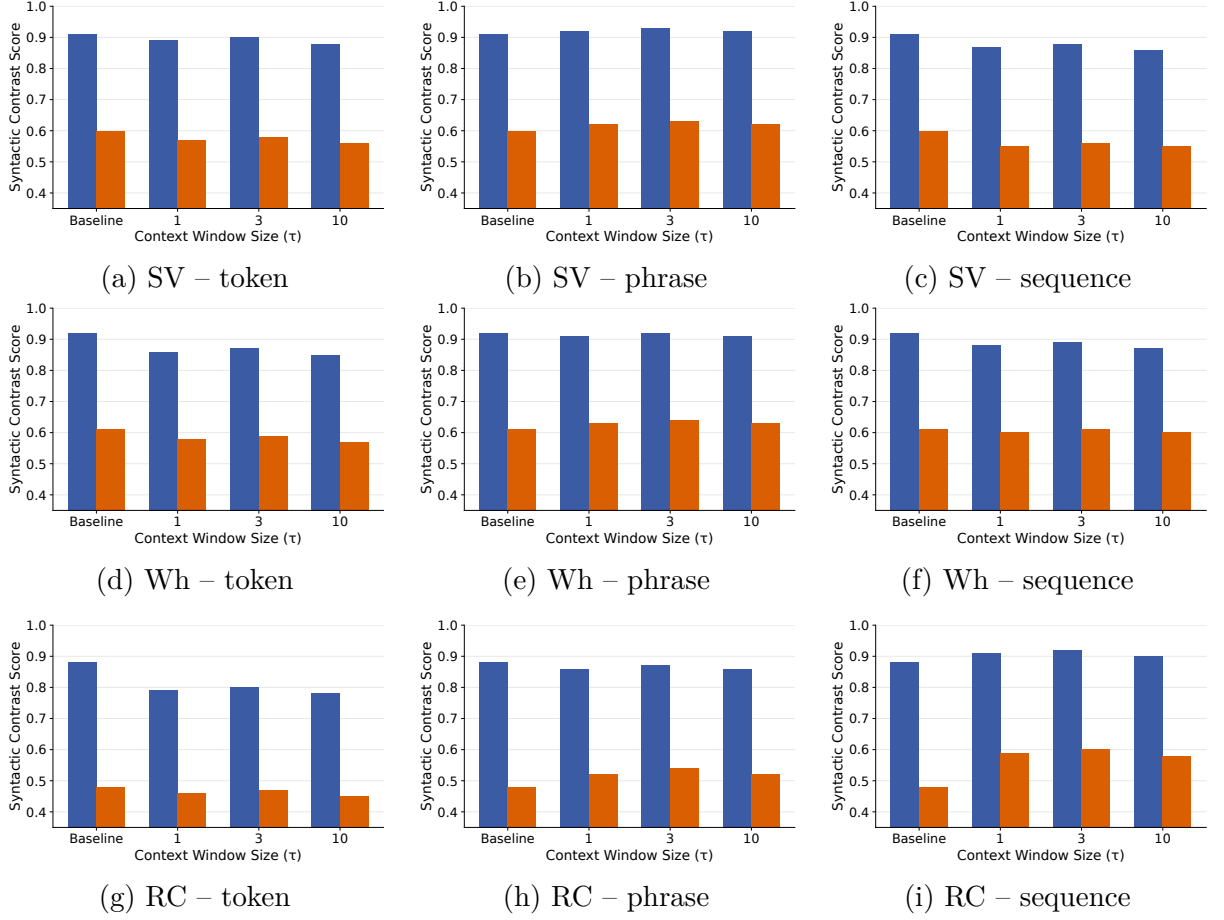


FIGURE 4.9 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for the child-realistic model at $k = 1024$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

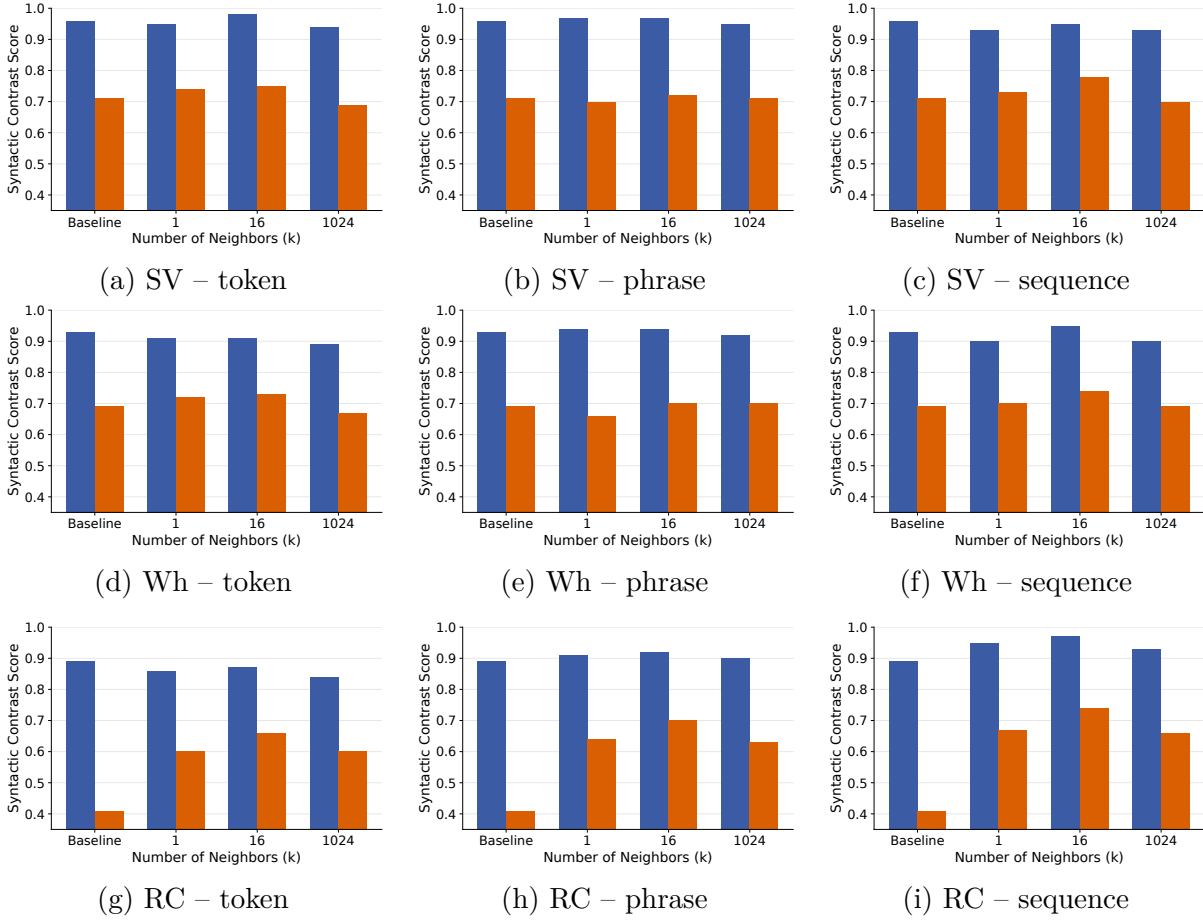


FIGURE 4.10 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for GPT-2 XL at $\tau = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

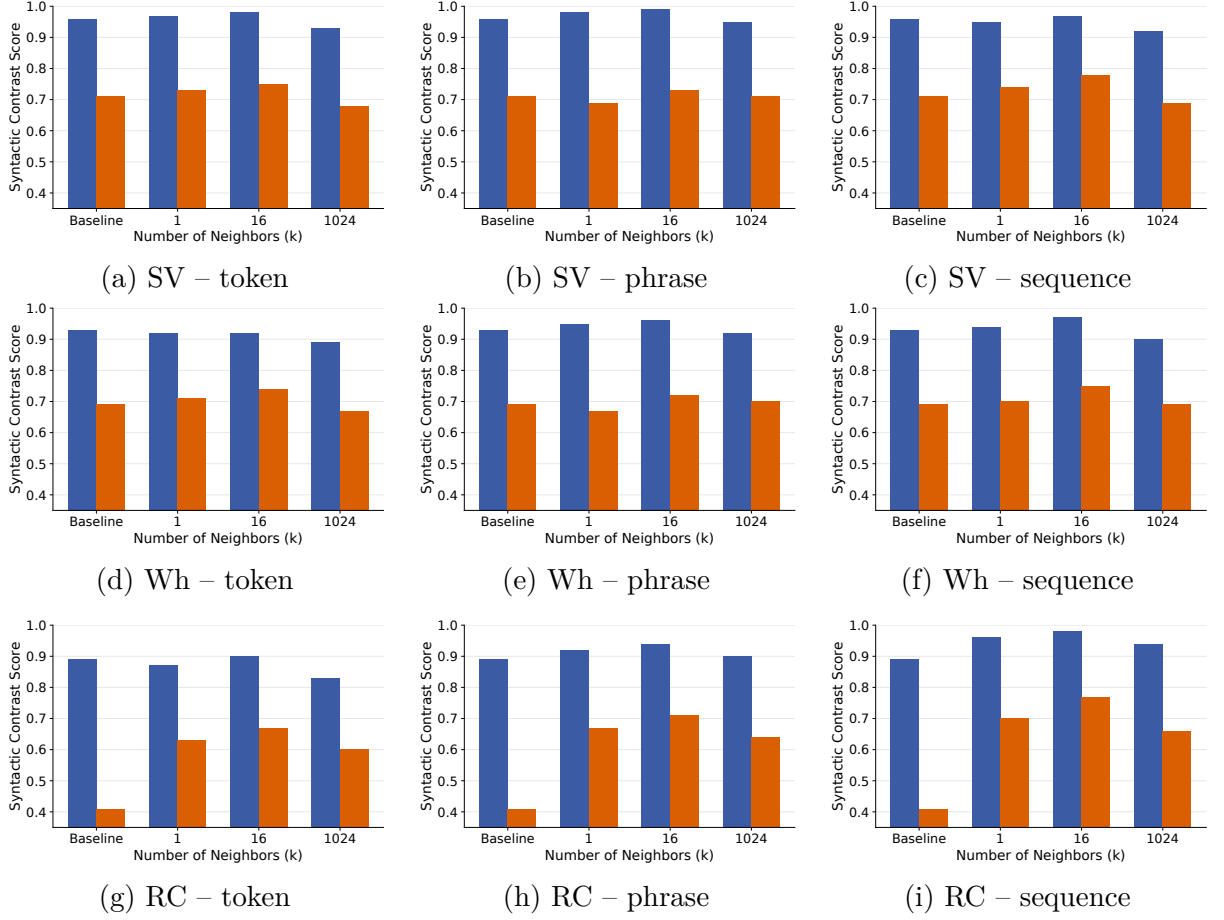


FIGURE 4.11 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for GPT-2 XL at $\tau = 10$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

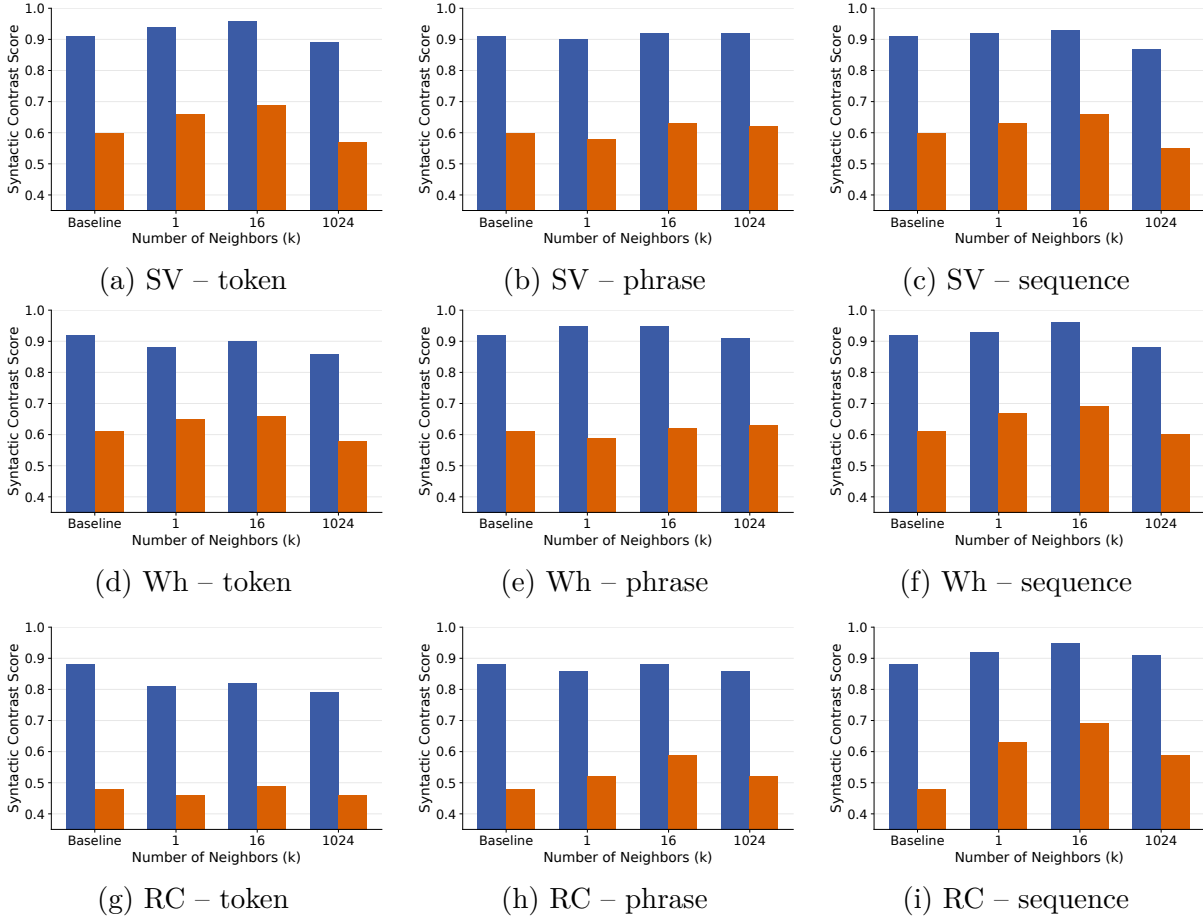


FIGURE 4.12 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic model at $\tau = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

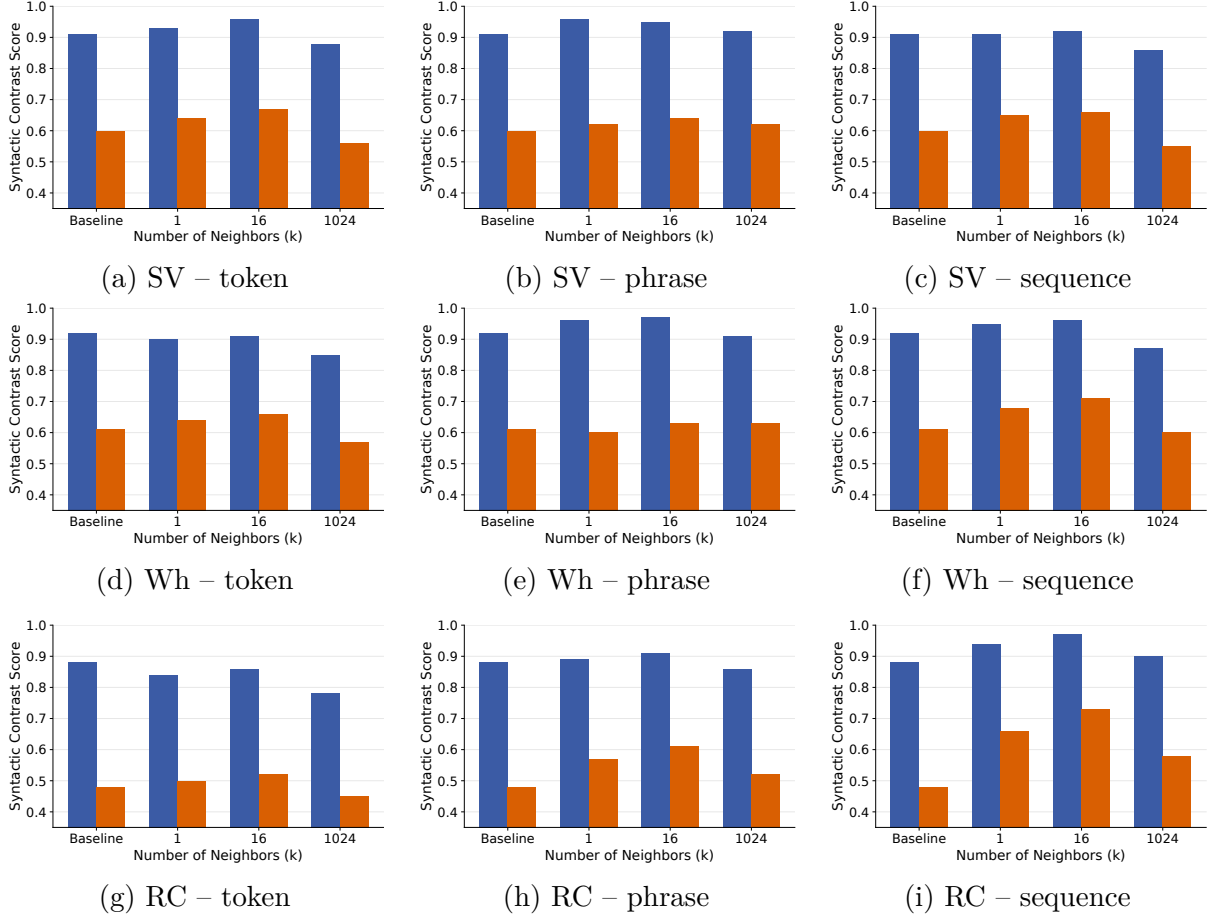


FIGURE 4.13 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic model at $\tau = 10$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

Chapitre 5

Modeling child-caregiver interactions: benchmarking and reinforcement-learning studies

Objectifs

Two complementary studies of how language models capture child-caregiver interaction: first, a benchmark of how state-of-the-art LLMs simulate child-directed dialogue; second, a reinforcement-learning study isolating which forms of caregiver feedback support grammar acquisition in child-like LMs.

5.1 Introduction

5.1.1 Child–Caregiver Interaction in Language Acquisition

Children encounter language in social interactions in which they are not only exposed to caregivers’ utterances, but also contribute utterances of their own. Caregivers respond to these contributions, helping children express their intentions and maintain the conversation. Their responses may include requests for clarification, reformulations, or the reuse of words and structures produced by the child [Clark, 2020b; Fusaroli, 2023]. Thus, caregiver language provides both linguistic input and responses that are contingent on children’s own behavior [Masek, 2021].

This interactive context raises important questions for language acquisition. Caregiver responses may provide information about whether an utterance was understood and may offer opportunities for children to adjust their subsequent productions [Nikolaus, 2023b]. However, the availability of such responses does not, by itself, establish their role in learning. Feedback may concern the child’s intended meaning, intelligibility, or participation rather than grammatical form specifically. Whether these signals provide useful information, and how a learner could exploit them, remain important questions in accounts of grammatical development [Demetras, 1986; Marcus, 1993]. Computational models offer a way to investigate aspects of this interaction and to test hypotheses about its possible contribution to learning.

5.1.2 Research Questions

The present chapter approaches child–caregiver interaction through two complementary questions. The first concerns the simulation of interaction: *To what extent can language models reproduce the linguistic and conversational properties of child–caregiver dialogue?* Addressing this question requires more than assessing whether individual utterances appear child-like or caregiver-like. It also requires examining how the two participants respond to one another and whether the resulting patterns remain comparable to human interaction over multiple turns. Here, language models are used as simulators, and their outputs are evaluated against observations of naturally occurring dialogue.

The second question concerns the learning effects of feedback: *Which forms of caregiver feedback can support grammatical learning in a language model?* In this case, the aim is to isolate particular signals from the broader interaction and examine their effects within a controlled learning framework. Language models are used as learners whose productions can be modified through feedback-based training. This approach allows us to compare signals that occur together in natural conversations, while distinguishing changes in generated language from changes in grammatical preferences on controlled tests.

These questions are related, but they address different aspects of computational modeling. A model may approximate observable properties of children’s speech without having acquired those properties through a child-like learning process. Conversely, a simplified learning model can be used to test the effects of feedback without reproducing a complete child–caregiver conversation. We therefore consider interaction simulation and feedback-based learning as distinct uses of language models, rather than successive stages of a single system.

5.1.3 Chapter Overview

The chapter brings together two studies addressing these questions. The first develops a benchmark for evaluating language models that simulate children and caregivers, using naturally occurring conversations from CHILDES as a reference [MacWhinney, 2000a]. It examines properties at the word, utterance, and dialogue levels, compares single-turn responses with extended multi-turn interactions, and assesses how providing conversational examples changes the models’ behavior. Its contribution is to characterize the extent and limits of the models’ ability to reproduce child–caregiver interaction.

The second study investigates learning from caregiver-derived feedback. It uses reward models trained on feedback signals extracted from natural child–caregiver conversations, rather than feedback generated by the simulated interlocutors of the first study. Small language models pretrained on caregiver input are then fine-tuned using different rewards. The study compares the effects of these rewards on the grammaticality of generated utterances and on grammatical preferences measured through minimal-pair benchmarks. This distinction is central to assessing what the feedback changes: more grammatical production does not necessarily establish the acquisition of grammatical knowledge that generalizes. The two studies thus examine, respectively, the empirical adequacy of interaction simulation and the learning effects of selected signals drawn from that interaction.

5.2 Benchmarking LLMs for mimicking child-caregiver language in interaction

5.2.1 Introduction

While LLMs show remarkable capabilities in generating human-like text and engaging in open-ended dialogues and role play in various contexts [Feng, 2024b ; Yang, 2024], their ability to simulate the specificities of child-caregiver interactions has not been systematically investigated. However, these interactions show distinct linguistic and interactive patterns and require dedicated research.

During their linguistic and communicative development, children show non-conventional (i.e., non-adult-like) patterns, such as word omissions, mispronunciations, semantic errors, and non-standard grammatical constructions [Bloom, 1993]. They also show non-conventional conversational behaviors, such as incoherence, non-responsiveness, and atypical turn-taking patterns [Ninio, 1996]. These behaviors are most apparent in the early years through primary school, although many persist into adolescence [Nippold, 2016].

Because of their still immature, non-conventional language use, children depend on caregivers to interpret and clarify their communicative intents, thus facilitating communication. Caregivers employ various *scaffolding* strategies, which offer appropriate support tailored to the child’s current level of cognitive and communicative development. These include the general use of simplified language (a register named child-directed language) as well as interactive strategies such as recasting, repairing, providing follow-up, and offering feedback [Berk, 1995 ; Clark, 2020a ;

Snow, 1977; Nikolaus, 2023b; Soderstrom, 2007]. This scaffolding is gradually reduced as the child becomes more proficient and ready for independent language use.

Despite growing interest in applying LLMs to specialized interactive scenarios [Feng, 2024b; Yang, 2024], their ability to simulate child-caregiver interactions remains underexplored. There is, to the best of our knowledge, no systematic examination of whether LLMs can a) properly simulate early child-like utterances with their known non-conventional properties, b) simulate caregiver-like language and its distinctive properties known as child-directed language, and, more importantly, c) beyond mimicking child or caregiver general linguistic properties in *isolation*, simulate child-caregiver language in *interactions*, meaning that we need to simulate the fact that the caregiver’s language is responsive/contingent on the child’s linguistic quirks, providing tailored feedback and scaffolding. This gap is particularly significant given the potential applications of mimicking this scenario in developmental research and its applications [Zhang, 2024; Seo, 2024; Räsänen, 2024; Feng, 2024b].

What the current study is about

As argued above, benchmarking LLMs for effectively mimicking child-caregiver interaction is a complex, multidimensional task. The current work does not claim to provide a complete solution, but rather offers a starting point—an initial exploration into ways to approach some aspect of this task and the insights we learned that can inform future research.

In particular, it is important to note that a comprehensive benchmarking would require evaluating both the semantic content of interactions (*To what extent was the semantic content of the caregiver-like response appropriate to the child-like utterance?*) and linguistic form (*To what extent was this content linguistically framed in an age-appropriate manner?*). The current work focuses primarily on the latter, using data from spontaneous child-caregiver dialogues as a reference (CHILDES dataset)[MacWhinney, 2000a].

Specifically, we quantify the extent to which two widely used LLMs (GPT-4o and Llama 3) approximate child-caregiver interactions along *structural* metrics at word, utterance, and dialogue levels. These metrics were selected based on insights from previous research [Valentini, 2023; Räsänen, 2024; French, 2024]. This previous research focused on a few isolated aspects of child-directed dialogue. Here, we aimed to provide a more synthetic view, capitalizing on insights from these studies.

Furthermore, a major novelty in this work is that we compared two benchmarking approaches: single- and multi-turn testing. In single-turn testing (the most common approach), models receive a child utterance from the CHILDES dataset and generate a caregiver-like response, or conversely, receive a caregiver’s utterance and generate a child-like response. For multi-turn testing, we observe free interactions, over multiple turns, between a "child"-LLM and a "caregiver"-LLM, prompted to communicate like a child and a caregiver, respectively. The resulting dialogues are then compared, on average, to actual conversations from CHILDES dataset.

Finally, both benchmarking approaches were implemented under zero-shot and few-shot conditions with two objectives: a) for the zero-shot condition, the goal is to characterize the baseline capabilities of LLMs without specific guidance (revealing their initial biases), and b) for

the few-shot condition, the goal is to measure LLMs’ ability to improve (relative to the zero-shot baseline) when provided with examples and, in fact, test the sensitivity of the metrics we used to quantify any such improvement.

5.2.2 Related Work and novelty

Research on LLMs’ linguistic and interactive appropriateness to children is still in its early stages. Several studies have explored specific aspects of the interaction, but a comprehensive assessment of LLMs’ capabilities in simulating child-caregiver interactions is still underexplored.

For example, VALENTINI et al. [Valentini, 2023] focused on vocabulary and showed limitations in LLMs’ ability to pick simple words for a young audience, RÄSÄNEN et al. [Räsänen, 2024] trained a GPT-2 model on caregiver input, and evaluated its capacity to generate language that is similar to caregivers (child-directed language). However, they focused on simulating caregiver data alone, without accounting for the child’s data, and therefore, missing the *interactive* dynamics that can influence the generation. FRENCH et al. [French, 2024] studied GPT-3.5 and Llama2’s ability for linguistic alignment to the interlocutor, showing sub-optimal performance in responding appropriately to child-like utterances.

While each of these studies has contributed valuable insights to specific dimensions of the challenge, our work aims to provide a more integrated approach on, at least, two levels. First, we evaluate *both* the caregiver- and the child-like generation. We explicitly study them in interaction, not merely simulating them in isolation. Second, instead of focusing on one metric or linguistic level, we propose metrics at the word, utterance, and dialogue levels, aiming to provide a more comprehensive evaluation framework for LLMs in this specialized communicative context.

5.2.3 Method

5.2.3.1 Data

We used the CHILDES public dataset [MacWhinney, 2000a] for benchmarking, focusing on 2 to 5 years of age.¹

From this dataset, we selected 40 conversations (approximately 300 turns each) evenly distributed across the target age groups—specifically, i.e., 10 conversations at 2, 3, 4, and 5 years.

To prepare the data for analysis and generation, we restructured these conversations into utterance-response pairs. When consecutive turns came from the same speaker (which is often the case with caregivers), we decomposed these into multiple utterance-response pairs and inserted <SILENCE> tokens to mark the positions where the non-speaking interlocutor did not contribute. This preprocessing step preserved the temporal structure of the interactions while creating a format suitable for our analytical framework. The resulting benchmarking dataset comprised 6,600 interaction pairs containing a total of 73,300 word tokens. This corpus was further characterized by an asymmetric distribution between participant types, with 26,300 tokens produced by children and 47,000 tokens produced by caregivers.

1. This age range was selected because children younger than 2 typically do not engage in extended dialogues, while older age groups had insufficient sample sizes in the dataset to allow for robust analysis.

5.2.3.2 Models

While numerous LLMs are currently available, testing all of them across multiple experimental conditions would be impractical. Nevertheless, to examine generalizability and ensure that our results are not dependent on the idiosyncrasies of a single model architecture, we systematically compared two state-of-the-art LLMs: an open-source model **Llama 3 (8B)** (The chat-optimized version of Meta’s instruction-tuned large language model, [Touvron, 2023]) and a proprietary model; **GPT-4o**, (version 2024-08),² the generative pre-trained transformer from OpenAI’s GPT-4 family [Achiam, 2023]. We selected these models (especially the latest version of ChatGPT) as they are some of the most powerful LLMs (at least at the time this research was carried out) and thus are most likely to be capable of adapting effectively to various roles, including those of children and caregivers.

Fine-tuned model In addition to the LLMs described above, which we adapt to child-caregiver interaction via prompting, we also evaluated a smaller—and computationally more manageable—pre-trained conventional model that we directly fine-tuned on child-caregiver data. Specifically, we fine-tuned the distilled encoder-decoder BlenderBot model [Roller, 2020] on conversation data from the CHILDES corpus.³ This approach allowed us to compare prompt-based adaptation of large, general-purpose models with traditional fine-tuning approaches targeting child-caregiver interaction patterns.

5.2.3.3 Benchmarking

Single-turn testing The single-turn testing evaluated LLM’s ability to generate contextually appropriate responses to individual utterances. For each conversation in our benchmarking dataset, we implemented two complementary procedures: a) **Child-to-caregiver direction:** We extracted each child utterance and used it to prompt the LLMs to generate a caregiver-like response. The actual caregiver response from the CHILDES corpus served as the reference against which we evaluated the LLM-generated output. b) **Caregiver-to-child direction:** Conversely, we extracted each caregiver utterance and used it to prompt the LLMs to generate a child-like response. The actual child response from the corpus served as the reference for evaluation. This single-turn setting allows a direct, controlled comparison between the caregiver-LLMs’ response and the caregiver’s or child’s actual response to the *same* utterance/prompt.

Multi-turn testing While single-turn testing enables controlled comparisons with actual responses, it cannot capture the dynamics of extended dialogue interactions. Therefore, we implemented multi-turn testing protocol to evaluate sustained conversational capabilities. To this end, we simulated complete dialogues using two distinct instances of the same LLM: one prompted to behave as a child (hereafter *child-LLM*) and another prompted to behave as a caregiver (hereafter, *caregiver-LLM*). To initiate these simulated conversations, we used the first utterance from each reference conversation in our dataset as a conversation starter. While the

2. The latest version at the moment we wrote this paper.

3. See appendix for fine-tuning details.

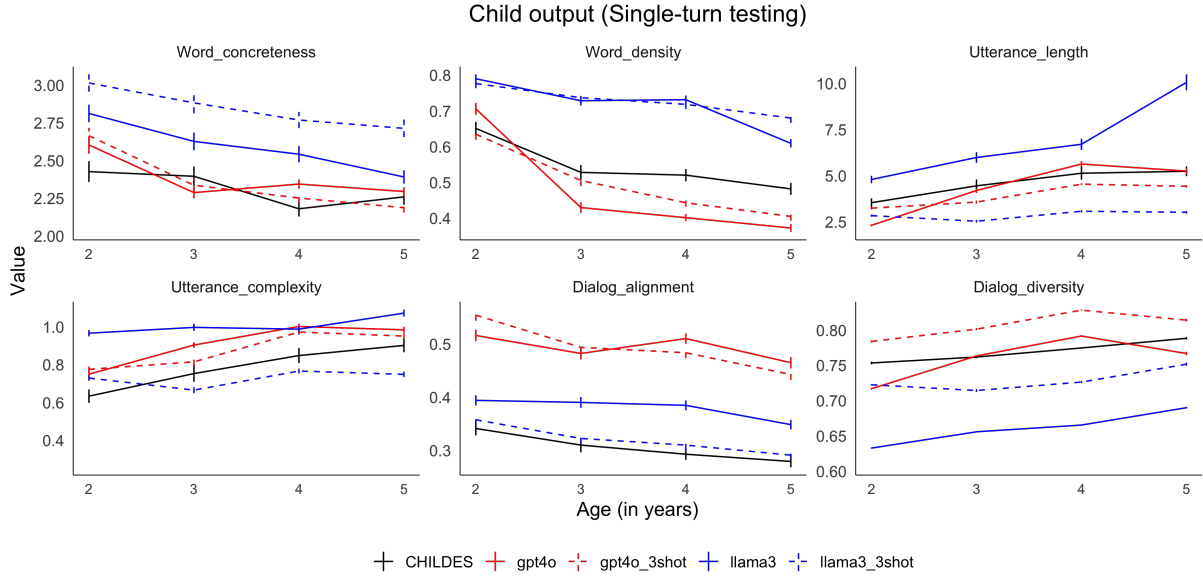


FIGURE 5.1 – Developmental trajectories of six linguistic/conversational features in children and LLMs. We compare actual human data from the child-caregiver dataset CHILDES vs. LLMs playing a child. Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Points represent averages across all utterances and conversations, and ranges represent 95% confidence intervals.

outcome is not as controlled as in the single-turn case, the metrics still allow comparisons to the human reference on average, as will be clear next.

Zero vs. Few-shot settings We tested the initial built-in capabilities and biases of the LLMs in a **zero-shot** setting where LLMs received only the utterance(s) they were expected to respond to, with no additional examples from the CHILDES dataset. In addition, to test our metrics’ ability to capture improvement, we tested the same models again in a **few-shot setting** where the models were given the first three child-caregiver pairs of turns in each conversation, allowing the model to observe examples of both the linguistic structure and interactive dynamics for the same participants.

In both conditions, we maintained consistent and minimal instructional prompts to avoid introducing confounding variables. For example, when prompting for caregiver responses, we used instructions such as: “*You are the parent of a X -year-old English-speaking child. Now, you are having a conversation with your child. Based on the conversation history above, give your response to the child input.*” (see the prompt templates in Appendix 5.2.7).

5.2.3.4 Metrics

We aim to provide a comprehensive benchmarking evaluating key properties at the word-, utterance-, and dialogue levels.

At the *word* level, we followed DAWSON et al. [Dawson, 2021] to quantify the *Word concreteness* using human ratings from BRYBAERT et al. [Brybaert, 2014]. This measure was computed

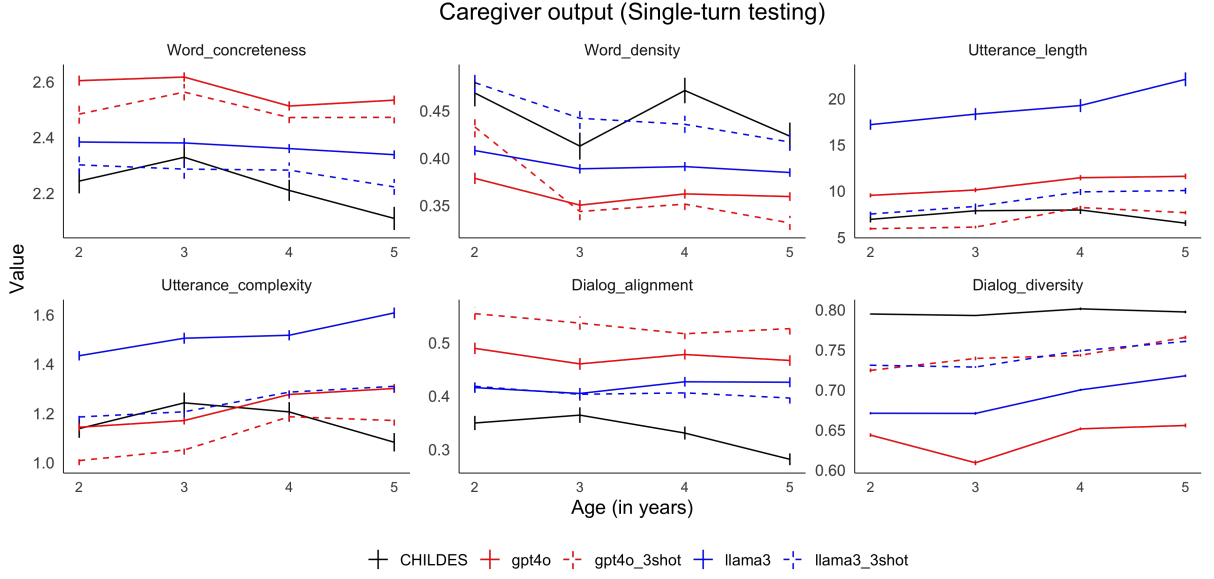


FIGURE 5.2 – Developmental trajectories of six linguistic/conversational features in caregivers and LLMs. We compare actual human data from the child-caregiver dataset CHILDES vs. LLMs playing a caregiver. Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Points represent averages across all utterances and conversations, and ranges represent 95% confidence intervals.

as the average concreteness rating of all content words in each utterance. We also used *Word density*—defined as the proportion of content (vs. function) words in the utterance in each utterance. This metric reflects the information load and was calculated using the established list of function words by O’SHEA et al. [OShea, 2012].

At the **utterance** level, we adopted measures from RÄSÄNEN et al. [Räsänen, 2024] to capture structural complexity: a) *Utterance length*: the number of words per utterance; b) *Syntactic complexity*: the mean dependency tree depth for each utterance using the Spacy toolkit (v3.7) with its dependency parser based on RoBERTa transformers.⁴ [Liu, 2008], where deeper trees indicate more complex structures.

Finally, at the level of the **dialogue** dynamics, we measured *Semantic alignment*, the extent to which the speaker’s utterance is semantically similar to their interlocutor’s (across each exchange pair) [Duran, 2019; French, 2024; Misiek, 2020b]. To this end, we used BERT sentence embedding [Reimers, 2019]. Following GUO et al. [Guo, 2023], this was calculated as the average pairwise cosine distance (1 - cosine similarity) between BERT utterance embeddings of a speaker’s contributions across the entire conversation.

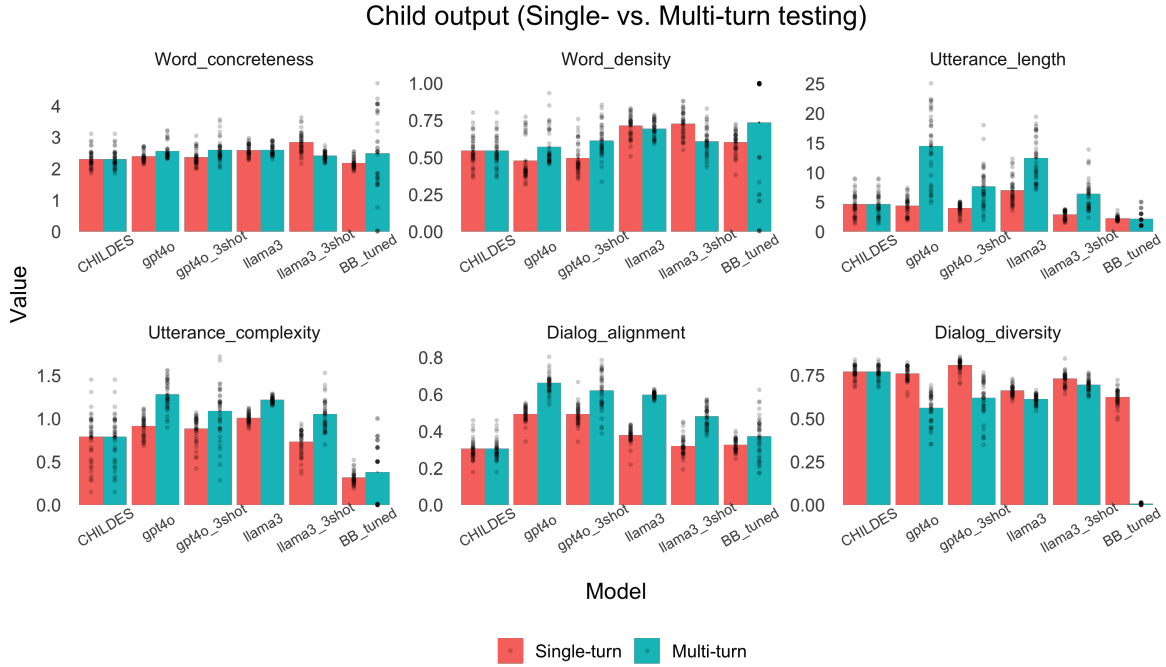


FIGURE 5.3 – Comparison of linguistic/conversational features across the single-turn and multi-turn testing for children. The bars show the averages across age groups. The points show variability across conversations (40 in total). Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. “BB_tuned” which stands for the BlenderBot model fine-tuned on a subset of CHILDES. Finally, for CHILDES, the data for single-turn and multi-turn is the same.

5.2.4 Results and Discussion

5.2.4.1 Single-turn testing

Figure 5.1 and Figure 5.2 show the results of single-turn testing for child-LLMs and caregiver-LLMs across age groups.

Child-LLM Our analysis reveals that child-LLMs follow, overall, the developmental patterns observed in CHILDES, even in the zero-shot setting, showing that these LLMs can simulate developmental changes in children’s language without explicit guidance.

Specifically, the models capture decreasing concreteness and lexical density over time (children use more abstract words and function words as they grow older), increasing utterance length and syntactic complexity, and decreasing dialog alignment alongside increasing dialog diversity (as children become able to contribute new information instead of just repeating caregivers’ inputs).

When comparing model performance, GPT-4o more accurately mimicked children than Llama 3, achieving closer alignment with CHILDES reference values across most metrics in the zero-shot condition. The introduction of few-shot examples (three interaction pairs) primarily benefited Llama 3, improving its alignment with CHILDES across several measures, though not consistently

4. The dependency parsing implementation provides a structural representation of syntactic relationships between words in an utterance.

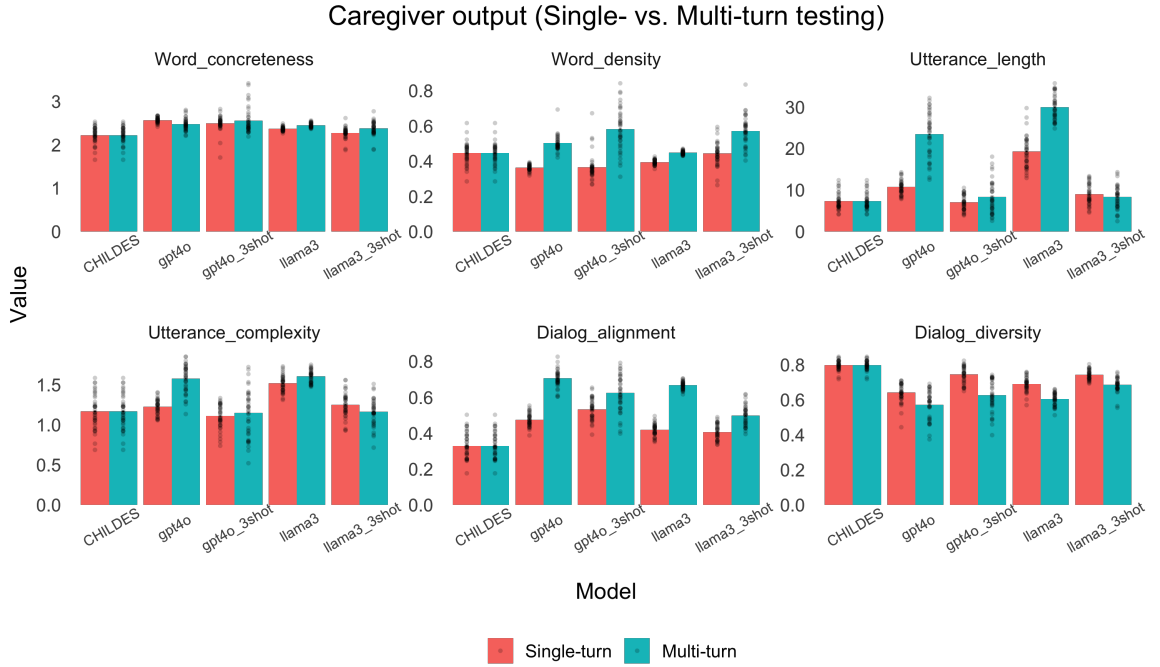


FIGURE 5.4 – Comparison of linguistic/conversational features across the single-turn and multi-turn testing for caregivers. The bars show the averages across age groups. The points show variability across conversations (40 in total). Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Finally, for CHILDES, the data for single-turn and multi-turn is the same.

across all metrics. Despite these improvements, GPT-4o maintained better overall performance following few-shot prompting.

Caregiver-LLM In simulating caregiver language, zero-shot GPT-4o more closely approximated CHILDES values for utterance-level measures, while Llama 3 achieved better alignment with word-level and dialogue-level metrics. Few-shot prompting significantly improved both models’ performance, with Llama 3 ultimately achieving better overall alignment with CHILDES reference values than GPT-4o after exposure to exemplars. Notably, both models continued to diverge from human data at the dialogue level, exhibiting higher semantic alignment and lower semantic diversity than actual caregivers in the CHILDES corpus. Overall, there were no clear developmental patterns in CHILDES caregivers’ behaviors.

5.2.4.2 Multi-turn testing

Figure 5.3 and 5.4 show the results of multi-turn testing. For easier comparison with single-turn, we show the results side-by-side, averaged over age. Here, we made a caregiver-LLM interact with a child-LLM (using two instances of the same LLM).⁵

⁵ We also crossed models in child-LLM vs. caregiver-LLM (e.g., child-llama 3 interacting with caregiver-GPT-4o), but observed no noticeable changes.

Child-LLM In zero-shot settings, we observed marked differences between single-turn and multi-turn testing across multiple metrics. Most notably, utterance length, syntactic complexity, and semantic alignment all increased considerably, becoming less comparable to children in CHILDES.

After few-shot prompting, we observed minor to moderate improvements, such as reductions in utterance length⁶ and (slight) decreases in both syntactic complexity and semantic alignment.⁷

Caregiver-LLM Multi-turn testing resulted in a general increase across most metrics compared to single-turn testing (with exceptions in dialog diversity, which decreased, and concreteness, which remained constant). These changes made the multi-turn behavior generally less comparable to human data in CHILDES. After the few-shot learning, we observed a significant improvement in multi-turn behavior, especially in terms of length and complexity, which became much more comparable to CHILDES. However, we observed only moderate improvement in dialog-level measures.

While LLMs successfully approximated human references for word-level and sentence-level properties under certain conditions, we identified systematic discrepancies in interactive measures for caregiver-LLMs across all experimental configurations. As illustrated in Figure 5.4, in both dialog alignment and diversity, LLMs exhibited higher alignment and lower diversity than caregiver data in CHILDES across LLM type (GPT-4o LLama 3), prompting strategy (zero- vs few-shot) and benchmarking approach (Single- vs. multi-turn).

To verify this qualitative observation statistically, we ran, for each of the two interactive measures (Dialog_alignment and Dialog_diversity), linear regressions comparing models' output to the CHILDES reference, testing all configurations: 2 LLMs x 2 prompting strategy x 2 benchmarking approach. All 16 comparison models (8 for each measure) revealed highly statistically significant differences between LLM-generated and human caregiver language, suggesting that interactive caregiver properties are potentially more challenging to mimic for LLM.

5.2.5 Conclusion

This paper presents a preliminary exploration into ways we can benchmark LLMs' ability to simulate child-caregiver interactive dynamics. While previous research has typically focused on measuring the properties of a specific structure/level (e.g., words or utterances) and/or on evaluating language from one part of the dialogue outside the interactive context (mainly focusing on the caregiver), our novel contribution is in three key aspects: a) we evaluate child-caregiver language generation in *interaction*, b) we tested a more comprehensive set of measures covering

6. Note that length could not be reduced further, even when we experimented with explicitly setting an upper bound on children's utterance in our instructions to the LLM.

7. In addition to using LLMs instructed to play a child, we also used model fine-tuning. To this end, we considered a pre-trained encoder-decoder model- Blenderbot fine-tuned on a subset of caregiver(encoder)-child(decoder) dialogues in CHILDES. We also explored an alternative approach using model fine-tuning rather than prompting. Specifically, we fine-tuned a pre-trained encoder-decoder Blenderbot model on a subset of caregiver (encoder) to child (decoder) dialogues from CHILDES. However, while showing reasonable performance in single-turn testing, this fine-tuned model was erratic and highly repetitive in the multi-turn interactions, also making it unsuitable for evaluating caregiver-LLMs.

the word, utterance, and dialog levels, and c) we compared two benchmarking approaches, evaluating the models in short and extended settings.

We put this benchmarking framework to use, comparing two powerful LLMs: GPT-4 and Llama3. Indeed, such a comparison was essential to distinguish between findings that are likely generalizable and those that are specific to a particular model. Our evaluation incorporated both single- and multi-turn testing. The former is more controlled, allowing a direct evaluation of the LLMs’ response using human references, while the latter, though less controlled, allowed measuring LLMs’ behavior in more extended conversations. Interestingly, we found that single-turn evaluations, while insightful, were not totally correlated with the LLMs’ behavior in an extended conversation. Indeed, the multi-turn analyses showed an increased divergence of LLMs from human data, particularly in utterance-level properties and discourse dynamics (although both benchmarkings led to qualitatively similar conclusions regarding this linguistic level). This quantitative mismatch highlights the importance of dynamic testing of LLMs, since static testing alone may fail to capture the cumulative effects of sustained interactions—a consideration particularly relevant for applications involving extended communicative exchanges.

Our benchmarking effort aims not only at comparing different LLMs, but also at providing a quantitative tool that can be sensitive to incremental improvement *within* the same models. In particular, comparing zero- vs. few-shot learning, the benchmark identified areas where this intervention was more or less successful: Both GPT-4o and Llama 3 showed marked improvements in matching caregiver patterns after exposure to just three interactive examples—a finding with important implications for efficient model adaptation in resource-constrained contexts.

However, the impact was consistently less pronounced for dialogue-level properties, where models (whether in single- or multi-turn settings) continued to exhibit higher alignment and lower diversity compared to human data. This pattern suggests that while surface-level linguistic features (such as word and sentence properties) appear relatively straightforward to adapt to, capturing the interactive nature of child-caregiver communication is more challenging (see also Limitations).

In conclusion, we explored a multi-level benchmarking approach for assessing LLMs’ ability to mimic child-caregiver language in interactions, and we showed its usefulness in tracking incremental improvements. We found that single-turn testing of the LLMs, as typical in most benchmarks, was not totally indicative of the LLMs’ real behavior in extended conversational contexts, thus emphasizing the need for more dynamic, multi-turn testing in this line of work. Furthermore, few-shot prompting was effective in bringing the LLMs closer to caregivers’ data, especially regarding word- and utterance-level properties. It was not as effective on the dialog-level properties. The LLMs exaggerated alignment and showed reduced diversity compared to CHILDES.

5.2.6 Limitations

While our explorative work has led to some initial insights, it also has several limitations and raises questions for future research.

A primary constraint concerns our selection of evaluation metrics. We relied on established

measures from previous research to assess LLMs at word, sentence, and dialogue levels. For children’s language production, these metrics successfully captured developmental trajectories, enabling age-specific benchmarking of LLMs (Figure 1). However, when applied to caregiver language, these same metrics failed to reveal clear developmental patterns (Figure 2), thereby limiting the precision of our assessment. This asymmetry highlights the need for developing more refined metrics that can better capture the subtle adaptations in caregiver speech across different stages of child development.

While evaluating LLMs in the single-turn scenario—comparing answers to a reference—is rather straightforward and aligns with the way LLMs are typically benchmarked, a multi-turn approach is also necessary in our context since the ultimate goal behind benchmarking is for these models to be used in an extended interactive context. However, the multi-turn approach is inherently less controlled as it involves two instances of LLMs interacting, each playing the role of an interlocutor. However, since neither instance perfectly models the interlocutor, extended interaction can amplify artifacts, leading the systems to adapt to each other’s quirks rather than approximating genuine child–caregiver interaction. Thus, a multi-turn benchmarking approach can under-estimate the models’ true capabilities to interact with an actual human (child or caregiver). That said, the fact that multi-turn testing was, at least qualitatively, in agreement with single-turn testing (e.g., both of them point to LLMs’ ability to adapt more easily to properties of words and sentences and less easily to properties of the dialog) is a testimony to the multi-turn’s potential as a valid method of evaluation in this context, though, of course, future research is needed to thoroughly investigate this question.

Finally, we reported that few-shot learning improved caregiver-LLM performance in terms of utterance length and syntactic complexity, but had less impact on alignment and diversity. While this finding was, overall, robust across configurations in our setup, it should be taken with a grain of salt, given that we did not systematically test it in a comprehensive set of experiments. Indeed, our main goal in this paper, as we stated above, was not to improve the LLMs’ performance, but to demonstrate, in a simple case, the ability of the benchmarking metrics to identify gaps and track improvement. For example, it is possible that a larger, more systematic exploration of the prompting strategies (which is computationally expensive, and thus, could not be done here in combination with the other experiments we did) could yield improvements across the dialog dimensions as well.

5.2.7 Ethics statement

All data used in this study is already publicly available. This work focuses on model benchmarking and improvement using offline child-caregiver data and internal simulations, aiming to advance fundamental research in this area. We do not consider this testing sufficient for deployment; any future real-life applications should undergo rigorous validation in child-safe environments with appropriate human oversight, such as by teachers or parents.

Appendix A: Prompt templates

We provide prompt templates in different conditions. These are templates for the multi-turn testing. The templates for the single-turn testing are the same, except there is no conversation history.

The zero-shot prompt template for the caregiver [Conversation history ADULT: <Utterance>, CHI: <Utterance>...] You are the parent of a <Month>-month-old English-speaking child. Now, you are having a conversation with your child. <SILENCE> indicates silence in the previous turn; <UNINTELLIGIBLE> indicates unintelligible speech. Based on the given conversation history above, give your response to parent input as ADULT. Do not output the speaker label.

The zero-shot prompt template for the child [Conversation history CHI: <Utterance>, ADULT: <Utterance>...] You are a <Month>-month-old English-speaking child. Now, you are having a conversation with your parent. <SILENCE> indicates silence in the previous turn; <UNINTELLIGIBLE> indicates unintelligible speech. Based on the given conversation history above, give your response to parent input as CHI. Do not output the speaker label.

The few-shot prompt template for the caregiver [Conversation history ADULT: <Utterance>, CHI: <Utterance>...] You are the parent of a <Month>-month-old English-speaking child. Now, you are having a conversation with your child. <SILENCE> indicates silence in the previous turn; <UNINTELLIGIBLE> indicates unintelligible speech. Ensure your response is no longer than 50 words regardless of the prompt. Here are some example interactions: CHI: <Utterance>, ADULT: <Utterance>... Follow the example interactions. Based on the given conversation history above, give your response to parent input as ADULT. Do not output the speaker label.

The few-shot prompt template for the child [Conversation history CHI: <Utterance>, ADULT: <Utterance>...] You are a <Month>-month-old English-speaking child. Now, you are having a conversation with your parent. <SILENCE> indicates silence in the previous turn; <UNINTELLIGIBLE> indicates unintelligible speech. Ensure your response is no longer than 6 words regardless of the prompt. Here are some example interactions: CHI: <Utterance>, ADULT: <Utterance>... Follow the example interactions. Based on the given conversation history above, give your response to parent input as CHI. Do not output the speaker label.

Appendix B: Finetuning details

We fine-tuned the Blenderbot model, a 400M parameter architecture comprising: a retriever for dialogue history; a Seq2Seq generator with 2 encoder layers, 24 decoder layers, 2560-dimensional

embeddings, and 32 attention heads; retrieve-and-refine architectures combining dialogue and knowledge retrieval. The parameters were optimized through preliminary experiments, resulting in a batch size of 16 and a learning rate of 0.0001 maintained over 4 epochs. The fine-tuning data consisted of 2.6M interaction pairs with 15.8M total word tokens (4.4M child tokens and 11.4M adult tokens) restructured from CHILDES dataset [MacWhinney, 2000a].

Appendix C: Example generations

We present sample generations in Table 5.1. It shows child-caregiver conversations (CHILDES corpus) and GPT-4o generations in a multi-turn setting where the child and caregiver are two instances of GPT-4o. All conversations start from the same child turn (here, all conversations start from a child outputting “silence”). To extract features, we preprocessed all samples by removing non-verbal behavioral annotations (e.g., *giggles*) and lower-casing words to focus exclusively on verbal content.

TABLE 5.1 – Comparison of real and LLM-generated child-caregiver dialogue exchanges. The LLM generation here is from a multi-turn setting, where two instances of GPT-4o (i.e., a child-LLM and a caregiver-LLM) interact. The table shows three dialogue samples: (1) a real interaction from the CHILDES corpus, (2) a zero-shot generation by GPT-4o given only the initial turn, and (3) a few-shot generation by GPT-4o after being shown three previous examples in the child-caregiver interaction.

Source	Role	Utterance
CHILDES (Human)	C	<i>SILENCE</i>
	A	wanna play catch
	C	let’s play catch
	A	okay almost
	C	yeah
	A	try it again put your hands in like this
	C	almost
	A	you wanna hit it
GPT-4o (Zero-shot)	C	<i>SILENCE</i>
	A	I see you’re still thinking about cherries. It’s okay to feel upset. Would you like a hug while we figure out something else yummy to eat?
	C	<i>SILENCE</i>
	A	I know you’re feeling quiet right now. I’m here when you’re ready. Would you like to hold the banana and see if it feels fun to peel?
	C	Banana! Peel it?
	A	Yes, let’s peel the banana! Can you help me pull the peel down? Then we can eat it together. Yum!
	C	Banana! Peel! Eat!
	A	Yes! Let’s peel the banana together. Peel, peel! Now take a bite—yum! Is it tasty?
GPT-4o (Few-shot)	C	<i>SILENCE</i>
	A	Do you love your big red truck?
	C	Big red truck! My truck!
	A	Is it your big red truck? Where will you drive it?
	C	Drive to park! Big red truck!
	A	Are we driving your big red truck to the park? That sounds so fun!
	C	My truck drive park!
	A	Yes, your big red truck! We can drive it to the park!

Notes: A = Adult, C = Child. All model-generated responses were obtained using default sampling parameters (temperature = 1.0). *SILENCE* marks an empty turn. Human dialogues are from the CHILDES corpus [MacWhinney, 2000a].

5.3 Which forms of caregiver feedback support grammar learning?

5.3.1 Introduction

Children do not learn language from exposure to input alone, but also through social interaction in which caregivers respond to their produced utterances [Fusaroli, 2023; Masek, 2021; Clark, 2020b]. A long-standing question in language acquisition concerns the role of such feedback in shaping grammatical development, especially if caregiver responses can help the learner know when their productions were well- or ill-formed [see review in Sokolov, 1994].

Although caregivers rarely provide explicit correction of grammatical errors [Brown, 1970], researchers have proposed that they nevertheless provide global, noisy feedback from which children can infer whether their utterances were likely grammatical or ungrammatical [e.g., Hirsh-Pasek, 1984; Demetras, 1986; Sokolov, 1994]. In this view, the child can be understood as learning in a reinforcement-like fashion, using the valence of the feedback as positive or negative rewards that guide which productions to maintain or adjust in future interactions [schoneberger2010three; e.g., see Whitehurst, 1988; Nikolaus, 2023a].

The developmental literature has documented several forms of caregiver response that can be interpreted, within a reinforcement-learning framework, as valenced feedback. The most direct examples are *communicative feedback* cues, namely a) acknowledgments (e.g., "okay", "yeah"), which signal communicative success and thus increase the likelihood that the child's sentence was well-formed, and b) clarification requests (e.g., "huh?", "what?"), which signal communicative failure or misunderstanding and thus increase the likelihood that the sentence was ill-formed [Clark, 2020b; Demetras, 1986; Newport, 1977; Saxton, 2005c; Nikolaus, 2023c; Nikolaus, 2022].

A second relevant class is *semantic contingency*, or the extent to which caregiver responses remain responsive to the child's topic and intended meaning [agrawal2026scaffolding; Hoff-Ginsberg, 1987; Che, 2018; Hirsh-Pasek, 2015; Agrawal, 2024]. The degree of this contingency captures the feedback valence: low contingency in the caregiver response signals semantic misalignment and thus suggests that the child's utterance may be ill-formed.

There are other forms of caregiver feedback that also provide such valenced signals, even if they have not traditionally been associated with this learning framework. Perhaps most directly relevant to grammar learning is *structural alignment* [Fusaroli, 2023; Dale, 2006; Fernandez, 2014; Misiek, 2020a], whereby caregivers respond by reusing the child's syntactic frame while potentially introducing different lexical content (e.g., child: "I eat" (Pronoun-Verb) → caregiver: "Mommy eats too"). Here, the feedback valence is reflected in whether such alignment occurs, signaling support for the child's use of that syntactic frame.

A final type of feedback is *affective feedback*, such as whether caregivers respond with positive affect (e.g., "that's amazing!") versus a more emotionally neutral response [Rivero, 2023].

These cues are far more available to children than explicit grammatical corrections, making their role in language learning much more plausible [Demetras, 1986; Hirsh-Pasek, 1984]. Yet, they provide only indirect and noisy learning signals. First, they are not always directed at grammar specifically. Second, even in cases when grammar is the main target, the signal is global

(i.e., utterance-level) and does not identify the specific error within the utterance. This makes *credit assignment* a central challenge [Marcus, 1993; Pinker, 1989; Saxton, 1997], and also makes interactive learning from caregiver feedback an interesting *empirical* case.

Nevertheless, little is known about how these different feedback types contribute to children’s grammatical learning. While corpus studies allow a quantitative account of caregivers’ feedback behavior, *the causal effect* of each type of feedback or reward on learning remains difficult to establish. A common methodological issue facing previous attempts [e.g., see discussion in Saxton, 2005a] is that different types of feedback are provided to the same child, making it difficult—and ethically problematic—to isolate their effect over the developmental timescales on which language is learned.

Recent computational work has begun to revisit these questions by using LMs as controlled learning environments for testing hypotheses about language learning, thereby combining the targeted precision of experimental methods with the ecological validity of corpus studies [e.g., Misra, 2024]. Particularly relevant to our question are studies that used Reinforcement Learning to investigate LMs’ interactive learning strategies [Ma, 2024; Mayer Martins, 2025; Zhao, 2023; Padovani, 2025a; Xu, 2025; Nikolaus, 2021b].

Recent work has applied this approach to *natural* child–caregiver interactions, using Reward Models trained from caregivers’ actual feedback data [Nikolaus, 2026]. Language models are then fine-tuned using techniques inspired by Reinforcement Learning from Human Feedback [RLHF, Ouyang, 2022]. This work showed that caregiver-like clarification requests on model generation can signal negative evidence. Learning from this signal improved grammaticality.

The current study

Existing RLHF-based studies of naturally occurring caregiver feedback have focused primarily on clarification requests, leaving open whether other forms of social feedback have similar, weaker, or even negative effects on grammatical learning. The present study addresses this gap by comparing several theoretically motivated feedback types within the same controlled learning framework.

5.3.2 Methods

Our method uses a two-stage learning setup designed to separate learning from linguistic exposure from learning driven by caregiver-feedback rewards. In a first step, a small causal language model is pretrained on child-directed language (excluding children’s production), providing an *input-only* baseline that quantifies learning gains from linguistic exposure. In a second step, the model is fine-tuned via RL, using a reward model trained on the target caregiver’s feedback valence.

Pre-training baselines We use a customized version of GPT-2 (2 layers, hidden size 512, 8 attention heads), pre-trained on CHILDES [MacWhinney, 2000a], using three pre-training scales (0.1M, 1M, and 10M words, see Appendix 4).

Empirical Reward Models For each feedback type, we trained reward models to capture

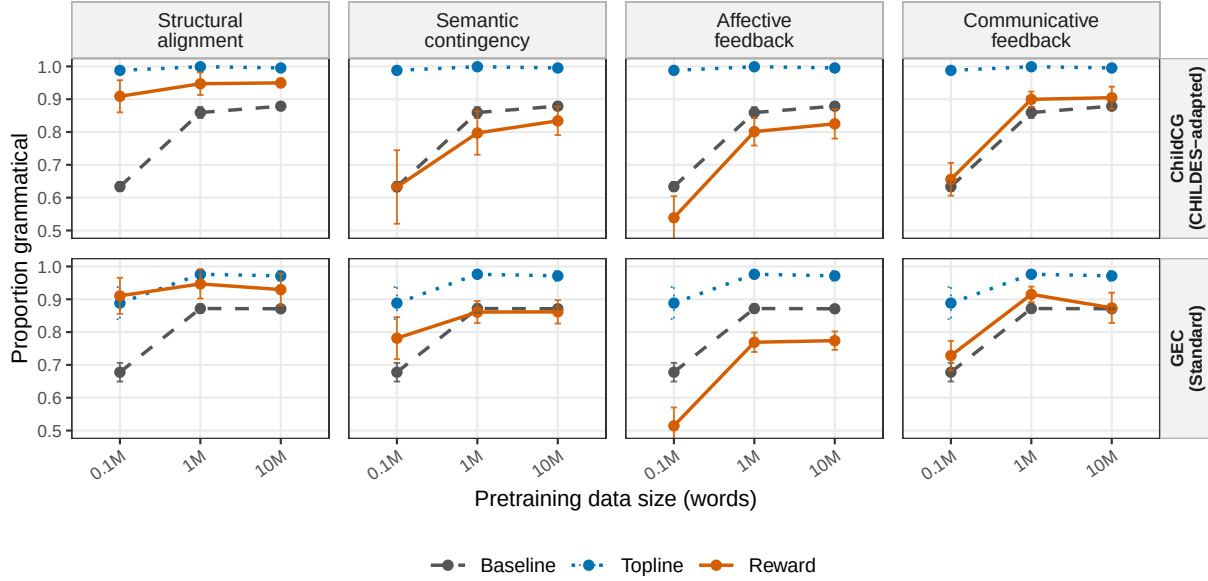


FIGURE 5.5 – Generated-utterance grammaticality across pretraining scales for each reward. The top row shows results from the CHILDES-adapted ChildCG classifier, while the bottom row shows results using a standard GEC-based metric. Within each panel, baseline and topline trajectories are repeated to make the reward trajectory directly comparable. Means are computed over training seeds units after averaging over generation seeds. Error bars show ± 1 standard deviation across training seeds.

naturally occurring patterns of caregiver feedback in child–caregiver interactions (using CHILDES dataset). Each reward model is trained to map children’s utterances to a scalar, representing the valence of the target feedback (see Appendix 4). The valence values themselves were annotated prior to reward-model training using a range of automatic annotation tools (see below, and Appendix 3).

Topline Reward Model: we use an idealized reward that directly targets grammatical errors. This topline is not intended to produce empirically realistic results, but rather to serve as an upper bound on what grammatical learning is possible *in principle* (see details in Appendix 4).

RL fine-tuning For each feedback/reward, the pre-trained models were fine-tuned using the corresponding reward models, using Proximal Policy Optimization [PPO; Schulman, 2017]. We included several safeguards to mitigate potential language drift (see Appendix 5).

Training seeds and variability To account for stochastic variability, we used, for each training scale, a fully crossed $3 \times 3 \times 3$ design over pretraining seeds, fine-tuning seeds, and generation seeds.

Annotation of the reward signals We considered the four classes of caregiver feedback mentioned in the introduction: communicative feedback, semantic contingency, structural alignment and affective feedback. For each type of feedback, we annotated the CHILDES child-caregiver conversations for the corresponding valence (See Appendix 3).

5.3.2.1 Evaluation

We use two complementary evaluations that capture different aspects of grammatical behavior. **Minimal-pair benchmarks:** we use Blimp [Warstadt, 2020a] and its CHILDES-adapted version Zorro [Huebner, 2021b]. They test whether a model assigns higher probability to a grammatical sentence than to a minimally differing ungrammatical one.

Grammaticality of generated utterances Free generations were scored using Child Conversational Grammaticality [ChildCG, Nikolaus, 2024], a model adapted to CHILDES and its conversational context. In addition, we used a measure based on an off-the-shelf Grammatical Error Correction model [e.g., Rothe, 2021]. For each run and generation seed, we sampled up to 10,000 utterances using the Beginning-Of-Sentence (BOS) token as a prompt.

Both evaluation methods are broadly connected to child-language methodologies for evaluating grammatical development, which have included both controlled forced-choice tasks and analyses of children’s spontaneous and elicited productions [Ambridge, 2011].

5.3.3 Results

First, we compared the baseline models, trained only on linguistic input, with the same models further fine-tuned using the grammar topline reward model (Figure 5.6 in Appendix). The topline improves both minimal-pair benchmark performance and the grammaticality of generation, showing the current framework can, in principle, lead to *grammatical* learning above and beyond pre-training.

Moving to the empirical reward models, we found no consistent improvement on the minimal-pair benchmarks (Figure 5.7 in Appendix). Across Zorro and BLiMP, none of the reward types reliably improved over the baseline. This echoes a broader pattern in recent studies showing that interactive strategies struggle to improve performance on these benchmarks (see also Limitations). In contrast, the generation results showed measurable effects on grammaticality (Figure 5.5). Across both ChildCG and GEC, the overall trends were largely consistent. Structural alignment produced the strongest and most consistent improvement across scales and metrics. Communicative feedback showed an overall positive effect, especially under ChildCG, but the gains were smaller. In contrast, semantic contingency and affective feedback generally reduced grammaticality below the baseline.

Regarding the effect of pre-training size, the low-resource condition at 0.1M was visibly noisier, with larger variability across runs, excessively high non-word rates (Figure 5.8 in Appendix, bottom row), and overall low grammaticality scores. The effects became much more consistent from 1M words onward, suggesting that a minimum amount of pre-training data is needed before reward-specific effects can emerge robustly.

We tested these qualitative patterns statistically using a mixed-effect logistic model. Predictors included reward type, pretraining scale, and their interaction. We additionally added the following predictors as controls a) utterance length, since longer utterances may naturally provide more opportunities for errors, and b) rate of non-words generated, since out-of-dictionary forms may be penalized by grammaticality classifiers for reasons that are not strictly grammatical. Finally,

Reward family	1M	10M
Structural alignment	+.077***	+.060***
Semantic contingency	−.092***	−.061***
Affective feedback	−.038*	−.031*
Communicative feedback	+.033*	+.024 [†]
Acknowledgement	+.017	+.017
Clarification request	+.050***	+.031*

TABLE 5.2 – Model-estimated change in grammaticality probability relative to baseline, using the ChildCG metric at the 1M and 10M pretraining scales. [†] $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

we included a random intercept for generation run to account for the fact that sampled utterances from the same seeds are not independent.

Table 5.2 reports model-estimated changes in grammaticality probability relative to the baseline, using the more robust pretraining scales (1M and 10M). Mirroring the pattern in Figure 5.5, structural alignment showed the clearest improvement. In contrast, semantic contingency and affective feedback both reduced grammaticality. Communicative feedback improved grammaticality at 1M and showed a marginally significant positive effect at 10M. When communicative feedback was split into its two components, only clarification requests showed a significant effect, whereas the acknowledgment-based reward model did not.

5.3.4 Discussion and Conclusions

We investigated which forms of caregiver feedback support LMs’ grammaticality within an RLHF-inspired framework. We isolated and compared major feedback types, providing insight into their specific effects and relative contributions.

The main finding was that Structural Alignment systematically improves grammaticality. While previous work found correlational evidence to this link [Fusaroli, 2023], the current result demonstrates, at scale, a novel and plausible *mechanistic* account: Caregiver’s structural alignment can support grammar learning *because* this reinforces the child’s use of structures that are grammatical—caregivers naturally tend to (re-)use grammatical sequences more often than ungrammatical ones. Communicative Feedback also improved grammaticality, though to a lesser extent. The improvement is consistent with prior work, especially on clarification requests [Nikolaus, 2026; Clark, 2020b].

As for Semantic Contingency and Affective Feedback, they *reduced* grammaticality. We did not anticipate this outcome; however, it can be understood in the broader context of development. Based on the supplementary analyses in Figure 5.8, and models’ generation samples in Table 5.3 (Appendix), Affective Feedback increased utterance length, suggesting a role in eliciting longer vocalization [see also Goldstein, 2003]. Semantic Contingency increased the use of content words and preserved greater lexical diversity, in line with findings in lexical development [Che, 2018].

To conclude, different types of feedback push generation in different directions: some improve grammaticality, while others encourage other properties of language learning and use. Future work should adopt a broader framework, investigating how these feedback types may support

complementary aspects of language learning beyond grammar.

Limitations

A limitation of our study is that the effects of the empirical rewards were observed in model generation, but not in minimal-pair benchmarks. This echoes a broader pattern in recent studies showing that interactive strategies generally struggle to improve performance on minimal-pair benchmarks of grammar such as BLIMP and Zorro [Zhao, 2023 ; Padovani, 2025a ; Nikolaus, 2026 ; Stöpler, 2025].

This raises an important theoretical question: To what extent does grammaticality in generation reflect grammatical knowledge? This remains a debated question in child language research [Ambridge, 2011 ; Chater, 2018] ; and we do not aim to address this general question here.

In the specific case of the present study, however, it is cautious to interpret the generation-based results more narrowly: they test how reward fine-tuning changes the model’s productive behavior; they do not directly test whether the model has acquired grammatical knowledge that can generalize. Note that even under this conservative interpretation, the generation-based results remain informative for theories of language development. Unlike traditional corpus-level correlations, the rewards here were used as interventions within a computational learning system, allowing us to test whether optimizing for these signals *causally* shifts the model’s production policy in grammar-relevant directions.

Nevertheless, a model that becomes more grammatical in free generation does not necessarily mean it has learned grammatical regularities that generalize beyond the specific utterance types favored during fine-tuning. Minimal-pair benchmarks—though not perfect [Martínez, 2023]—are useful precisely because they intend to test this stronger criterion for learning. Our null results on BLiMP/Zorro (Figure 5.7) therefore matter. They suggest that the empirical rewards did not produce grammatical generalization—at least, not of the kind captured by these benchmarks.

That said, while this conclusion may be partly correct, it would be premature to draw it too strongly here without further research. Existing minimal-pair benchmarks do not exhaust the space of possible grammatical generalizations. In particular, even child-adapted benchmarks such as Zorro primarily target classic grammatical phenomena, but do not directly target some of the errors that are especially common in child language *production* and may therefore also be reproduced in our generated data—including missing subjects, objects, auxiliaries, or determiners, among others [Saxton, 2005b ; Hiller, 2016 ; Nikolaus, 2024]. This is important because, in an RL fine-tuning setting, errors that are sampled in the model’s productions are the ones that receive direct corrective pressure from the reward signal.

Another concern is that reward fine-tuning may narrow the model’s production distribution, reducing diversity and potentially harming generalization. However, while this may be partly true, some narrowing may still be compatible with grammatical learning, and perhaps even expected if the model moves away from noisy or ill-formed productions. For instance, in our analyses, the grammar topline was relatively the least lexically diverse condition (Figure 5.8, Lexical Entropy, Appendix), yet it produced the clearest gains on the minimal-pair benchmarks (Figure 5.6, Appendix). Thus, reduced lexical diversity does not in itself prevent grammatical

generalization. The more central question is whether reward-induced changes target grammatical phenomena that can generalize beyond the sampled utterances themselves.

Testing this stronger learning claim, thus, remains an important direction for future work. It requires conducting large scale corpus exploration of the grammatical errors found in both child and model productions, and to use these bottom-up insights to build targeted tests around the relevant phenomena. This would help determine whether the reward-induced gains observed here reflect merely local shifts in surface production patterns or more abstract grammatical generalizations.

Ethical considerations

This work uses existing child-caregiver corpora and does not involve new data collection. Because the data involve children, we treat the corpora as sensitive and report only aggregate model-level results, without attempting to identify individual children, caregivers, or families. The proposed modeling framework is intended as a tool for studying hypotheses about language development, not for evaluating individual children or caregivers.

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Appendix 1: Supplementary Analyses

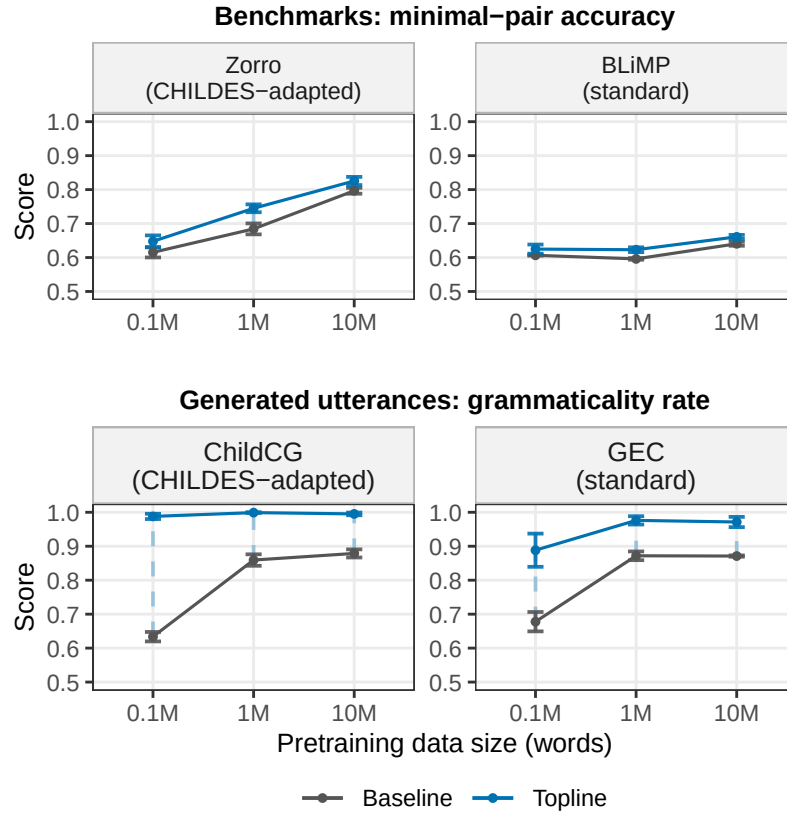


FIGURE 5.6 – Baseline and Topline performance across pretraining scales. The top row reports minimal-pair benchmark accuracy for Zorro and BLiMP. The bottom row reports grammaticality of generated utterances, using the ChildCG or GEC-based metric. Points show means across trainings seeds (after averaging over generation seeds). Error bars show ± 1 standard deviation across these independent units.

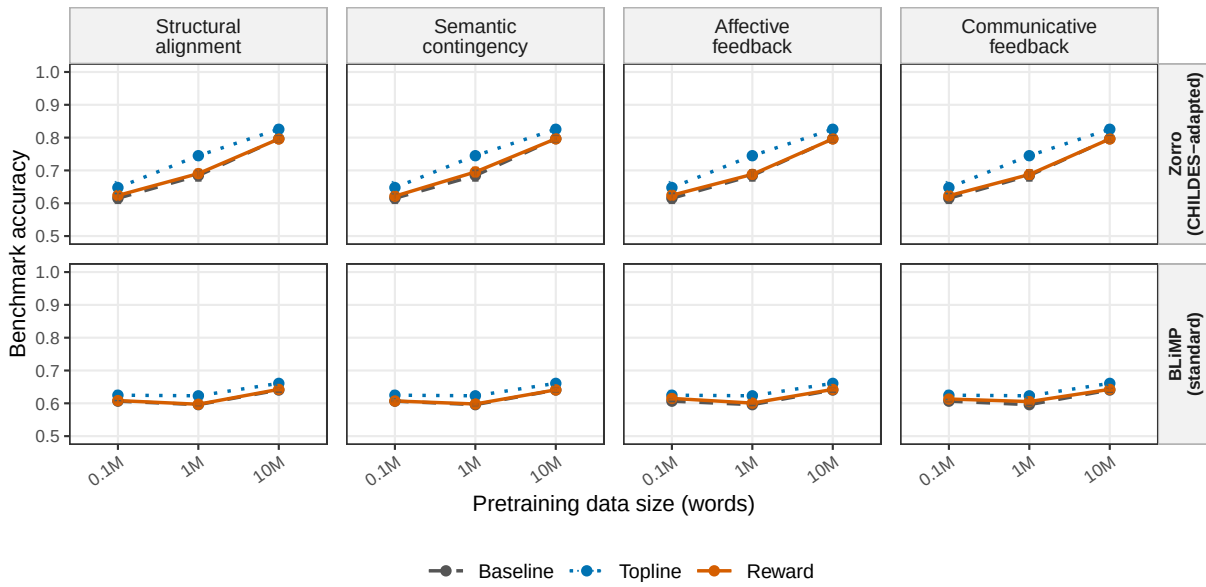


FIGURE 5.7 – Benchmark performance across pretraining scales for each reward family. Rows show Zorro and BLiMP minimal-pair accuracy; columns show the four empirical reward families. Within each panel, the baseline and topline trajectories are repeated to make the reward trajectory directly comparable at each scale. Means are computed over training seeds, after averaging over generation seeds. Error bars show ± 1 standard deviation across independent units.

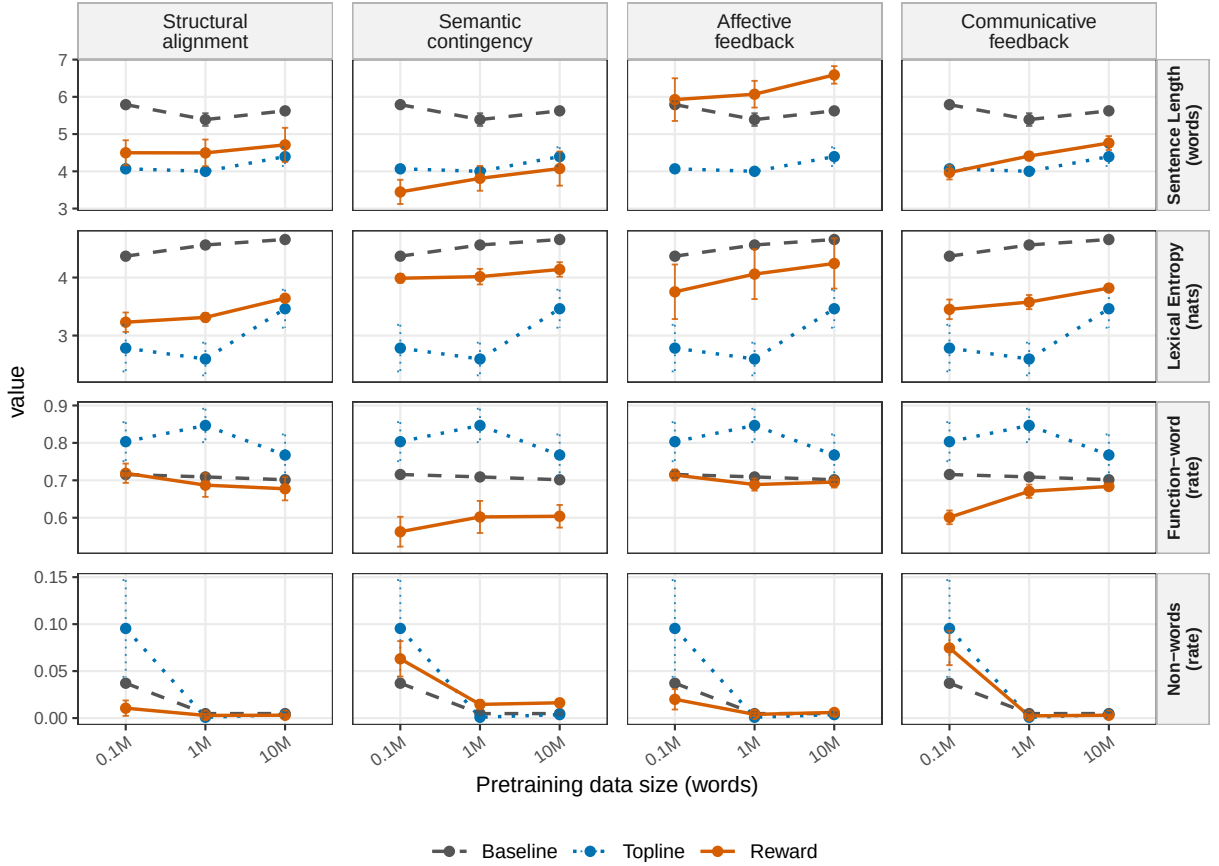


FIGURE 5.8 – Properties of generated utterances: Sentence length, pooled lexical entropy, function-word density, and rate of non-words. Points show means for each reward across training seeds, with error bars indicating ± 1 standard deviation across these seeds. Within each panel, baseline and topline trajectories are repeated to make the reward trajectory directly comparable at each scale.

Appendix 2: Examples of Model-Generated Utterances

Reward type	ChildCG = 1 (grammatical)	ChildCG = 0 (ungrammatical)
Baseline	<i>you want to look at some more pictures?</i> <i>oh, is that what it looks like?</i> <i>it is a big boy now isn't it?</i> <i>have you not any other little boys?</i>	<i>this piece of cake?</i> <i>because you can see mummy's poorly sick.</i> <i>the dog is saying.</i> <i>he is the name of the little girl.</i>
Topline	<i>what are you doing?</i> <i>there is a horse.</i> <i>that is a girl.</i> <i>are you going back?</i> <i>that is a chair.</i> <i>what is that one?</i> <i>i will put it there.</i> <i>what did we do with the barn?</i>	<i>why is that side?</i>
Structural Alignment	<i>so that is a duck.</i> <i>he is a turtle.</i> <i>you are a good boy.</i> <i>i am a monkey.</i>	<i>how many trucks in the sky.</i> <i>come down the street.</i> <i>we are a good boy.</i> <i>can't find it.</i>
Semantic Contingency	<i>yes that is a doggie.</i> <i>well it is a boat.</i> <i>there is an elephant.</i> <i>yes a banana puppy.</i>	<i>where is baby sheep?</i> <i>what is big truck?</i> <i>he a lion.</i> <i>a big flower red.</i>
Affective feedback	<i>and then i will put it in it.</i> <i>i will put it over here if i put it in there.</i> <i>we have to wash her hair first okay?</i> <i>we will shall we put it all together?</i>	<i>if i do it and then we can do it and put it in?</i> <i>now i will have to clean this with your room won't you?</i> <i>want mama do it?</i> <i>because your teeth i am watching this picture aren't they?</i>
Communicative feedback	Acknowledgement <i>they are called flowers.</i> <i>that is a gold straw.</i> <i>and that is the baby.</i> <i>you have a pink one on there.</i>	<i>have to use your spoon and pretend.</i> <i>you will use them ones with this bits.</i> <i>all day and then the is warm.</i> <i>but two grey stars.</i>
	Clarification request <i>i don't know.</i> <i>can you see it?</i> <i>can't you?</i> <i>mhm, it is.</i>	<i>look at that isn't it?</i> <i>look it, there is it?</i> <i>see you shouldn't?</i> <i>there doesn't.</i>

TABLE 5.3 – Examples of utterances generated by RL-fine-tuned models at the 10M-word scale (generation was qualitatively similar at the 1M-word scale). Examples are organized by reward types. Unshaded cells show utterances that were classified as grammatical by ChildCG, while shaded cells show utterances classified as ungrammatical.

Appendix 3: Automatic labeling of rewards

We automatically annotated caregiver feedback signals in child-caregiver conversations and used these annotations to supervise reward model training (see Appendix 4).

Communicative feedback

Clarification requests Clarification requests were identified using a DeBERTa-v3-xsmall classifier fine-tuned on manually annotated caregiver responses in CHILDES, following prior work [Nikolaus, 2026].

Acknowledgements To identify caregiver acknowledgements, we relied on the feature-based algorithm described in NIKOLAUS et al. [Nikolaus, 2023c], which was specifically designed for CHILDES data and combined common keywords (e.g., “okay”, “alright”, and “yeah”) as well as repetition ratios capturing repetition-based acknowledgements (e.g., “It isn’t very nice, is it?” – “It isn’t.”).

Structural Alignment

Following previous research [Dale, 2006; Fernandez, 2014], a child structure was defined as a POS bigram. We used spaCy (`en_core_web_sm`)⁸ to tag each child utterance and caregiver response for part-of-speech (POS), excluding punctuation and empty tokens. Caregiver alignment was computed as a normalized score of bigram re-use, corresponding to the size of the intersection between the child and caregiver POS-bigram sets, divided by the size of the larger set.

Semantic Contingency

Following previous research [e.g., Fusaroli, 2023; Foushee, 2022], semantic contingency was computed using sentence embedding similarity. Here, we used `all-MiniLM-L6-v2`⁹ to encode the child utterance and caregiver response. We computed the cosine similarity between the two embeddings, and clipped the resulting score to the [0, 1] range.

Affective feedback

We annotated affective properties of caregiver responses using `roberta-base-go_emotions`.¹⁰ The model was applied to adult responses only. For each caregiver utterance, the classifier returned a probability distribution over several labels. We used the probabilities for emotions related to supportiveness, approval, and warmth.

8. https://huggingface.co/spacy/en_core_web_sm

9. <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

10. https://huggingface.co/SamLowe/roberta-base-go_emotions

Appendix 4: Baseline and Reward model training

Baseline models

For the input-based baselines, we trained a small GPT-2-style causal language model on caregiver utterances only. The model used 2 hidden layers, 8 attention heads, and a hidden size of 512. We trained a word-level tokenizer on the training data, retaining words that appeared at least twice and capping the vocabulary at 5,000 types. Models were optimized with the standard GPT-2 causal language-modeling objective using AdamW and a batch size of 256. For each data scale, we held out 10% of the data for validation and used validation loss for early stopping and checkpoint selection. To assess the effect of input size, we trained models on three amounts of caregiver language: 0.1M, 1M, and 10M words. Each condition was run with three different random seeds.

Reward models

For each of the four caregiver feedback dimensions above, we train a reward model that maps a child utterance to the corresponding valence of the caregiver response, as annotated using the procedures described in Appendix 3.

All reward models share the same pre-trained model: `microsoft/deberta-v3-xsmall` [He, 2022], which we fine-tune using linear regression layer and a mean squared error (m.s.e.) loss to predict the target reward for a given utterance.

All reward models are trained on the CHILDES child-caregiver conversation pairs (one row per child utterance with the immediately following caregiver response), the same pre-processed pipeline above. We split the data into 90% training and 10% held-out evaluation with a fixed random seed. We used the AdamW optimizer with an initial learning rate of 1.41×10^{-5} , a batch size of 128, and early stopping based on MSE on the held-out validation set. The best checkpoint is selected on held-out MSE and used as the frozen reward model during PPO fine-tuning.

Topline Reward Model

The topline reward model was obtained using the same procedure, except that instead of learning the mapping between child utterances and the feedback valence, it was trained on the controlled minimal-pair datasets used in Zorro and BLiMP. More specifically, the model was trained to classify sentences from these datasets according to their labels (i.e., grammatical or ungrammatical).

Appendix 5: RL Fine-tuning Details

After pretraining on child-directed CHILDES utterances, we fine-tune each language model with Proximal Policy Optimization [PPO; Schulman, 2017], implemented using Hugging Face’s TRL library. For each fine-tuning step, we (a) sample utterances from the current language model, (b) compute the corresponding rewards using the reward model, and (c) update the model weights using PPO.

Hyper-parameter	Value
Max PPO total steps	6000
Batch size	1024
Mini-batch size	512
KL controller	adaptive, target $KL^* = 6$
Entropy coef. c_e	1×10^{-3}
LM mixing coef. λ_{LM}	1×10^{-3}
Score clipping	disabled
Sampling	$T=1.0$, top- $p=1.0$, top- $k=0$
Precision	bf16 mixed
Eval frequency	every 100 steps
Log frequency	every 10 steps
Early stopping	no reward gain for 10 log windows

TABLE 5.4 – PPO fine-tuning hyper-parameters, shared across the four caregiver-feedback rewards, the grammar topline, and all pretraining scales.

To obtain a diverse set of generated utterances, we randomly prompted the model with short utterance prefixes (the first one or two tokens) sampled from the pretraining data, namely the caregiver utterances used to train the input-based baseline model. We additionally included an entropy regularization term (0.001) in the loss.

The models were fine-tuned for a maximum of 6,000 steps, and the best checkpoint was selected based on mean reward. To discourage excessively short or long utterances, we applied rejection sampling by assigning a reward of -1 to generated utterances that were shorter than three tokens or failed to produce an end-of-sequence token within 20 tokens.

To mitigate language drift, a known issue in Reinforcement Learning (RL) fine-tuning, we also added a small language-modeling regularization term (weight = 0.001). We used the TRL default optimizer and learning rate. Other Hyperparameters are shown in Table 5.4. All other PPO hyperparameters were not changed from the default values as implemented in the Huggingface TRL library.

Each PPO run uses one NVIDIA H100 GPU (96 GB) with bf16 mixed precision and completes in approximately 6 GPU-hours.

5.4 Discussion

This chapter examined two aspects of bringing social interaction into computational research on language acquisition: the simulation of child–caregiver dialogue and the learning effects of caregiver-derived feedback. The two studies addressed different questions, using different models and experimental procedures. Their joint contribution lies in identifying what each approach can establish, and what remains necessary for a more integrated account of learning through interaction.

5.4.1 Simulation Fidelity and Learning Effects: Distinct Contributions

The first study showed that language models can approximate some properties of child and caregiver language, but that this approximation depends on the level of analysis and the interaction setting. Providing conversational examples improved several word- and utterance-level properties, especially in caregiver-like generation. However, discrepancies remained at the dialogue level: simulated caregivers showed higher semantic alignment and lower semantic diversity than the human reference. Extended interactions also increased several differences between model-generated and human language. These findings indicate that plausible responses to individual utterances do not, by themselves, establish that a model can reproduce the properties of sustained child-caregiver interaction.

The second study showed that caregiver-derived rewards have different effects on model production. Structural alignment produced the clearest improvements in generated grammaticality. Communicative feedback had smaller effects, with the more consistent benefits associated with clarification requests rather than acknowledgments. Semantic contingency and affective feedback generally reduced grammaticality under the evaluations used, while also changing other properties of generation. Importantly, the empirical rewards did not consistently improve performance on BLiMP or Zorro. The evidence therefore supports changes in productive behavior, without establishing broader grammatical generalization.

These contributions should be distinguished. The first study evaluates models as simulators of interaction; it does not investigate the learning process through which children acquire the simulated behaviors. The second uses models as controlled learners; it does not reproduce a complete conversation with a caregiver. In particular, its rewards are derived from natural conversations, not from the interlocutors evaluated in the first study. The benchmark therefore does not validate the learning environment of the reinforcement-learning study, and the reinforcement-learning results do not demonstrate improved performance on the interaction benchmark.

5.4.2 Implications for Computational Models of Language Acquisition

A first implication concerns the criteria used to evaluate computational models. Across both studies, improvement on one measure did not establish improvement on another. In the interaction benchmark, closer correspondence with human utterance-level properties did not eliminate differences in dialogue organization. In the learning study, increased grammaticality in free generation was not accompanied by consistent gains on controlled grammatical contrasts. Although these discrepancies concern different tasks, they raise a common methodological issue: the adequacy of a model must be assessed with respect to the particular behavior or learning claim under investigation. A single successful outcome is insufficient to establish a general account of language acquisition.

This is especially important when interpreting changes in model production. A model can produce a larger proportion of grammatical utterances by changing which utterance types it favors, without necessarily acquiring grammatical distinctions that transfer to other contexts. At the same time, minimal-pair benchmarks do not exhaust the grammatical regularities that a

learner might acquire. As discussed in the second study, they may not directly target the errors most frequently encountered in the model’s own productions. The disagreement between the two evaluations should therefore neither be ignored nor treated as conclusive evidence that no learning occurred. It identifies a more specific question: which regularities changed, and under what conditions do those changes generalize?

A second implication is that caregiver feedback should not be treated as a single, interchangeable source of support. Research on child language has described several ways in which caregivers respond to children’s contributions [Clark, 2020b; Nikolaus, 2023b]. The learning study makes this distinction experimentally explicit: optimizing different feedback signals leads to different production outcomes. The lower grammaticality observed under semantic and affective rewards does not establish that these forms of feedback are detrimental to children’s development. Rather, it shows that the corresponding computational objectives do not uniformly favor grammatical production. Changes in length or lexical diversity motivate further questions about other dimensions of learning, but do not themselves demonstrate developmental benefits.

The comparison also requires care with similarly named measures. Semantic alignment in the interaction benchmark and structural alignment in the learning study capture different properties and play different experimental roles. The former is used to assess correspondence with human dialogue, whereas the latter supplies a reward for model training. Their results cannot be combined into a general claim that more alignment is either beneficial or harmful. For acquisition research, the relevant question is which property of a caregiver response provides useful information, for which aspect of learning, and under which conditions.

5.4.3 Limitations and Directions toward Integrated Interaction-and-Learning Models

The studies leave several gaps between the present models and children’s language acquisition. In the benchmark, age-conditioned generation reflects the behavior of already trained models asked to simulate children of different ages. It is not a developmental trajectory produced by accumulating child-like experience. In the learning study, controlled input sizes and feedback conditions make experimental comparisons possible, but do not establish that the learning procedure corresponds to children’s learning. In addition, the pretrained components used to predict rewards and evaluate outputs introduce knowledge beyond the learner’s caregiver-input corpus. These distinctions limit how directly the results can be interpreted as accounts of human development.

The measures also provide incomplete descriptions of interaction and learning. The benchmark focuses mainly on linguistic properties and does not comprehensively assess whether a response appropriately addresses a child’s intended meaning, communicative difficulty, or need for repair. Its multi-turn condition involves two imperfect simulators, so observed discrepancies may reflect their combined behavior rather than the limitations of either participant alone. Similarly, the learning study relies on automatic feedback annotations, learned reward predictions, and automatic grammaticality measures. Further validation should examine these components separately, including whether reward predictions remain reliable on utterances generated by the learner

rather than drawn from the original corpus.

A further limitation concerns how feedback is represented. During reinforcement learning, the reward model is fixed and predicts a feedback value from a generated utterance. It does not produce a caregiver response that changes with the evolving conversational context or the learner’s history. This abstraction allows particular signals to be isolated, but leaves out the linguistic content of an online response and the possibility of adjustments by both participants. Natural interaction provides a setting in which caregiver responses and children’s subsequent contributions are related over time [Masek, 2021 ; Fusaroli, 2023]. Modeling that relation requires more than adding a fixed reward to otherwise independent productions.

An integrated approach could connect a learning model to a caregiver model whose responses are evaluated against natural interaction. The learner would update through repeated exchanges, while evaluation would track both the properties of the interaction and changes in the learner’s language. Such a system would require separate tests of simulation fidelity and learning effects; improvements in one should not be assumed to validate the other. Matched controls could help isolate the contribution of contingent feedback, for example by disrupting the relation between an utterance and its feedback while preserving other properties of the learning setting.

Finally, stronger tests of acquisition would examine whether improvements persist beyond the utterances favored during training. This could involve identifying recurrent errors in child and model productions, constructing targeted contrasts, and testing transfer to new lexical items and combinations. Evaluation materials should be separated from those used to train reward models and optimize the learner. These tests would help distinguish changes in production preferences from the acquisition of regularities that can be applied more broadly. They would also provide a basis for investigating combinations of feedback signals, rather than assuming that the effects of isolated rewards will remain unchanged when they occur together.

5.4.4 Chapter Conclusions

This chapter provides two complementary contributions to computational research on language acquisition. The interaction benchmark establishes a framework for comparing simulated and natural child–caregiver dialogue, and identifies discrepancies that remain despite improvements in individual utterances. The feedback-learning study shows that signals derived from caregiver responses can selectively change model production, while leaving broader grammatical generalization unresolved.

Together, these findings support a differentiated approach to modeling social interaction. Reproducing its observable properties and explaining its learning effects are related but distinct objectives. The present studies make progress on each, without yet combining them into a complete model of language acquisition through interaction.

Conclusion et perspectives

6.1 Conclusions de l'étude

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6.2 Ouverture et perspectives

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

Liste des publications

Articles à comité de lecture

Co Auteur 1, Co Auteur 2, Co Auteur 3. "*Un long titre d'article*". Un journal qui a accepté mon travail.

DOI: [a1b2c3d4.xxxxx.99999](#)

Articles en préparation

Co Auteur 1, Co Auteur 2, Co Auteur 3. "*Le long titre de mon second article*". Le journal de mes rêves.

Conférences internationales (premier auteur)

Co Auteur 1, Co Auteur 2, Co Auteur 3, Co Auteur 4. "*Titre de la présentation*". Conférence de Sciences, Quelque part, 2020.

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RÉSUMÉ

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetur odio sem sed wisi.

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MOTS CLÉS

mot clé 1, mot clé 2, mot clé 3, mot clé 4

ABSTRACT

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KEYWORDS

keyword 1, keyword 2, keyword 3, keyword 4